Program overview

19-May-2025 12:54

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer and Embedded-Systems Engineering**

Code	Omschrijving	ECTS
Master CESE 2024	**Master Computer & Embedded Systems Engineering 2024**	
Compulsory Courses 2024		
CESE4000	Software Fundamentals	5
CESE4005	Hardware Fundamentals	5
CESE4015	Software Systems	5
CESE4020	Effective and Responsible Engineering	5
CESE4025	Real-time Systems	5
CESE4030	Embedded Systems Laboratory	5
CESE4040	Processor Design Project	5
CESE4085	Modern Computer Architectures	5
CESE4130	Computer Engineering	5
EE4C11	Systems Engineering	5
Specialisation Courses 2024		
Specialisation courses Software 2024		
CESE4075	Supercomputing for Big Data	5
CESE4120	Smart Phone Sensing	5
CS4205	Evolutionary Algorithms	5
CS4430	Network Security	5
CS4505	Software Architecture	5
CS4520	Security and Cryptography	5
CS4530	Modelling and Problem Solving	5
CS4535	Constraint Solving	5
CS4545	Distributed Algorithms	5
CS4555	Compiler Construction	5
DSAIT4005	Machine and Deep Learning	5
DSAIT4010	Probabilistic AI and Reasoning	5
DSAIT4020	Elements of Statistical Learning	5
DSAIT4025	Alternative Learning Strategies	5
WI4049TU	Introduction to High Performance Computing	6
Specialisation courses Computer Architecture 2024		
CESE4010	Advanced Computing Systems	5
CESE4035	Computer Arithmetic	5
CESE4075	Supercomputing for Big Data	5
CESE4090	Reconfigurable Computing Design	5
CESE4115	Embedded Computer Architecture 2	5
CS4555	Compiler Construction	5
EE4610	Digital IC Design	3
EE4615	Digital IC Design II	3
EE4690	Hardware Architectures for Artificial Intelligence	5
EE4695	Hardware Dependability	5
EE4700	Modelling, Algorithms and Data Structures	5
ET4351	Digital VLSI Systems on Chip	4
ET4362	High Speed Digital Design for Embedded Systems	5
QIST4400	Quantum Computer Architecture	5
Specialisation courses Networking 2024		
CESE4045	High-performance data networking	5
CESE4055	Ad hoc and Sensor Networks	5
CESE4060	Wireless IoT and Local Area Networks	5
CESE4065	Advanced Practical I.o.T. and Seminar	5
CESE4120	Smart Phone Sensing	5
CS4090	Quantum Communication and Cryptography	5
CS4430	Network Security	5
EE4396	Mobile Networks	5
EE4630	Telecommunication Network Architectures	3
EE4C06	Networking	5
ET4034	Telecom Business Architectures and Models	4
ET4358	Fundamentals of Wireless Communications	5
IN4341	Performance Analysis	5
SC42101	Networked and Distributed Control Systems	4
Specialisation courses Control 2024		
RO47019	Intelligent Control Systems	4

SC42001	Control System Design	5
SC42015	Control Theory	6
SC42025	Filtering & Identification	6
SC42056	Optimization for Systems and Control	3
SC42061	Nonlinear Control Systems	3
SC42075	Modeling and Control of Hybrid Systems	3
SC42095	Digital Control	3
SC42101	Networked and Distributed Control Systems	4
SC42110	Dynamic Programming and Stochastic Control	5
SC42125	Model Predictive Control	4
SC42130	Fault Diagnosis and Fault Tolerant Control	4
SC42145	Robust Control	3
SC42150	Statistical Signal Processing	3
SC42155	Modelling of Dynamical Systems	3

Thesis Project 2024

CESE5000	Thesis Project	45

Suggestion free elective space: Internship & Projects 2024

CESE5010	Industry Internship	15
CESE5020	Research Internship (10-15 EC)	10
IFEEMCS520201	Advanced Interdisciplinary Artificial Intelligence Project	15
IFM4040	Joint Interdisciplinary Project	15

Language Courses 2024

TPM018A	English Grammar for the University	2
TPM302B	Spoken English for Academic Purposes Advanced	2
TPM305A	Writing a Masters Thesis in English	2
TPM306A	Writing Academic English using AI	2
WM1115TU	Dutch Elementary 1	3
WM1116TU	Dutch Elementary 2	3
WM1117TU	Dutch Intermediate 1	3

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer and Embedded-Systems Engineering**

Master CESE 2024	
Introduction 1	The individual examination programme (IEP) has a study load of 120 credits and consists of: A. Common core courses worth 25 credits, including a homologation course worth 5 credits B. Compulsory integration set courses worth 20 credits C. Courses from one of the specialisation lists, worth at least 15 credits, D. Free elective courses worth at least 15 credits E. A thesis project worth 45 credits (CESE5000 Final Project). If the total number of credits under subsection A to subsection E is lower than 120 credits the IEP should be completed with free electives to a minimum of 120 credits.

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer and Embedded-Systems Engineering**

Compulsory Courses 2024	
Introduction 1	Common core courses worth 25 credits, including a homologation course worth 5 credits Compulsory integration set courses worth 20 credits. Students taking 15EC in the specialisation Control can opt to take an additional 10 EC of Control specialisation courses instead of the CE Integration set CESE4085, CSES4040.
Introduction 2	CESE4000 Homologation course for students with an Electrical Engineering Bachelors degree Electrical Engineering CESE4005 Homologation course for students with a Computer Science bBachelors degree Computer Science

CESE4000	Software Fundamentals	5
Responsible Instructor	Prof.dr. K.G. Langendoen	
Instructor	Ir. J.B. Dönszelmann	
Contact Hours / Week x/x/x/x	12/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	This course serves as homologation for non-computer science students that enrolled in the Computer and Embedded-Systems Engineering programme. This course provides students with the fundamentals needed to successfully develop software systems in Rust, both as a single developer and when working in a team.	
Study Goals	Students will be able to: * Explain the programming language concepts followed in Rust. * Design, implement and debug a small software system from scratch in Rust following the language standard including proper coding style. * Set up a project and build environment, using the Rust ecosystem. * Use Git to version and share source code contributions for collaborative development. * Evaluate and integrate code contributions of other team members.	
Education Method	Lectures + lab	
Assessment	This is a skills-based course, so the grade will be composed of lab results (individual assignments + group project). When the end result is not a mark 6 or higher a (individual) repair option will be provided. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Enrolment / Application	The capacity is limited and -as this is a compulsory course for CESE students- they get preference over other MSc students. Note: this is a homologation course, so extra restrictions may apply depending on the specific MSc program.	

CESE4005	Hardware Fundamentals	5
Responsible Instructor	Prof.dr.ir. G. Gaydadjiev	
Instructor	Dr.ing. A.B. Gebregiorgis	
Contact Hours / Week x/x/x/x	8/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	This purpose of this course it to familiarize students with a Computer Science-focused background in topics from the field of Electrical and Control Engineering. It serves as a homologation course.	
Study Goals	The course covers three main areas: * Transistors as the basic devices in digital computers; * Signals and Systems (Discrete and Continuous); * Basics of Control System Theory. For each area, we have defined learning objectives. In the following, we list them per area. In the area of "Transistors for Digital Computers", after following this course, the student is able to: * explain MOSFET characteristics; * distinguish between combinational and sequential circuit implementations; * analyse simple logic functions in terms of area, delay and power/energy usage; * describe simple hardware functions both at the register-transfer level (RTL) and using a hardware description language such as VERILOG; * discuss the relationships between the transistor properties and logic gate characteristics. In the area of " Discrete Signals and Systems", after following this course, the student is able to: * describe and mathematically model discrete and continuous systems; * describe system properties and classifications; * linearize non-linear ODEs systems; and * describe and mathematically model simple physical systems: ODEs, TF, Bode plots. In the area of "Control Systems", after following this course, the student is able to: * describe stability notions; * design and analyse feedback controllers for linear systems, e.g.,: PID; and * connect to Python commands and low level software implementations.	
Education Method	The course will be taught via a collection of lectures and four (PASS/FAIL) labs. Detailed overviews can be found on the Brightspace and CESE pages of the course.	
Assessment	The course is assessed as follows: * all labs must be successfully completed - there is a minimum threshold per lab that must be reached * there is a concluding exam (graded 1-10) for which the student must achieve a passing grade, i.e., 6 or higher. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Enrolment / Application	The capacity is limited and -as this is a compulsory course for ES and CE students- they get preference over other MSc students.	
Co-Instructor	Prof.dr.ir. S. Hamdioui	
Co-Instructor	Dr. R. Ishihara	
Co-Instructor	Ing. A.M.J. Slats	
Co-Instructor	Ir. S.C. Bregman	

CESE4015	Software Systems	5
Responsible Instructor	Prof.dr. K.G. Langendoen	
Instructor	Ir. J.B. Dönszelmann	
Instructor	R. Corvino	
Instructor	Dr. G. Lan	
Contact Hours / Week x/x/x/x	0/8/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Expected prior knowledge	Literacy in Rust (CESE4000 or equivalent)	
Course Contents	This course aims to acquaint students with modern software development methods for large complex systems, in particular advanced programming language concepts (in Rust) and Model-Based Software Engineering (MBSE).	
Study Goals	At the end of the course the students will be able to: * Apply advanced programming language concepts followed in Rust: 1) at a high level, working with concurrency and synchronization; and 2) at a low level, programming device drivers for embedded systems and designing safe abstractions over unsafe code. * Apply MBSE (Model-Based Software Engineering) techniques and tools, including UML, Finite-State Machines, and Domain-Specific Languages. * Analyse the performance implications of program development at a high level of abstraction using profiling tools.	
Education Method	Lectures and labs	
Assessment	Graded assignments where both the Rust and MBSE parts need a grade >= 5.0 In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: Graded assignments = Re-submit deliverable Graded report = Re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

CESE4020	Effective and Responsible Engineering	5
Responsible Instructor	Drs. J.A. Weitenberg	
Instructor	O.E. Popa	
Instructor	Drs. P.C. Post	
Contact Hours / Week x/x/x/x	0/0/0/5 Lectures (2x2h) Tutorial (1x2h)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	Academic Writing: The Academic Writing course teaches students to write about relevant scientific research on an academic level. They will write a literature review on a subject connected to their specialisation, in which they will review relevant scientific sources and learn to compare and contrast these. Students will learn to reflect on their work and the work of their peers by doing peer reviews and processing feedback in revision rounds. Ethics: Students will familiarise themselves with the method of value-sensitive design as a method of approaching ethical problems in technological innovations. They will master this method and apply it in a case study of their choosing. Oral Presentations: The Oral Presentations course teaches students to prepare and deliver a presentation on a professional level, on a case study concerning value-sensitive design. Students will practice presenting during all lectures and will build their skills in a step-by-step approach, in which there is room for differentiation between those with less and those with further developed starting levels. Students will reflect on their progress in Oral Presentations and Ethics, incorporating the feedback from both lecturer and peers. All activities in this course have compulsory attendance.	
Study Goals	At the end of this course, students will: Master basic skills in academic writing. They will be able to write a literature review on a subject connected to their specialisation. Master basic principles of ethics. They will be able to recognize social and ethical issues in engineering and technology and critically evaluate possible solutions for the social and ethical issues coming up in engineering, by taking into account multiple perspectives and stakeholders. Master a professional level of presenting. They will be able to present an ethical issue connected to CESE to a mixed academic audience.	
Study Goals continuation	Academic Writing: At the end of this course, students will master basic skills in academic writing. They will be able to write a literature review on a subject connected to their specialisation. To achieve this goal they will be able to: 1. formulate a main question (or aim/purpose statement) with subquestions 2. support writing sufficiently and correctly with literature 3. support writing with argumentation 4. structure a text according to the scientific conventions 5. implement knowledge about the specific elements of a scientific text and their generic requirements in their own text. 6. use visualisations, such as tables and figures, appropriately. 7. apply a correct formal scientific style 8. apply the general writing strategies: genre analysis, outlining and revising. 9. determine further learning goals for their thesis project.. Ethics: Students will be able to: recognize social and ethical issues in engineering and technology and conceptualise these issues in terms of value conflicts critically evaluate possible solutions for the social and ethical issues coming up in engineering, by taking into account multiple perspectives and stakeholders. Oral Presentations: At the end of this course, students will master a professional level of presenting. They will be able to present an ethical issue connected to CESE. To achieve this goal they will be able to: 1. adjust academic and technical information to a target audience; 2. use content and rhetorical techniques to make presentations relevant and attractive to the audience; 3. use voice and body language to support the presentation; 4. apply a clear structure to a presentation; 5. use visualisation tools supporting the content of the presentation and operate these; 6. deal professionally with questions from the audience; 7. reflect on their presenting skills and formulate further personal learning goals.	
Education Method	Academic writing: 6 small-scale workshops in which students discuss and practice aspects of writing. Students will write a literature review on a subject connected to their specialization. Students will receive feedback from the lecturer as well as from their fellow students (peer feedback) to enhance their writing skills. Ethics: 1 lectures and 2 workshop on ethics. Students will then work on a case study for their final presentation. Oral Presentations: 7 small-scale workshops in which students discuss and practice aspects of presenting. Students will receive feedback on several presentation exercises from their lecturer as well as their fellow students (peer feedback) to enhance their presentation skills and prepare their final presentation on an ethical case study. Students make two reflection assignments. Attendance is mandatory for all activities.	
Literature and Study Materials	Academic writing: Brightspace course TU Write Ethics: Materials will be made available on Brightspace Oral presentations: Brightspace course TU Present	
Assessment	The final grade of CESE4020 consists of the following components: - Academic Writing: Written paper (pass/fail); (2,5 EC). - Ethics: Written assignment (pass/fail); (1 EC). - Oral Presentations: Final presentation: 50% group presentation and 50% individual presentation techniques (pass/fail) 1,5 EC). Two reflection assignments count towards the final OP grade. Repair possibilities: - Academic Writing: Reparation written paper	

	- Ethics: reparation written assignment - Oral Presentations: reparation for group presentation, individual presentation techniques, and/or reflection assignment(s). Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).
Enrolment / Application	This course is only open to students with an active and valid registration for the M-CESE, M-CE and M-ES programme

CESE4025	**Real-time Systems**	**5**
Responsible Instructor	Dr. G. Iosifidis	
Instructor	Prof.dr. K.G. Langendoen	
Contact Hours / Week x/x/x/x	0/8/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The course covers the following topics: - basic concepts of RTS - worst case execution time estimation - scheduling policies - response-time analysis - jitter analysis - handling overloads - multiprocessor scheduling - reservation-based scheduling	
Study Goals	At the end of this course, the student is able to: 1. Explain the fundamental concepts and terminology of real-time systems 2. Construct task schedules using different scheduling policies under a given set of realistic system constraints 3. Analyze the timing behavior of a system for a given system model and scheduling policy 4. Discuss advantages and disadvantages of different scheduling policies for a given platform or system 5. Discuss the effect of hardware and software interferences on the timing behavior of a given system 6. Identify (reverse engineer) parameters of a scheduling scheme or a task set from output traces of the system 7. Derive (reverse engineer) the system specification from a given implementation (in the lab) 8. Evaluate the scheduling overheads of a given implementation (in the lab) 9. Implement event-based scheduling policies on a given microcontroller (in the lab)	
Education Method	lectures with exercises (32 hrs); self study (78 hrs); lab assignments (30 hrs)	
Literature and Study Materials	Hard Real-Time Computing Systems by G.C. Buttazzo, Springer 2011	
Prerequisites	Basic software engineering, C system programming, basic Linux operating system knowledge	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 60%) - Individual Assignment: lab work (weighting 40%) Final grade calculation = 0.6 * Written Exam + 0.4 * Individual Assignment A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 6 (5.8 and higher). In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit opportunity - Individual Assignment: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

CESE4030	Embedded Systems Laboratory	5
Responsible Instructor	Prof.dr. K.G. Langendoen	
Instructor	Ir. J.B. Dönszelmann	
Instructor	Dr. G. Lan	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This highly multi-disciplinary course comes with a lab project where teams of 4 students each will have to develop an embedded control unit for a tethered electrical model quad rotor aerial vehicle (the Quadrupel drone), in order to provide stabilization such that it can hover and (ideally!) fly, with only limited user control (one joystick). The control algorithm (which is given) must be mapped onto a home-brew PCB holding a modern RF SoC interfacing a sensor module and the motor controllers. The students will be exposed to simple physics, signal processing, sensors (gyros, accelerometers), actuators (motors, servos), and basic control principles. The embedded software running on the flight controller board (FCB) driving the drone must be written in the Rust programming language. The project work (including written report) covers the entire duration of the course period, and will take approximately 128 hours, of which 32 hours are spent at the lab facilities.	
Study Goals	Student is acquainted with real-time programming in an embedded context, along with a basic understanding of embedded systems, real-time communication, sensor data processing, actuator control, control theory, and simulation. Moreover, the student has had exposure to integrating the various multidisciplinary aspects at the system level. At the end of this course, the student is able to: 1. Explain and analyse real-time communication, sensor data processing, actuator control, control theory in the context of an embedded system 2. Consider various multidisciplinary aspects at the system level 3. Apply real-time programming concepts in an embedded context	
Education Method	Lectures (6*2hrs), lab work (8*4hrs), coding at home (8*12hrs), report (8hrs), so on average 2 days per week	
Prerequisites	Literacy in Rust (CESE4000 or equivalent, and CSE4015)	
Assessment	The final grade of the course consists of the following components: - Lab Project (weighting 75%) - Written Report (weighting 25%) Final grade calculation = 0.75 * Lab Project + 0.25 * Written Report A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Lab Project: Additional assignment - Written Report: Additional report Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Enrolment / Application	The capacity is limited and -as this is a compulsory course for CESE students- they get preference over other MSc students.	

CESE4040	Processor Design Project	5
Responsible Instructor	Dr.ir. J.S.S.M. Wong	
Responsible Instructor	Prof.dr.ir. G. Gaydadjiev	
Responsible Instructor	C. Galuzzi	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	To be able to complete the project the student has to have Digital Systems and Verilog knowledge. Moreover before embarking in this course she/he has to successfully complete the following courses: CESE4085 Modern Computer Architecture and CESE4130 Computer Engineering.	
Course Contents	This is an integration course that is aiming to let the students utilise computer architecture and computer engineering knowledge in the design improvements of a general purpose processor. In this line of reasoning they have to improve the performance of a RISC-V+LLVM-based computing platform by focusing on computer architecture and compiler relevant parts of it. As starting point students receive two archives: one which includes the RISC-V System on Chip VERILOG code and a set of benchmarks selected from the GMPbench and MiBench suites, as well as all the necessary files required for simulation and FPGA implementation, and a second archive with the LLVM RISC-V cross-compiler. To evaluate their new design, students are given a Zynq-7000 FPGA board per group.	
Study Goals	The course study goals are as follows: 1. The student can perform, assuming certain processor architecture and requirements, a design space exploration and select the most appropriate algorithms for the implementation of the basic functional units. 2. The student can integrate computer engineering, compilers and computer architecture knowledge and utilise them in order to design/optimize processors. 3. The student is able to evaluate processor performance based on benchmarks and take the appropriate architectural decisions in order to improve on various design metrics. 4. The student can design arithmetic/functional units and processors and implement them using FPGA technology. 5. The student is able to operate in a small team and collaborate for the successful completion of a design assignment. 6. The student can communicate his/her proposal, experimental results, and conclusions in English using the appropriate technical language in written as well as orally.	
Education Method	Lectures and Project.	
Literature and Study Materials	There is no dedicated study material but the students are advised however to utilise as supporting study material: the RISC-V Specifications https://riscv.org/technical/specifications/ , Computer Architecture: A Quantitative Approach, J.L. Hennessy and D.A. Patterson, 4th edition and Computer Arithmetic: Algorithms and Hardware Designs, Behrooz Parhami, Oxford University Press, NY, 2000. Moreover we encourage you to utilise the IEEE Explore library to find research publications that may help you in the successful project completion.	
Assessment	To complete the course students have to submit the project archive of the new core they developed, a report describing their approach and findings, and to give a short presentation of their project in the context of a symposium. The new core functionality is verified and all projects are checked for (between groups) plagiarism. Non functional designs do not pass the project. Plagiarism can also make you fail. Your final score for the project is determined based on the following criteria: The performance of your design (DP). The benchmark scores you report are important. The higher the benchmark score you get the better but this is not the only relevant aspect. In the performance evaluation we also take into consideration the other metrics with more emphasis on the compound ones. The technical merit of your approach (TM). Aspects as innovation level and implementation quality are considered. The report (R). Report organization, content, and language are important aspects at this point. The presentation (P). Here we also look at the capability to ask questions and to answer questions from the auditorium. The CESE4040 final grade is computed as: Grade = 0.35*DP + 0.35*TM + 0.20*R + 0.10*P, Repair option: Individual Assignment or Group Assignment = Repair: Re-submit deliverable Individual Project or Group Project = Repair: Re-submit deliverable Individual Presentation or Group Presentation = Resit (new presentation) Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

CESE4085	Modern Computer Architectures	5
Responsible Instructor	Dr.ir. J.S.S.M. Wong	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	The course emphasises on theoretical aspects and practical trade-offs of computer architecture and organization of the newest microprocessors currently on the market, e.g., RISC-V, and the latest developments in computer architecture and compiler research. Quantifying design decisions in terms of performance and cost. Computer architecture subjects include: basic principles, instruction set architectures, pipelining and pipelining consequences, multiple-issue (superscalar and VLIW) processors, out-of-order execution, branch prediction, speculative execution, advanced memory hierarchies, pre-fetching, multithreaded processors and multiprocessors, energy consumption, and reliability. Compiler subjects include: an overview of different compilers, LLVM, scanners and parsers, mapping of source code into intermediate representation (IR) as well as an overview of different IRs, different code optimizations, data flow analysis and scalar optimisation, back-end compilation, instruction selection and scheduling, and register allocation.	
Study Goals	1. The student can operate with concepts and notions related to: - Instruction sets: characteristics, functions, formats, addressing modes; - Processor structure, functions, and pipelining; - Instruction level parallelism and its static and dynamic exploitation; - Distributed memory hierarchy; - Multiprocessors; - Scanning, parsing, analysis, and code generation; - LLVM tool-chain; - Intermediate Representations (IRs); - Instruction selection and scheduling; - Instruction optimization. 2. The student can instantiate a modern RISC-V core and execute code on it. The student can perform measurements on the RISC-V core related to power, performance, and area. 3. The student can use LLVM to construct, optimize and/or produce intermediate and/or binary machine code for RISC-V architectures. 4. The student can apply the learned knowledge about computer architecture and compilers to perform design-space exploration for a (small) set of benchmarks. 5. The student can study recent advances in computer architecture and compilers, classify research papers, and report about them.	
Education Method	Lectures, lab work, and reading assignment.	
Literature and Study Materials	J.L. Hennessy and D.A. Patterson, Computer Architecture: A Quantitative Approach, 6th edition, Morgan Kaufman, 2017, Paperback ISBN: 9780128119051, eBook ISBN: 9780128119068. K. D. Cooper, L. Torczon, Engineering a Compiler, 3rd Ed., Morgan Kaufmann, 2022 A set of high-impact articles and presentations on computer architecture trends and practices available on Brightspace. IEEE Explore	
Assessment	This course is in transition due to recent restructuring of the computer engineering topics in the CESE programme. Consequently, the precise numerical breakdown of the assessment cannot be provided yet. However, the assessment will include the labs, course material (e.g., via a written exam), and potentially reading assignments. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

CESE4130	Computer Engineering	5
Responsible Instructor	Dr.ing. A.B. Gebregiorgis	
Instructor	Prof.dr.ir. G. Gaydadjiev	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	BSc in EE, CS or equivalent	
Course Contents	The course is designed to introduce students to the field of computer engineering. It starts from the basic foundations of computer engineering and equip students and key concepts and ideas used in the computer engineering field. Thus, the course covers three main areas of computer engineering: System software Hardware Design tools The main topics covered by this course are: Number representation: - This topic covers various schemes of number representation of a computer system such as, Signed Numbers, Redundant Number Systems, Residue Number Systems and encoding schemes used in Embedded Systems. Basic computer arithmetic: - This topic covers the basic binary arithmetic operations such as binary addition, counter, basic multiplication and division schemes, floating point representation and operation. Microarchitecture and computer architecture using RISC-V: - This topic covers the basics of microarchitecture and architecture design starting from basic logic gates, combinational and sequential circuits, storage elements, data-path computer organization of von-Neumann system such as processing unit, control unit, storage and registers, instruction format and instruction set architecture. System architecture: - In this topic the following concepts are covered: Assembly code and its role in the RISC-V architecture, extended program architecture and memory management. Parallel computing systems: - This topic introduces various parallelization and architectural level acceleration such as SIMD, multi and many core systems, GPU, TPU architectures. System modeling and management: - This topic covers basics on system software and operating system, virtualization, performance modeling and evaluation. Tools and methods for computer system design: - This topic covers concepts on compilation and interpretation, design automation methods and computer aided synthesis and optimization, hardware modeling and high-level synthesis.	
Study Goals	The goals of the course are: familiarize students from different background to the concepts and principles of computer engineering; Students to understand and explain how computing systems operate; Apply computing, mathematical and engineering principles to the design of computing system.	
Education Method	Lectures	
Literature and Study Materials	Lecture slides and reference materials	
Assessment	Obligatory Pass/Fail lab assignments. Written exam for a final grade. Students should pass the lab assignments to get their final grade. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Test and result scale	1-10	

EE4C11	Systems Engineering	5
Responsible Instructor	Prof.dr. K.G. Langendoen	
Responsible Instructor	S. Izadkhast	
Instructor	C. Galuzzi	
Instructor	C. Errando Herranz	
Instructor	Dr.ir. H. Vahedi	
Instructor	F.A. Muñoz Muñoz	
Instructor	M.A. Kitsak	
Instructor	Prof.dr. P.J. French	
Instructor	Dr. P. Pawelczak	
Instructor	P.P. Sundaramoorthy	
Instructor	S. Izadkhast	
Instructor	Dr.ir. R. Vismara	
Instructor	Dr. S.H. Tindemans	
Instructor	Dr. G. Lan	
Contact Hours / Week x/x/x/x	7/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	This course will introduce students to the V-model, and covers the following topics: - Problem Formulation - Requirements specification - Functional Decomposition - Means and Morphological Analysis - Decision Making - Simulation and modeling - Risk Analysis - Testing (plan)	
Study Goals	By the end of this course, you will be able to: - Recognize the need and relevance of systems engineering (SE) - Structure the key steps in Systems/Product Development - Systematically breakdown tasks and work in teams towards a common goal - Apply SE methodologies and SE tools in projects	
Education Method	The course is centered around the V-model commonly employed in systems engineering, in which a problem/project is gradually refined from high-level idea to detailed specification (followed by implementation and testing, which are not part of this course due to time limitations). The course includes 5 plenary instruction sessions focusing on subsequent steps in the V-model. The course takes a hands-on approach in which groups of about 6-8 students with different backgrounds work on a case study. Groups meet weekly with a TA to discuss progress and plan activities. In addition tutors from the teaching staff (case owners) will be available for assistance in problem formulation and formative feedback on deliverables. The course will be organized in 2 phases, the first phase ending in a formal review and the second phase ending with a final presentation and report. The course concludes with a symposium session in which all groups show their final (product) designs.	
Assessment	The final grade will be composed of a group grade based on - midterm review - case report - final presentation In addition, there will be formative assessment (i.e., not graded) in the form of feedback from the TA and Tutor, and peers. No resit will be offered; instead failing students will be assigned additional tasks to remedy any deficiency. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer and Embedded-Systems Engineering**

Specialisation Courses 2024	
Introduction 1	Courses from one of the specialisation lists, worth at least 13 credits: Students must select all specialisation courses in consultation with their prospective thesis advisor. All courses should relate to a Computer and Embedded Systems Engineering thesis subjectAll courses should relate to Computer and Embedded Systems Engineering to the thesis subject. IEPs containing courses that are not part of one of the mentioned lists require specific approval from the board of examiners. The four specialisations are: Computer Architecture Software Networking Control

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer and Embedded-Systems Engineering**

Specialisation courses Software 2024	
Introduction 1	Choose at least 15 EC on specialisation courses, from any of the 4 specialisation lists.

CESE4075	Supercomputing for Big Data	5
Responsible Instructor	Dr.ir. Z. Al-Ars	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Students are expected to have a strong background in programming. Experience in Java, Scala or Python is a plus due to their importance in the big data ecosystem.	
Course Contents	Big data is one of the hottest technology fields today, used to describe large and complex data sets that are difficult to process using traditional data processing systems. In this course, we will introduce the students to the most important concepts of big data analytics and the available tools and systems used to manage and process it. In a series of labs, the students will be working with a number of different big data applications, addressing such aspects as implementing big data algorithms using Spark and Hadoop on high performance computational systems, as well as using higher-level languages and system management tools to guide and monitor the performance of these systems. The students will have access to a big data cluster to evaluate the effectiveness of their implemented solution in practice.	
Study Goals	By the end of this course, the students will be able to - use basic big data processing systems like Hadoop - implement parallel algorithms using the in-memory Spark framework, and streaming using Kafka - use libraries to simplify implementing more complex algorithms - identify the relevant characteristics of a given computational platform to solve big data problems - utilize knowledge of hardware and software tools to produce an efficient implementation of the application	
Education Method	Lab assignments and reading material provided via Brightspace and online on the CognitiveClass.ai site	
Assessment	The students will be evaluated based on the quality of their implementation and the report they submit describing their solution approach. The course has three lab assignments: L1, L2 and L3. The final grade of the course will be evaluated as Grade = 0.5 * L1 + 0.3 * L2 + 0.2 * L3 In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Co-Instructor	Dr.ir. A.J. van Genderen	

CESE4120	Smart Phone Sensing	5
Responsible Instructor	M.A. Zuñiga Zamalloa	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Requirement 1: Students MUST either (1.1) have passed a JAVA programming course, or (1.2) have passed a C/C++ programming course and be familiar with JAVA, or (1.3) know Objective C (programming language for MACs). This requirement is equivalent to having passed the course CSE1100 in our first year Bachelor curriculum "Object-Oriented Programming" Requirement 2: Students MUST (2.1) have passed a basic course on Probability Theory. This requirement is equivalent to having passed the course CSE1210 in our first year Bachelor curriculum "Probability and Statistics". We will be refreshing some concepts on Probability, but we will not be refreshing concepts on Object Oriented Programming.	
Course Contents	The course provides an introduction to the current research trends in the area of smartphones. The course will be based on a programming project, where students will form groups of two to develop a smartphone application. This is not a programming course; students are expected to have already programming experience. To develop a smartphone application, a user needs to be familiar with (1) the signals and data that smartphones can gather, and (2) the mathematical tools necessary to process this data. This course will provide a solid background for the above two points. During the lectures we will analyze the latest research papers on this emerging field. We will dissect these papers to understand how techniques from algorithms, signal processing and machine learning are used to develop some exciting applications. The students will then use these basic technical tools to develop their own apps.	
Study Goals	The goals of this course are twofold. First, to expose students to the increasingly important area of mobile computing. Students will learn how mobile phones can be used to solve problems in areas ranging from health care and indoor localization to song recognition and traffic management. Second, to provide students with a basic set of tools to develop their own applications. For students aiming for industry, the course should enhance their ability to use theoretical tools to solve practical problems. For students involved in research activities, the course will provide them with the necessary background to use smartphones as a distributed sensing and processing unit that could be used to solve particular problems in their areas. At the end of this course, the student will be able to: (1) Explain the current applications, methods, and research trends in the area of indoor positioning with smartphones. (2) Apply key mathematical tools in the development of smartphone applications. (3) Analyze how a sensing and computing problem can be solved via the use of smartphones, and identify the steps required to design a solution. (4) Create a non--trivial and innovative smartphone application.	
Education Method	Lectures + Lab The project work, including the written report, covers the entire duration of the course period, and will take approximately 120 hours, of which 14 hours are spent on lectures, 10 hours preparing reports, 10 hours reading research papers, and the remaining part programming the App (the time spent in the Lab belong to this latter part).	
Literature and Study Materials	Research Papers and web tutorials	
Assessment	The final grade of the course consists of the following components: - Written Report 1: two intermediate reports (10% each, 2 pages each) (weighting 20%) - Written Report 2: final report (4 pages) (weighting 10%) - Oral Exam: final project demonstration (weighting 70%) The first two reports are due on the third and sixth week; and the final report, project, and oral exam are due on the ninth week. Final grade calculation = 0.2 * Written Report 1 + 0.1 * Written Report + 0.7 * Oral Exam A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Report 1: Resit about a new related project - Written Report 2: Resit about a new related project - Oral Exam: Resit about a new related project Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Enrolment / Application	1. You need to enrol in Brightspace 2. The first lecture will be compulsory 3. This course can only accommodate 60 students, with CESE students having a preference when demand exceeds capacity. If your program marks this course as required, you are guaranteed a spot.	

CS4205	Evolutionary Algorithms	5
Responsible Instructor	Prof.dr. P.A.N. Bosman	
Responsible Instructor	Dr.ir. P.A. Bouter	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	In this course we consider a specific subfield of Artificial Intelligence: Evolutionary Algorithms (EAs). These algorithms, sometimes also identified as being part of the class of bio-inspired algorithms, have as a metaphor the concept of natural evolution, i.e., the mechanisms by which, the fittest individuals in a population survive, reproduce, and in doing so, over time, change to be better equipped to thrive in their environment. Initiated in the 60s and 70s of the 20th century, research on EAs has progressed immensely. Today, EAs are being used to solve real-world problems in many areas, e.g. to optimize the layout of electrical wind farms, to automatically create radiation therapy treatment plans, and to optimize the architectures of deep neural networks. This course covers a spectrum of topics in EAs, ranging from basic concepts to advanced, recent, and state-of-the-art research, and ranging from theoretical to applied. In particular, topics include genetic algorithms, evolution strategies, genetic programming, estimation-of-distribution algorithms, optimal mixing evolutionary algorithms, multi-objective optimization, (GPU) parallelization, and real-world applications.	
Study Goals	At the end of this course, the student is able to: 1. Explain the key concepts underlying the main streams in Evolutionary Algorithm (EA) research, with in particular genetic algorithms, evolution strategies, genetic programming, estimation-of-distribution algorithms, and optimal mixing evolutionary algorithms. 2. Explain key ingredients underlying the rationale of when these algorithms work and when they do not work. In particular: schema analysis and how the match between the search bias of an EA and the fitness landscape is influenced by aspects such as variable dependencies and multi-modality. 3. Explain key research lines along which state-of-the-art research in EAs is done to achieve more robust, efficient, and effective EAs. 4. Identify good opportunities for using EAs, or hybrid versions thereof, in practice. 5. Properly (scientifically) experiment with EAs as well as program your own.	
Education Method	8-10 lectures 4 smaller (individual) projects 1 larger group project	
Literature and Study Materials	Various scientific papers, course-specific texts, and slides	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 60%) - Group Assignment (weighting 30%) - Individual Assignments: weekly assignments (weighting 10%) Final grade calculation = 0.6 * Written Exam + 0.3 * Group Assignments + 0.1 * Individual Assignments A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit opportunity - Group Assignments: Repair opportunity - Individual Assignments: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

CS4430	Network Security	5
Responsible Instructor	Dr.ir. H.J. Griffioen	
Responsible Instructor	G. Smaragdakis	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The course provides an overview of the most important concepts, methods, and best computer and network security practices. In this course, students will obtain the knowledge and hands-on experience to secure networking and communication systems. The course will primarily focus on technologies, protocols, attacks, and defenses. More precisely, starting from a review of common vulnerabilities and attack scenarios, the course will discuss the fundamentals of security engineering and their application in system design and finally review tools and methods to assess and test communication infrastructure from a security perspective. As a result, students will gain theoretical knowledge and hands-on experience in network attacks and defense methods. Knowledge activation and the transfer from conceptual understanding towards practical experience will be further facilitated by students implementing their own attack or defense tools on selected topics and conducting measurements on the effectiveness of attack and defense schemes. Topics covered in this course are: (1) physical layer security (2) link layer security (3) network layer security (4) transport layer security (5) application layer security (6) Internet security (7) cyber threats and countermeasures	
Study Goals	At the end of this course, the student is able to: 1. Design secure computer networks 2. Detect vulnerabilities in computer networks 3. Apply mitigation techniques for network attacks 4. Measure the effectiveness of network defence mechanisms 5. Explain and investigate security incidents 6. Design and implement network defence mechanisms	
Education Method	Lectures and lab assignments	
Literature and Study Materials	- Lecture slides - "Network Security Essentials: Applications and Standards Paperback", William Stallings, Pearson - "Computer Security: Principles and Practice", William Stallings, Pearson	
Assessment	The final grade of the course consists of the following components: Component 1: - Individual Assignment 1: lab assignment on Link Layer Security (weighting 10%) - Individual Assignment 2: lab assignment on Network Layer Security (weighting 10%) - Individual Assignment 3: lab assignment on Transport Layer Security (weighting 10%) - Individual Assignment 4: lab assignment on Application Layer Security (weighting 10%) Component 2: - Individual Assignment 5: lab assignment on Internet Security (weighting 25%) - Individual Assignment 6: lab assignment on Cyberthreat Intelligence (weighting 25%) Component 3: - Individual Exam (weighting 10%) Final grade calculation = 0.1 * Individual Assignment 1 + 0.1 * Individual Assignment 2 + 0.1 * Individual Assignment 3 + 0.1 * Individual Assignment 4 + 0.25 * Individual Assignment 5 + 0.25 * Individual Assignment 6 + 0.1 * Individual Exam A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Individual Assignment 1, 2, 3, 4: Repair opportunity - Individual Assignment 5, 6: Repair opportunity - Individual Exam: Resit opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

CS4505	Software Architecture	5
Responsible Instructor	Prof.dr. A. van Deursen	
Responsible Instructor	Prof.dr.ir. D. Spinellis	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The software architecture course offers students a chance to learn and experience the concepts of designing, modeling, analyzing and evaluating software design and software architectures. Furthermore, the course provides students with a discussion forum in which recent articles in the area of software architecture are presented and discussed. The course also features a number of guest lectures to show the state-of-the-art of software architecture in industry. Topics covered by this course are: fundamentals of software architectures, modeling and designing software architectures, architectural patterns and styles, architecture viewpoints and perspectives, the role of the software architect, analyzing and evaluating software architectures, component and plug-in frameworks, software product lines, service oriented architectures, code quality, technical debt, refactoring, and software architecture evolution. The course includes extensive labwork in groups of four. Each group is tasked to develop the architecture for a system of choice, covering problem analysis, a developed architecture, and a proof-of-concept implementation, and to describe these in written reports.	
Study Goals	Bring students into the position that they can (1) structure a systems problem space; (2) architect a software solution that meets stakeholder needs appropriately using models, views, and perspectives; (3) manage evolving needs, keeping architecture aligned, recognizing and addressing technical debt; (4) understand and assess architectures of existing software systems; (5) Use proof-of-concepts for architectural decision making; (6) communicate architectural decisions; (7) fulfil the role of an architect;	
Education Method	Interactive lectures, lab assignment, paper presentation, and discussion.	
Literature and Study Materials	The course uses the books: Cesare Pautasso. Software Architecture: Visual Lecture Notes. Leanpub, 2020; Additional reading material will be announced in the lectures.	
Assessment	The final grade of the course consists of the following components: - Group Report: Essay (weighting 60%) - Group Assignment: coding lab assignment (weighting 20%) - Group Presentation (weighting 20%) Final grade calculation = 0.6 * Group Report + 0.2 * Group Assignment + 0.2 * Group Presentation A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Group Report: Repair opportunity - Group Assignment: Repair opportunity - Group Presentation: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

CS4520	Security and Cryptography	5
Responsible Instructor	Dr. Z. Erkin	
Instructor	Dr. K. Liang	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Recommended: - Probability and Statistics - Programming skills	
Course Contents	Motivation: Computers are now found in every layer of society, and information is being communicated and processed automatically on a large scale. Examples include medical and financial files, automatic banking, video-phones, pay-tv, teleshopping and global computer networks. In all these cases there is a growing need for the protection of information to safeguard economic interests, to prevent fraud and to ensure privacy. Synopsis: Security and cryptography are essential components of any digital system. In this course, the fundamentals of secure data storage and transportation of information are described. In particular, classical (e.g. Caesar, Vigenere) and modern encryption schemes (RSA, DES, AES, Elliptic curves) are described along with their mathematical background such as number theory. Methods for authentication, data integrity and digital signatures are discussed in detail, as these are the main components of many security architectures. The course also investigates more advanced topics such as zero-knowledge proofs and secret-sharing schemes. Aim: It is the aim that at the end of the course one has a survey of the state of the art of both cryptographic algorithms and protocols for security and privacy, as well as is familiar with present applications. Among other things, the following topics are covered: -Classical systems -Information-theoretic security -Definition of Security notions -Symmetric encryption (e.g. DES, AES) -Asymmetric encryption (RSA, Elliptic Curves) -Hash functions -Random number generation -Key Management -Digital Signatures.	
Study Goals	At the end of this course, the student is able to: 1. Understand the basic concepts of cryptography, including those used in classical cryptography, modern secret cryptography and public key cryptography 2. Explain the fundamental concepts of security, including its importance in cryptography. 3. Understand the societal needs for confidentiality, integrity, and authenticity of information. 4. Understand the role of cryptographic primitives in securing information, including the differences between symmetric and asymmetric cryptography, the functions of hash functions, digital signatures, and PKI (Public Key Infrastructure). 5. Grasp advanced cryptographic topics relevant to modern society, where trust in entities cannot be guaranteed.	
Education Method	Lectures, assignments and weekly exercises. Through assignments, students are expected to have the chance to work on the topics covered in the lectures.	
Literature and Study Materials	- Introduction to Modern Cryptography, Katz and Lindell, CRC Press, Second Edition - Cryptography made simple, Nigel P. Smart, 2nd Edition, Springer, 2016 (PDF Available Online) - Handouts of lectures	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 70%) - Mandatory Assignments (weighting 30%) Final grade calculation = 0.7 * Written exam + 0.3 * Mandatory Assignments A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit opportunity - Mandatory Assignment: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

CS4530	Modelling and Problem Solving	5
Responsible Instructor	Dr. N. Yorke-Smith	
Instructor	S. Dumani	
Instructor	Dr. E. Demirovi	
Instructor	Ir. S.E. Verwer	
Instructor	Y. Murakami	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Required: bachelor course on Algorithm Design, programming Recommended: bachelor course on reasoning, complexity theory	
Course Contents	The course covers major (combinatorial) solving approaches, namely constraint programming, integer programming, SAT, decision diagrams, and local search. The course focuses on modelling practical problems, expressing them in the survey paradigms; algorithms for solving such problems are covered in the follow-up "Constraint Solving" course. Lastly, the course focuses on conducting proper empirical evaluations.	
Study Goals	At the end of this course, the student is able to: 1. Comprehensively explain main modelling paradigms such as constraint programming, mixed-integer programming, SAT and decision diagrams. 2. Motivate a choice for an appropriate modelling paradigm. 3. Model real-world problems in a suitable modelling paradigm. 4. Conduct proper empirical evaluation of their models and related algorithms.	
Education Method	Lectures and labs	
Literature and Study Materials	* Russell and Norvig, Artificial Intelligence: A Modern Approach, 4th (Global) Edition * provided study material	
Assessment	The final grade of the course consists of the following components: - Digital Exam (weighting 60%) - Group Assignment 1: lab assignment (weighting 10%) - Group Assignment 2: lab assignment (weighting 10%) - Group Assignment 3: lab assignment (weighting 10%) - Group Assignment 4: lab assignment (weighting 10%) Final grade calculation = 0.6 * Digital Exam + 0.1 * Group Assignment 1 + 0.1 * Group Assignment 2 + 0.1 * Group Assignment 3 + 0.1 * Group Assignment 4 A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Digital Exam: Resit opportunity - Group Assignment 1: Repair opportunity - Group Assignment 2: Repair opportunity - Group Assignment 3: Repair opportunity - Group Assignment 4: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Special Information	For personal administrative matters only: mps-cs-ewi@tudelft.nl For general questions: please use answers.ewi.tudelf.nl	
Tags	Artificial intelligence Group work Modelling	
Co-Instructor	Prof.dr. M.M. de Weerdt	

CS4535	Constraint Solving	5
Responsible Instructor	Dr. N. Yorke-Smith	
Responsible Instructor	Dr. E. Demirovi	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Covers state-of-the-art algorithms used in practice to solve combinatorial optimisation problems, focusing on (Max)SAT and (lazy clause generation) constraint programming. The course offers an in-depth view of the algorithms including advanced techniques (e.g., propagator design, conflict analysis, decomposition approaches), alongside techniques that can certify the correctness of the output of the algorithms beyond reasonable doubt. The assignments include a notable programming component with Rust as the programming language. After the course, students will have a principled understanding of the algorithms and will be able to use that knowledge to design their own tailored approaches to solve combinatorial optimisation problems.	
Study Goals	At the end of this course, the student is able to: 1. Explain how combinatorial optimisation paradigms based on (Max)SAT, (lazy clause generation) constraint programming, and to some extent mixed-integer linear programming work. 2. Understand the main mechanisms behind ensuring the correctness of the output of algorithms (certificates of unsatisfiability / optimality). 3. Analyse the inner workings of the algorithms to determine the reason for the effectiveness of one paradigm over the other on a given problem. 4. Design and implement algorithms to solve combinatorial optimisation problems.	
Education Method	Lectures, practical assignments, labs	
Literature and Study Materials	Lecture notes, lecture videos, academic papers, package documentation, various textbooks.	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 60%) - Group Assignment 1: lab assignment (weighting 10%) - Group Assignment 2: lab assignment (weighting 10%) - Group Assignment 3: lab assignment (weighting 10%) - Group Assignment 4: lab assignment (weighting 10%) Final grade calculation = 0.6 * Written Exam + 0.1 * Group Assignment 1 + 0.1 * Group Assignment 2 + 0.1 * Group Assignment 3 + 0.1 * Group Assignment 4 A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit opportunity - Group Assignment 1: Repair opportunity - Group Assignment 2: Repair opportunity - Group Assignment 3: Repair opportunity - Group Assignment 4: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

CS4545	Distributed Algorithms	5
Responsible Instructor	Dr.ir. J.A. Pouwelse	
Responsible Instructor	Dr. J.E.A.P. Decouchant	
Contact Hours / Week x/x/x/x	0/8/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Knowledge of Computer Networks (CSE1405) and Operating Systems (CSE2430) is useful as background for understanding this course.	
Course Contents	Introduction to distributed algorithms; notions of time and ordering of events; distributed algorithms for network overlays, message ordering, detecting global states, termination detection, deadlock detection, mutual exclusion, election, minimum-weight spanning trees, fault tolerance, broadcast, consensus, and agreement; blockchain technology and its relation with consensus.	
Study Goals	At the end of this course, the student is able to: 1. Understand the fundamental problems that distributed algorithms face 2. Understand the most important distributed algorithms that solve these problems 3. Reason about the execution of distributed algorithms 4. Program distributed algorithms	
Education Method	Lectures and lab. Lab work executed in groups of two students	
Literature and Study Materials	Additional resources are not necessary to attend and succeed in the course, which is entirely self-contained. However, the following list contains additional resources that might be helpful. - Distributed algorithms for message-passing Systems' book by Michel Raynal. - 'Introduction to secure and reliable distributed Programming' book by Cachin et al. - 'Distributed algorithms' book by Wan Fokkink. - 'Principles of Distributed Computing' lecture notes by Roger Wattenhoffer. - 'Introduction to Reliable and Secure Distributed Programming' book by Christian Cachin, Rachid Guerraoui, Luís Rodrigues. Springer. - 'Distributed Systems' book by Maarten van Steen, Andrew S. Tanenbaum.	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 100%) - Group Assignment: lab (Pass/Fail) Final grade calculation = Written Exam and need to pass the lab A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit opportunity - Group Assignment: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

CS4555	Compiler Construction	5
Responsible Instructor	Dr. S.S. Chakraborty	
Instructor	Dr. M.A. Costea	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Compilers translate the source code of programs in a high-level programming language into executable (virtual) machine code. In this course you will study the architecture of compilers, and important concepts and techniques underlying the components of that architecture. You will study these techniques by applying them in practice to implement your own compiler for a small language. The course covers the following topics: - Syntax and parsing - Static semantics and type checking - Virtual machines, assemby code, byte code - Interpretation vs. compilation - Code generation - Memory management, garbage collection - Data-flow analysis - Optimization	
Study Goals	At the end of this course, the student is able to: 1. Explain the architecture of a compiler and the role of each component of that architecture 2. Explain the algorithms and techniques for the implementation of compiler components and apply these techniques to examples 3. Implement a code generator that translates source language abstract syntax trees to target language instructions 4. Implement simple type checkers and data-flow analyzers 5. Integrate the course-relevant components into a working compiler	
Education Method	- Attending weekly lectures - Attending weekly lab sessions (which may start with a group tutorial) - Reading lecture material and papers - Making homework assignments (building a compiler)	
Literature and Study Materials	Required reading: We will use selected chapters from the book ""Essentials of Compilation"" by Jeremy Siek, available online: https://github.com/IUCompilerCourse/Essentials-of-Compilation Lecture slides and selected papers from the literature Optional reading: - We will use Scala to implement our compilers. The following book is optional reading which may be helpful in getting up to speed with programming in Scala: https://www.artima.com/shop/programming_in_scala_5ed (Additional) reading materials will be made available on Brightspace. We will use WebLab (https://weblab.tudelft.nl/cs4200/) for the submission of homework assignments.	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 80%) - Individual Homework Assignments (weighting 10%) - Groupwork Assignments (weighting 10%) Final grade calculation = 0.8 * Written Exam + 0.1 * Individual Homework Assignments + 0.1 * Groupwork Assignments A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit opportunity - Individual Homework Assignments: Complete and submit assignment after course ends - Group Assignments: Complete and submit assignment after course ends Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

DSAIT4005	Machine and Deep Learning	5
Responsible Instructor	Dr. D.M.J. Tax	
Responsible Instructor	Dr. J.C. van Gemert	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The range of problems that a computer scientist may encounter is very broad, from the very abstract (to determine if some problem can be solved by a computer) to very specific (to implement an electronic health record system). Due to the complexity and noisiness of the real world, a modern computer scientist has to rely more and more on the analysis and models of data. Machine Learning, and a recently successful extension of it, Deep Learning, deals with the analysis of data and the development of models. To be able to apply these algorithms, the computer scientist needs to have a good understanding of the foundations of Machine Learning. This course provides a recapitulation of (un)supervised learning, classification, decision theory and overfitting. It introduces some classical statistical learning classifiers and discusses their complexity, and ways to adapt their complexity (regularisation). The course focuses further on Deep Learning, including backpropagation, Optimization, Convolutional Neural Networks, Recurrent Neural Networks, self-attention and Transformers. Finally, the course discusses biases in datasets, and the design and analysis of Machine Learning experiments. This makes the student aware how the Machine and Deep Learning models can be applied in a responsible manner to real world problems.	
Study Goals	At the end of this course, the student is able to: 1. Explain the fundamental concepts and assumptions in the fields of Machine and Deep Learning, like the classical supervised learning, generalization, overfitting, model selection, and specialized architectures such as CNNs, RNNs, Transformers, etc. 2. Reflect on the strengths and limitations of classical and modern Machine and Deep Learning concepts 3. Apply the covered techniques, with reasons about their effectiveness 4. Implement key Machine Learning and Deep Learning techniques	
Education Method	In-person lectures with additional labs/practical projects	
Literature and Study Materials	- Bishop, "Pattern Recognition and Machine Learning" - Goodfellow et al., "Deep Learning"	
Assessment	The final grade of the course consists of the following components: - Digital Exam (weighting 100%) Final grade calculation = 1 * Digital Exam A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Digital Exam: Resit opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

DSAIT4010	Probabilistic AI and Reasoning	5
Responsible Instructor	Dr. N. Yorke-Smith	
Responsible Instructor	Dr. F.A. Oliehoek	
Instructor	S. Dumani	
Instructor	Prof.dr. M.M. de Weerdt	
Contact Hours / Week x/x/x/x	5/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Artificial intelligence and data science deal with a variety of complex problems ranging from face recognition to inventory management and controlling humanoid robots. For such tasks it is not immediately clear how to manually program good solutions. Instead, many approaches to successfully dealing with such complex problems are based on modelling the problem in a formal framework (such as logic, a graphical model, or Bayesian network), and applying corresponding reasoning or optimization techniques to find solutions. This course aims to introduce a number of such frameworks and how they can be used to model real-world problems and often also the uncertainties that arise in them. This includes discussing relations between these models and their associated methods like search, inference, learning and optimization. The specific topics include logical modelling, constraint satisfaction problems, graphical models, Bayesian networks, as well as temporal reasoning and learning via Markov decision processes and reinforcement learning.	
Study Goals	At the end of this course, the student is able to: 1. Comprehensively summarize the covered topics (what is AI?, modelling, graphical models and reasoning, bayesian networks, MDPs, reinforcement learning, etc.) and relevant aspects. 2. Explain the relations between search, inference, learning and optimization, as well as explain the main concepts and techniques. 3. Appraise the role of modelling, reasoning and learning in AI, based on examples of specific modelling approaches. 4. Apply covered modelling approaches (such as graphical models, bayesian networks, and markov decision processes) to practical real-world problems. 5. Reason about the effectiveness of certain solution (search/learning/inference/etc.) approaches for specific types of problems.	
Education Method	Lectures, homework exercises	
Literature and Study Materials	* Russell & Norvig. Artificial Intelligence: A Modern Approach, 4th Global ed., 2021 * provided materials and readings	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 100%) Final grade calculation = 1 * Written Exam A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Special Information	Course email address: dsait-pair-cs-ewi@tudelft.nl	
Tags	Artificial intelligence	

DSAIT4020	Elements of Statistical Learning	5
Responsible Instructor	Dr. D.M.J. Tax	
Responsible Instructor	T.J. Viering	
Instructor	Dr. J. Sun	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2 Summer Holidays	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Required: Linear Algebra, Probability Theory and Statistics, Machine and Deep Learning core course	
Course Contents	This course covers all the basic concepts of Statistical Learning, focusing on the classical techniques of Machine Learning before the era of Deep Learning. These concepts include classification, (ridge) regression, (hierarchical) clustering, feature reduction and extraction, model selection and bootstrapping, fairness in ML. The emphasis is on the concepts rather than the mathematical details.	
Study Goals	At the end of this course, the student is able to: 1. Explain the fundamental concepts and assumptions in the field of Statistical Learning, like the classical supervised learning, but also regression, clustering and dimensionality reduction 2. Reflect on the strengths and limitations of classical and modern Statistical Learning techniques. 3. Apply the covered learning techniques to analyse a dataset and reason about their effectiveness	
Education Method	There are lectures introducing the topics, a lab course and a final project where a dataset is analysed.	
Literature and Study Materials	"Elements of Statistical Learning' Hastie et al.	
Assessment	The final grade of the course consists of the following components: - Digital Exam (weighting 60%) - Group Report: project report on analysis of a dataset (weighting 40%) Final grade calculation = 0.6 * Digital Exam + 0.4 * Group Report A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Digital Exam: Resit opportunity - Group Report: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

DSAIT4025	Alternative Learning Strategies	5
Responsible Instructor	Dr.ir. J.H. Krijthe	
Responsible Instructor	T.J. Viering	
Instructor	Dr. N.M. Gürel	
Instructor	N. Tömen	
Instructor	Dr. D.M.J. Tax	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Recommended: Machine and Deep Learning core course, Elements of Statistical Learning course	
Course Contents	Many applications of machine learning do not fit exactly into the classical supervised learning setting. This course covers machine learning scenarios and strategies beyond the classical setting, which are relevant to contemporary machine learning research and applications. Examples are causal, meta, adversarial, multiple instance and semi-supervised learning.	
Study Goals	At the end of this course, the student is able to: 1. Identify which setting beyond (un)supervised learning a problem belongs to 2. Explain the essential problem that (adversarial/semi-supervised/causal/meta-/active/online/...) learning addresses and how it differs from the classical ML setting 3. Evaluate methods that are proposed in these learning scenarios, both theoretically and empirically 4. Mathematically prove basic properties of methods in these scenarios 5. Apply common methods in these learning scenarios on real-world datasets 6. Develop machine learning techniques adapted for a particular learning task	
Education Method	Weekly lecture (2 hours) and tutorial session (2 hours), reading material (including scientific papers) and exercises. Parts of these exercises constitute the 3 formative assignments that have to be handed in, where students have to critically reflect on research papers, program small simulations, prove properties of methods and apply methods to real-world datasets	
Literature and Study Materials	We use recent monographs and papers that are relevant for each of the topics. For example, for the causal machine learning we use a mix of material from books and papers: - Hernán, Miguel A, and James M Robins. Causal Inference: What If. CRC Press, 2020 - Pearl, J., Glymour, M., & Jewell, N. P. (2016). Causal Inference in Statistics: A Primer. Wiley. - Künzel, S. R., Sekhon, J. S., Bickel, P. J., & Yu, B. (2019). Metalearners for estimating heterogeneous treatment effects using machine learning. Proceedings of the National Academy of Sciences, 116(10), 41564165. https://doi.org/10.1073/pnas.1804597116.	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 100%) - Individual Assignment: assignment theoretically and empirically evaluating and developing methods and critiquing a research paper (pass/fail) Final grade calculation = Written Exam Grade (if pass for all three individual assignments) A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Witten Exam: Resit opportunity - Individual Assignment: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

WI4049TU	Introduction to High Performance Computing	6
Responsible Instructor	Prof.dr.ir. H.X. Lin	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Required for	Required: Linear algebra (matrix and vector operations), numerical analysis (solution of a system of linear equations Strongly advised: some experience with a programming language (e.g., C) is preferred.	
Course Contents	This course is intended for students who are interested in computing-intensive research. In the course, a number of algorithms that are being used within a diversity of research areas are considered. The scaling behavior of these algorithms in case of increasing problem size and/or an increasing number of processors is analyzed. Attention is paid to those aspects of computer architectures that are important to understanding the resulting performance, such as the memory hierarchy and the interconnection network. By analyzing their computing-intensive characters, determine the possible bottlenecks of several case studies (applications). Based on performance analysis, it will be indicated how the effect of those bottlenecks can be reduced. The goal is to learn how to achieve high performance with the available hardware architecture. The lab exercises will be done on the supercomputer at TU Delft. The emphasis will be on designing efficient parallel algorithms and optimizing for high performance. During the lab exercises, the following types of problems will be elaborated on: a parallel Poisson solver, a parallel finite element simulation, a parallel N-body simulation, and matrix computation on GPUs. More information, such as handouts and slides, can be found on Brightspace. Note: This course is the same as the previous ocurse IN4049TU. Starting from 2024/2025, lectures and exams will be given under the new course code WI4049TU (which is equivalent to IN4049TU)	
Study Goals	1. Can categorize high-performance computer systems 2. Can explain and classify parallel and distributed architectures, and programming models; 3. Can explain the concepts of data decomposition and parallel algorithms; 4. Can apply various parallelization methods to benefit from the potential parallel computer systems; 5. Capable to implement parallel programs (using MPI) on parallel computers and GPU (using Cuda); 6. Capable to evaluate and analyze the performance of parallel programs.	
Study Goals continuation	High Performance Computing, parallel programming, parallel algorithm	
Education Method	Lectures, computer lab exercise using MPI. As an option, answers to the bi-weekly quizzes can be handed in, and a maximum of one bonus point to the exam grade can be obtained.	
Computer Use	Lab exercises (mandatory): Given the sequential numerical simulation programs, make parallel implementations with C, MPI, and/or Cuda on the Delftblue supercomputer.	
Literature and Study Materials	An online book will be used for parts of the course. Lecture notes and lab exercises Will be made available throughout the course and can be downloaded from the Brightspace.	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 50%) Lab work (weighting 50%) Quizzes, a bonus of max 1 point can be obtained (only valid on the first examination occasion after the lectures) Final grade calculation: 0.5 * Written exam + 0.5 * lab work You can only pass the course if both exam and lab work scores are at least a 5.0, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: Resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Lab work: re-submit the lab report, the repair grade will be capped at 6.0 Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Enrolment / Application	Via Brightspace	
Co-Instructor	Dr. D. Palagin	

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer and Embedded-Systems Engineering**

Specialisation courses Computer Architecture 2024	
Introduction 1	Choose at least 15 EC on specialisation courses, from any of the 4 specialisation lists.

CESE4010	Advanced Computing Systems	5
Responsible Instructor	Dr.ir. Z. Al-Ars	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Students are expected to have a strong background in programming, to have knowledge of the Linux OS, and to have followed a bachelor-level course in computer architecture. Experience in C or C++ is a plus since this course focuses on low level (closer to hardware) programming.	
Course Contents	Recent developments in computing systems have resulted in the emergence of a number of different computational platforms that provide various performance, cost and power advantages in different application domains. This course discusses the most widely used computational platforms (CPUs, GPUs, FPGAs and DSPs), while addressing the theoretical and practical trade-offs in computer system organization and the latest developments and trends in computer architecture. The course will help the students in quantifying architectural design decisions in terms of performance, cost and power. An accompanying lab aids the students in applying this knowledge to create powerful heterogeneous (CPU, GPU, FPGA and/or DSP) computational solutions in computationally intensive application domains, such as multimedia and scientific computing.	
Study Goals	After completing this course, students are able to: 1. Analyze the computational characteristics of the application (available parallelism and memory access). This helps students to understand the algorithmic limitations of running them on a multicore platform. 2. Identify the components of a multicore architecture (processing elements, memory system, interconnect network). This helps students to estimate how their application will perform on an architecture built with these components. 3. Determine the potential performance of implementing an application on various available multicore architectures. This helps students to select an optimal architecture for their application. 4. Use profiling tools to identify bottlenecks in running applications on a system. This helps students to optimize their implementation. 5. Select a specific multicore system and implement the application using various techniques such as OpenMP, CUDA/OpenCL, and SIMD extensions. This helps students to select optimal multicore platforms for their target applications.	
Education Method	Lectures for 2 hours per week in addition to lab sessions of 2 hours a week. Students are expected to spend 10 hours a week to prepare for the lab, read material and solve homework assignments at home.	
Literature and Study Materials	Publications and reading material provided via Brightspace	
Assessment	1. L1 and L2: lab assignments to use advanced computing techniques 2. L3: a lab assignment to accelerate a specific application on heterogeneous multicore platforms (CPU and GPU) 3. H: 5 homework assignments based on reading material 4. C: a multiple choice exam used to assess individual competence Final grade = 0.3 * H + C * (0.1 * L1 + 0.2 * L2 + 0.4 * L3) L1, L2, L3, H is in the range [0, 10] C is in the range [0, 1.1] In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Remarks	Students are expected to be engaged in course activities throughout the course duration. Therefore, some students can experience a higher study intensity in this course compared to other courses in the curriculum. There will be a possibility to repeat the exam of this course. However, there is no possibility to repeat the homework or the lab. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr.ir. A.J. van Genderen	

CESE4035	Computer Arithmetic	5
Responsible Instructor	Dr. S.D. Cotofana	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The course emphasises on advanced theoretical and practical aspects of computer arithmetic. It covers various topics related to fixed and floating-point number systems, algorithms, and implementations for addition, multiplication, division, square root, and other high order arithmetic operations.	
Study Goals	The study goals for the course are as follows: 1. Allow the student to extend her/his knowledge beyond the computer arithmetic fundamentals on: (i) number representation systems; (ii) algorithms and implementations for basic integer and floating point arithmetic operations and function evaluation. 2. Assuming a certain processor architecture and design requirements she/he can perform design space exploration and select the most appropriate algorithms for functional units implementations. 3. She/he can operate with advanced concepts related to floating point systems and operations, elementary function evaluation, e.g., CORDIC, and error analysis. 4. She/he can design and evaluate arithmetic units and application specific (co-)processors, optimized for speed, area, power consumption, or combinations of those.	
Education Method	Lectures and lab assignments.	
Literature and Study Materials	Text book: Computer Arithmetic: Algorithms and Hardware Designs, Behrooz Parhami, Oxford University Press, NY, 2000, ISBN 0-19-512583-5. Example exams with solutions available on Brightspace.	
Assessment	Exam and lab assignments contribute to the final grade as follows: written open book exam - 40% of the final grade; lab assignments - 60% of the final grade. Repair option: Written exam = resit; Lab assignment = Re-submit deliverables. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Permitted Materials during Tests	Exam is open book. Books on computer arithmetic,computer architecture, and logic design, and lecture slides are allowed during the exams.	

CESE4075	Supercomputing for Big Data	5
Responsible Instructor	Dr.ir. Z. Al-Ars	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Students are expected to have a strong background in programming. Experience in Java, Scala or Python is a plus due to their importance in the big data ecosystem.	
Course Contents	Big data is one of the hottest technology fields today, used to describe large and complex data sets that are difficult to process using traditional data processing systems. In this course, we will introduce the students to the most important concepts of big data analytics and the available tools and systems used to manage and process it. In a series of labs, the students will be working with a number of different big data applications, addressing such aspects as implementing big data algorithms using Spark and Hadoop on high performance computational systems, as well as using higher-level languages and system management tools to guide and monitor the performance of these systems. The students will have access to a big data cluster to evaluate the effectiveness of their implemented solution in practice.	
Study Goals	By the end of this course, the students will be able to - use basic big data processing systems like Hadoop - implement parallel algorithms using the in-memory Spark framework, and streaming using Kafka - use libraries to simplify implementing more complex algorithms - identify the relevant characteristics of a given computational platform to solve big data problems - utilize knowledge of hardware and software tools to produce an efficient implementation of the application	
Education Method	Lab assignments and reading material provided via Brightspace and online on the CognitiveClass.ai site	
Assessment	The students will be evaluated based on the quality of their implementation and the report they submit describing their solution approach. The course has three lab assignments: L1, L2 and L3. The final grade of the course will be evaluated as Grade = 0.5 * L1 + 0.3 * L2 + 0.2 * L3 In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Co-Instructor	Dr.ir. A.J. van Genderen	

CESE4090	Reconfigurable Computing Design	5
Responsible Instructor	Dr.ir. J.S.S.M. Wong	
Responsible Instructor	G.K. Kuzmanov	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	ET4170/CESE4035 Computer Arithmetic and ET4171/CESE4040 Processor Design Project	
Course Contents	This course introduces the students to the field of Reconfigurable Computing. The course will provide a balanced insight of both theoretical trends and practical hands-on experience with reconfigurable technology - Field-programmable gate arrays (FPGAs) is the most prominent nowadays. In addition to the basic concepts, the students are taught advanced topics in reconfigurable technologies, organizations, architectures, programming paradigms, design tools and runtime systems. The particular topics to be covered are, but not limited to: fine and coarse grain reconfigurable technologies, partial and runtime reconfiguration, context switching, runtime algorithms for reconfigurable resource management, tools and methods for system-level and RTL synthesis, reconfigurable-specific compiler optimizations. Special attention is paid on application-specific acceleration in the context of reconfigurable technology. For example, the following application domains will be given special attention: image processing, streaming applications, encryption, compression, bioinformatics, supercomputing, etc. The lab attached to this course will focus on high-level synthesis (HLS) that is an increasingly more common way to design for FPGAs.	
Study Goals	- The student understands the basics of reconfigurable technology; - The student is able to design efficiently for reconfigurable technology (in terms of performance and resource utilization); - The student can identify computationally intensive parts of applications that are suitable for acceleration using reconfigurable computing; - The student masters state-of-the-art EDA tools, in particular HLS tools, for reconfigurable computing; - The student will be able to approach the design problems using a top-down methodology (starting from an application, to an architecture, down to a (re)configuration bitstream);	
Education Method	Lectures and laboratory work	
Literature and Study Materials	Handouts and research papers	
Prerequisites	CESE4085 (ET4074)	
Assessment	Open questions written test (50%) + Lab work (50%) In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

CESE4115	Embedded Computer Architecture 2	5
Responsible Instructor	Dr.ir. A.J. van Genderen	
Responsible Instructor	Dr.ir. Z. Al-Ars	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Tasks to be executed by digital hardware become more and more complex. For that reason, it is desirable to have a structured design process to come from task (algorithm) to hardware. Besides that, powerful concepts to express the algorithms in a formal way are required. This course aims at providing in-depth knowledge of the relation between an algorithm and the possible architectures that execute the algorithm. We will discuss methods to transform a sequentially expressed algorithm via single assignment code and a dependency graph into a signal flow graph. From there on various hardware implementations will be developed in the form of array processors or systolic arrays. In case of an irregular dependency graph, the hardware will take the form of a simple von Neumann architecture with a dedicated instruction set. Topics included are: - Linear (static) scheduling - Pipelining - Re-timing and design transformations. A second aim of the course is to specify the resulting hardware alternatives in a formal way, by means of mathematical equations. Using the evaluation mechanisms of a functional programming language, these equations can be directly evaluated leading to a simulation of the hardware. Besides, these specifications open the possibility to prove that the functionality of the various hardware alternatives is the same. Theory concerning the structured design process and the specification of hardware in a formal way will be discussed during lectures. The students gain hands-on experience in a project assignment where an algorithm is described in a formal way and synthesized toward hardware. See also https://osiris.utwente.nl/student/OnderwijsCatalogusSelect.do?selectie=cursus&cursus=192130250&collegejaar=2024&taal=en	
Study Goals	The student after completing the course will be able to: - derive recurrent relations from mathematical expressions; - design a data dependency graph based on the recurrent relations; - transform the data dependency graph into a signal flow graph using projection and scheduling; - describe mathematical expressions that specify data dependencies and operations in a graph by means of a functional programming (FP) language; - synthesize hardware circuits using the FP descriptions; - describe the results of the project assignment in a report.	
Education Method	Lectures will be streamed and recorded via Canvas. All the lecture materials will be published on Canvas. Link to Canvas: https://www.utwente.nl/en/educational-systems/about-the-applications/canvas/	
Literature and Study Materials	Will be made available via the CANVAS internet platform in Twente University: https://www.utwente.nl/en/educational-systems/about-the-applications/canvas/.	
Assessment	The exam can also be taken at Delft and Eindhoven and will be made available via the CANVAS internet platform in Twente University Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Enrolment / Application	Students from Delft or Eindhoven need to register as a subsidiary student (Dutch: 'bijvakstudent') and send the form to the UT Student Services, studentservices@utwente.nl. Link to form: https://www.utwente.nl/onderwijs/student-services/ik-word-student/inschrijving/bijvakkers-formulier-2024-2025.pdf Once you have a UT student account, you can self-enroll for the course on https://osiris.utwente.nl.	

CS4555	Compiler Construction	5
Responsible Instructor	Dr. S.S. Chakraborty	
Instructor	Dr. M.A. Costea	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Compilers translate the source code of programs in a high-level programming language into executable (virtual) machine code. In this course you will study the architecture of compilers, and important concepts and techniques underlying the components of that architecture. You will study these techniques by applying them in practice to implement your own compiler for a small language. The course covers the following topics: - Syntax and parsing - Static semantics and type checking - Virtual machines, assemby code, byte code - Interpretation vs. compilation - Code generation - Memory management, garbage collection - Data-flow analysis - Optimization	
Study Goals	At the end of this course, the student is able to: 1. Explain the architecture of a compiler and the role of each component of that architecture 2. Explain the algorithms and techniques for the implementation of compiler components and apply these techniques to examples 3. Implement a code generator that translates source language abstract syntax trees to target language instructions 4. Implement simple type checkers and data-flow analyzers 5. Integrate the course-relevant components into a working compiler	
Education Method	- Attending weekly lectures - Attending weekly lab sessions (which may start with a group tutorial) - Reading lecture material and papers - Making homework assignments (building a compiler)	
Literature and Study Materials	Required reading: We will use selected chapters from the book ""Essentials of Compilation"" by Jeremy Siek, available online: https://github.com/IUCompilerCourse/Essentials-of-Compilation Lecture slides and selected papers from the literature Optional reading: - We will use Scala to implement our compilers. The following book is optional reading which may be helpful in getting up to speed with programming in Scala: https://www.artima.com/shop/programming_in_scala_5ed (Additional) reading materials will be made available on Brightspace. We will use WebLab (https://weblab.tudelft.nl/cs4200/) for the submission of homework assignments.	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 80%) - Individual Homework Assignments (weighting 10%) - Groupwork Assignments (weighting 10%) Final grade calculation = 0.8 * Written Exam + 0.1 * Individual Homework Assignments + 0.1 * Groupwork Assignments A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit opportunity - Individual Homework Assignments: Complete and submit assignment after course ends - Group Assignments: Complete and submit assignment after course ends Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

EE4610	Digital IC Design	3
Responsible Instructor	Dr. S. Du	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Linear circuit theory. Digital circuits basics. Helpfull: prior / BSc level knowledge of CMOS technology and design	
Course Contents	This course will present a broad yet thorough overview of the subject of digital VLSI design, spanning both circuit and system abstractions. This complete picture is the only way to make the right tradeoffs, find the most suitable optimizations and the best implementation strategies for very large scale integrated circuits in standard deep-submicron CMOS technologies. After an introduction to technology, devices and interconnect, combinational logic gates and sequential elements are studied. This is followed by system level perspectives of interconnect issues, timing issues and the design of macro blocks. At each level, the opportunities and limitations of the physical implementation are considered for finding better solutions and trade-offs. This includes the consequences of the analog behavior of digital systems with respect to e.g. cross-talk noise and signal waveforms, that generally tend to become more influential with each new technology generation.	
Study Goals	Be able to analyze, design, and verify a VLSI component with full comprehension of how its performance, power dissipation, size and reliability relates to the physical implementation. Be able to use this knowledge to make suitable tradeoffs and implementation choices. Be able to manage the complexity of scale of a VLSI system as well as the complexity of behavior of its individual components.	
Education Method	Lectures and design assignment	
Computer Use	Practice assignment (Spectre and Cadence)	
Literature and Study Materials	"Digital Integrated Circuits: A Design Perspective", 2nd edition, J.M. Rabaey et al., Prentice Hall / Pearson, 2003.	
Assessment	Two assignments, Written examination, open book In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Co-Instructor	Dr.ir. N.P. van der Meijs	

EE4615	Digital IC Design II	3
Responsible Instructor	Dr. S.M. Alavi	
Responsible Instructor	M. Babaie	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	This course will consolidate the knowledge acquired in EE4610 by practicing the design of an advanced complex digital VLSI design, where high-level specifications are translated to transistor sizes, going through several abstraction levels. At each level, the opportunities and limitations of the physical implementation will be considered for finding better solutions and tradeoffs, by means optimization and the search of new topologies, while keeping complexity to a minimum to achieve the desired functionality. Parasitics and the other effects deriving from the analog behavior of digital circuits will be taken into account using four-corner analysis, as well as simplifications and modeling at each abstraction level and other semi-automatic tools.	
Study Goals	The main goal of the course is to give students the tools to successfully design complex, scalable circuits and systems. The students will also learn how to work in and coordinate teams of designers around a complex project from high-level specifications to the actual transistor-level implementation and post-layout verification.	
Education Method	The course will entirely be based in the lab where the target design will be implemented by students in groups of 2-3. Only based on students/groups wish, the final design will be laid out in a commercial CMOS process and possibly fabricated and tested.	
Assessment	The course will be graded based on the target design. The following criteria will be taken into account: 1. Functionality achieved; 2. High-level specifications met; 3. Four-corner analysis completed; Bonus points will be given to those students that will exceed the specifications and create a robust, reproducible design. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4690	Hardware Architectures for Artificial Intelligence	5
Responsible Instructor	Dr.ing. R.K. Bishnoi	
Instructor	Dr.ing. A.B. Gebregiorgis	
Instructor	Prof.dr.ir. S. Hamdioui	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Basics of Digital design	
Course Contents	Recent advancements in the field of Artificial Intelligence (AI) have demonstrated its great potential in solving exceptionally complex problems, that can even surpass humans. To support such AI algorithms, the demand of developing new and efficient hardware is growing with rapid pace. In particular, for a variety of edge/IoT devices, hardware platform locally deploys these algorithms that decreases the exchange of information with the cloud which significantly reduces the networking costs and boost the overall performance of the devices. Hardware architectures play a critical role in executing AI algorithms. This course gives an overview of AI from hardware perspective and discusses close to the state-of-the-art architectures and training mechanisms. In this course, you will learn the concept of advanced forms of AI (i.e. Deep Neural Networks) and how they can solve the complex problems. You will also learn, what are the challenges associated with the underlying hardware and how new architectures can address those. Moreover, the course will teach you the concept of brain-inspired computing using in-memory processing hardware platform. Technically, this course covers following topics: 1) Machine inference models (i.e. Support Vector Machine, Decision tree, Regression analysis, clustering, etc.), machine learning algorithms (i.e., Supervised, Unsupervised, Semi-supervised, Reinforcement learning models) and their associated challenges. 2) Deep Neural Network (DNN), Classification of DNNs, feed-forward, back propagation and learning mechanism (such as Gradient decent and Stochastic learning). 3) Spiking Neural Network, Spike Time Dependent Plasticity training algorithms and some existing hardware architectures (such as Intels Loihi, SpinNaker, IBMs TrueNorth, Neurogrid, etc.). 4) Emerging technologies for neuromorphic computing (using mersister devices such as RRAM, MRAM & PCRAM) and non-conventional computing paradigm (such as computing-in-memory, hybrid-design approach, crossbar arrays).	
Study Goals	The main aims of the course are: 1. Describe the concept of AI, inference and training models as well as their associated challenges from hardware perspective. 2. Understand the hardware AI requirements for the data-centers as well as edge devices. 3. Discuss the neuromorphic computing and its advantages with respect to conventional computing paradigms. 4. State details about Artificial Neural Network and their training algorithms. 5. Develop understanding about Spiking Neural Network and their training mechanisms and also, comparison with Artificial Neural Network and their evaluations based on design metrics such as accuracy, power, speed, size, etc. 6. Describe state-of-the-art neuromorphic hardware architectures and discuss their design implementation details. 7. Study the advanced topics of neural networks based-on computation-in-memory concept using emerging non-volatile memories. 8. Become an AI engineer with good understanding about hardware implementation.	
Education Method	Lectures, assignments and lab-work	
Assessment	Oral and lab In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Elective	Yes	
Tags	Artificial intelligence	

EE4695	Hardware Dependability	5
Responsible Instructor	Dr.ir. M. Taouil	
Instructor	Prof.dr.ir. S. Hamdioui	
Responsible for assignments	Dr.ir. M.C.R. Fieback	
Contact Hours / Week x/x/x/x	0,0,4,0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A course on VHDL/Verilog design such as one of the following: - Digital Systems B (EE1D21) - System Design with HDLs (ET4272)	
Course Contents	Integrated circuits (ICs) are integral part of any electronics system we see around. Developing, designing and deploying of such chips is by far more than just getting your SPICE simulation showing a correct behavior. Ensuring high quality, reliable and secure manufactured chips is a critical and crucial step in the whole IC process. A chip targeting children toys could cost 0.5 while a similar chip (in terms of functionality) could cost 100X to 1000X more when used for critical application such as military. Hardware dependability (e.g., quality, reliability, security) is a key differentiator. This course will introduce you to hardware dependability concepts and aspects. You will learn that ensuring such dependability needs the design of appropriate countermeasures, both during the design phase (i.e., part of the design) and in field/ during the lifetime of the application (i.e., continue monitoring and acting when needed). The course will teach you how to guarantee that a chip that may consists of billions of transistors and connections can be tested in less than one second, why the warranty you get for your phone is just two years rather than e.g., 5 years, how hackers can retrieve secret information from your device or smart phone and how to prevent it from happening, etc. Technically, the topics that will be covered are: a) Test/Quality: manufacturing defects, fault modeling, test development, design -for-testability, memory test, build-in-self test, etc; b) Reliability: reliability failure mechanism, modeling, design-for-reliability, testing, etc.; c) Security: hardware attacks and countermeasures, IP attacks, functionality Attacks, Data Attack, design-for-Security, etc.	
Study Goals	The main aims of the course are: 1. Describe the importance of Hardware Dependability (i.e., testability, reliability, security, etc.) and understand its relevance to electronic design and its overall impact. . 2. Point out the strong correlation between VLSI Design, Test, Reliablity and security 3. State the trends and challenges in test, reliability, security domain 4. Describe transistor and interconnects defect mechanisms and the way they behave at the electrical/functional level and how they are tested using fault models and test algorithms 5. Examine different test methodologies for logic and sequential circuits, their advantages, disadvantages, cost, limitations, etc and how they impact the security 6. Analyze and propose different test algorithms and Design-for-Testability DFT methodologies, their advantages, disadvantages, cost and limitations and how they impact the security 7. Implement block ciphers in hardware and evaluate their weaknesses 8. Analyse hardware attacks and evaluate the countermeasures that can be used against them 9. Develop fault injection and side channel attacks and design countermeasures for them 10. Become a better VLSI designer, a better test engineer/ product engineer, and a better hardware security engineer.	
Education Method	Lectures, homework assignments and lab	
Literature and Study Materials	- Lecture Notes - Lab manual - Book: Essentials of Electronic Testing (by M.L. Bushnell and V.D. Agrawal, ISBN 0-7923-799-1-8) - Syllabus containing material from online resources and material from the following books: - Introduction to Hardware Security and Trust - Trustworthy Reconfigurable Systems - Trusted Computing for Embedded Systems - Security in Embedded Devices	
Assessment	Assignments + lab + "group" oral examination In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

EE4700	Modelling, Algorithms and Data Structures	5
Responsible Instructor	C. Galuzzi	
Instructor	Prof.dr.ir. S. Hamdioui	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	- Basics of digital technology - Boolean logic and reasoning - General concepts of programming languages	
Course Contents	To solve an electrical engineering (EE) problem, we typically begin by developing a suitable model for the problem. Once modeled using an appropriate data structure, we can concentrate on solving the modeled problem rather than the original EE problem. Thus, selecting the right data structure is crucial for generating efficient and optimal solutions. Solutions to these problems generally involve algorithms. Therefore, analyzing these algorithms in terms of efficiency, time complexity, required memory, and other factors is a critical aspect of addressing electrical engineering challenges. This course focuses on three key aspects of solving electrical engineering problems: modeling, algorithms, and data structures. Among others, the following topics will be discussed: sets and logic; proofs; functions, sequences, and relations; algorithms and complexity; introduction to number theory; counting methods; recurrence relations; graph theory; trees; and network models.	
Study Goals	By the end of the course, students will be able to: LO1: Develop models for specific problems and select appropriate data structures. LO2: Prove theorems and statements using various proof methods: a) Classify different proof methods. b) Apply these methods to prove theorems/statements/etc. LO3: Work with algorithms: a) Develop algorithmic solutions for given problems. b) Analyze the efficiency and effectiveness of these algorithms. LO4: Apply counting methods to solve discrete problems. LO5: Use graph theory to address and solve specific problems.	
Education Method	Lectures	
Literature and Study Materials	- R. Johnsonbaugh, Discrete Mathematics, Pearson Prentice Hall, 7th Edition. - Lecture slides and other material, such as scientific papers, will be distributed.	
Assessment	Written exam. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

ET4351	Digital VLSI Systems on Chip	4
Responsible Instructor	Dr.ing. C.P. Frenkel	
Responsible Instructor	C. Gao	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Basics of digital design (logic gates, combinational logic and sequential logic, finite state machines). Knowledge of hardware description languages (Verilog or VHDL), including how to write proper synthesizable code, is necessary. Previous exposure to C Code and microprocessor architecture is recommended.	
Course Contents	Todays large-scale digital integrated circuits and systems-on-a-chip, such as modern general processing units (GPUs), can >50 billion transistors. Two things were instrumental in enabling this massive scalability: electronic design automation (EDA) tools and intellectual property (IP) blocks allowing for hierarchical design. In this course, you will learn how to use such EDA tools and IP blocks to go from the register-transfer level (RTL) description of your digital design to a final layout that is clean for tapeout and meets your specifications. To do so, you will also learn about power/performance/area tradeoff optimization, hardware-algorithm co-design, and memory hierarchy. These tools and concepts will be practiced and applied in a project, in which you will explore and apply your own design choices to meet a well-defined design target in a digital system-on-chip.	
Study Goals	After successful completion of the course, you will be able to: LO1. Design modern digital systems-on-a-chip with automated synthesis and place-and-route tools by a) identifying the key design targets and tradeoffs, b) selecting a proper IP to meet design targets, c) translating the design constraints and tradeoffs to tool commands, d) applying the tools to meet design targets. LO2. Optimize, in a rigorous and creative manner, the power/performance/area tradeoff associated with a given application use case. LO3. Link algorithmic requirements to hardware constraints. LO4. Extract the key performance metrics of a digital design by a) verifying that simulation results can be trusted and that the design is clean for tapeout, b) communicating these results effectively and rigorously.	
Education Method	Lectures, labs	
Literature and Study Materials	* Course material (slides will be provided) * (Recommended) Digital VLSI Chip Design with Cadence and Synopsys CAD Tools, by Erik Brunvand, Addison-Wesley, ISBN-10: 0321547993	
Assessment	The final course grade consists of two parts: * Final design report (50%) * Individual written exam (50%) The results of the two parts must have been obtained within the same academic year, and students must have a grade of 5.8 or higher for both the final design report and the individual written exam to pass the course. In case the final design report is failed, two repair options exist for the associated 50% of the total grade: - if the project grade for the group is insufficient, another group assignment will be provided (maximum score for this assignment: grade 6); - if the project grade was not received by a student due to dropping out from the project or not contributing sufficiently, an individual assignment will be provided (maximum score for this assignment: grade 6). In case the individual written exam is failed, a resit individual written exam must be taken for the associated 50% of the total grade. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

ET4362	High Speed Digital Design for Embedded Systems	5
Responsible Instructor	Prof.dr.ir. G. Gaydadjiev	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	The course covers many important topics computer engineers and embedded system designers should be aware of. First, the physical effects of high-frequency signal propagation and transport are discussed. Next, the state of the art for high bandwidth/low latency digital communication and memory systems is presented. Special attention is given to parallel memory systems. Thereafter, low level circuit design and analysis principles relevant to high-speed digital design are explained. Clock distribution, metastability and skew-tolerant design are emphasized. Advanced techniques for high-speed combinational and sequential logic structures are considered in detail. More precisely, approaches like domino logic, pulsed latches and dual-rail logic will be discussed. The next part discusses advanced design techniques such as globally asynchronous locally synchronous (GALS) systems, systolic arrays, streaming dataflow, wave-pipelined circuits and fully asynchronous design. The last part concentrates on understanding the true theoretical bounds in, e.g., the energy consumption, of computers. The necessary theory and some practically realizable adiabatic logic design styles will be introduced. When time allows, brief introduction to the Gödel's incompleteness theorem and its impact on computers will be presented.	
Study Goals	The purpose of this course is to familiarize students with the latest developments in the field of high-speed digital design. The emphasis is on novel, non-traditional design concepts, techniques and methodologies.	
Education Method	Lectures and individual assignments.	
Literature and Study Materials	Presentations and selected research papers.	
Assessment	Written report on a selected research topic and oral exam on appointment. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

QIST4400	Quantum Computer Architecture	5
Responsible Instructor	Dr. F. Sebastiano	
Responsible Instructor	Dr. S. Feld	
Instructor	Dr.ir. T.H. Oosterkamp	
Instructor	Dr. R. Ishihara	
Contact Hours / Week x/x/x/x	0/0/4/0/	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	The course will assume knowledge of fundamentals of quantum information (see course QIST 4310 or equivalent). Knowledge about qubit dynamics and quantum control (e.g., QIST 4300) and quantum programming (e.g., QIST 4320) is a nice-to-have. Basic knowledge of Python is also required.	
Course Contents	The goal of this course is to introduce the students to the overall system of a quantum computer, focusing on the classical hardware and software infrastructure required to build a quantum computer together with the quantum hardware. Covered topics include: - Quantum computer architecture; quantum languages; compilers; QISA; microarchitecture. - Quantum simulator (QX simulator); - Quantum error correction: quantum error correction codes; encoding; logical operations. - Mapping of quantum circuits; placement of qubits; scheduling of operations; routing of qubits; - Classical hardware for quantum computing: classical controller and quantum processor; qubit hardware and the need for cryogenic electronics; - Receiver and transmitter architectures for controlling qubits: frequency up/down-conversion, modulation schemes, frequency generation. - Hardware building blocks: amplifiers, analog-to-digital converters, digital-to-analog converters, digital processing, FPGA - Cryo-CMOS: device characteristics, state-of-the-art, design principles. Besides, the course covers the aspect of efficiently reading and understanding scientific papers.	
Study Goals	By the end of this course, students are able to: - Efficiently read and understand scientific papers. - Explain the different layers that are required to build a quantum computer. - Discuss the basic functioning of quantum error correction (QEC). - Carry out the mapping step of simple quantum circuits onto specific physical device layouts. - Interpret the results of different quantum circuits written in language QASM and executed using simulator QX, including the following: Basic quantum algorithms; Fundamental QEC codes; Surface code, especially for a Ninja star. - Identify and describe the different hardware blocks in the classical controller in a quantum computer. - Derive the specifications of the individual components in a classical controller form the requirements of the whole controller. - Compare different hardware blocks based on their specifications and can select the appropriate component for target functionality and specifications. - Describe the main features and advantages/disadvantages of the adoption of cryogenic electronics in general and cryo-CMOS specifically.	
Education Method	- Slide decks including small exercises - Interactive classes including activities - Homework assignments - Workshop on paper reading including processing their content	
Assessment	The final grade consists of two parts: - Part 1: Weekly homework, will be submitted electronically. Individual submission. Students are allowed to omit one homework. - Part 2: Written exam. The global course grade is an average of each part's grade (50% part 1 + 50% part 2). The resit exam will be an oral exam. The weekly homework can be repaired via an oral exam.	

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer and Embedded-Systems Engineering**

Specialisation courses Networking 2024	
Introduction 1	Choose at least 15 EC on specialisation courses, from any of the 4 specialisation lists.

CESE4045	High-performance data networking	5
Responsible Instructor	Prof.dr.ir. F.A. Kuipers	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Strongly advised: Basic understanding of networking and programming (ideally Python).	
Course Contents	The Internet plays a critical role in modern society, but the complexity of large-scale networks and the diversity of protocols make network management increasingly challenging. As networks become more programmable, the distinction between compute systems and network elements is blurring, leading to a paradigm shift in network operations. The high-performance data networking course is an advanced networking course that will introduce you to the concept of network operations, programmability and management. The course will examine different networking protocols and techniques used across various organisational domains, from edge computing to Internet-scale systems. The course will cover essential topics such as Quality of Service, Internet architecture/organisation, and network measurements.	
Study Goals	At the end of the course, the student is able to: 1. Understand advanced networking concepts, such as Software-Defined Networking, Inter/Intra-domain routing, etc. 2. Understand network management and operation techniques applied in internet, edge, and cellular (5G) domains 3. Evaluate network performance via a network emulator (Mininet) and understand real-world Internet architecture via measurements. 4. Create network programs in the domain-specific language P4.	
Education Method	Approximately 50% of the course will consist of lectures and selfstudy and 50% focuses on (homework) exercises and instruction classes.	
Literature and Study Materials	Slides and a reader containing the exercise material.	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 100%) Final grade calculation = 1 * Written Exam A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Co-Instructor	Dr. N. Mohan	

CESE4055	Ad hoc and Sensor Networks	5
Responsible Instructor	Dr. R.R. Venkatesha Prasad	
Contact Hours / Week x/x/x/x	3/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Wireless communications and networking Computer communication principles, Layering principle of Computer Networks. Digital communication.	
Course Contents	IMPORTANT NOTICE ------------------------- Please note that the prevailing conditions of the COVID19 may force us to modify the study guide information. It may change depending on the developments around the coronavirus. We may have to make changes to the teaching methodology, assessment, practical work, assignment and group activities. This will be instructed by the faculty/university management based on the orders of the government from time to time to protect the faculty and students. The above applies to all the fields in this coursebase for this course. The course will be offered in PHYSICAL format. -- Ad-hoc networks are formed in situations where mobile computing devices require networking applications when a fixed network infrastructure is not available or not preferred to be used. In such cases, mobile devices may possibly set up an ad hoc network themselves. Ad-hoc networks are decentralized, self-organizing networks and are capable of forming a communication network without relying on any fixed infrastructure. Ad-hoc networks form a relatively new field of research. In this lecture, besides general introduction to ad-hoc networks and their applications, we will focus on state-of-the-art methods and technologies for forming an ad-hoc network and maintaining its stability despite the dynamics of the network. The contents of the course are as follows: Positioning and applications (Chapter 1, 2 & 3 of the textbook, these topics are basics & pre-requisites; And Chapter 5) o Definition of ad-hoc networks o Comparison with infrastructure based systems o Typical applications o Advantages and challenges o Radio technologies for ad-hoc networks o Wi-Fi, Zigbee, Bluetooth Modelling ad-hoc networks o Propagation models o Topology models based on graph theory o Degree and hopcount o Connectivity theorems MAC protocols for ad-hoc networks (Chapter 6, 10 of the textbook) o Introduction to MAC protocols o Issues and design goals o Classification o Directional, muli-channel MAC protocols o Energy efficiency in MAC protocols o Quality of service Self organisation and Routing (Chapter 7, 8, 11 of the textbook) o Flooding o Node discovery, neighbour discovery o Route establishment o Topology maintenance, localisation o Proactive, reactive and hybrid routing o Typical protocols o Energy efficiency in routing o Broadcast and multicast o Effects of mobility on connectivity and capacity o Effect of nodes joining and leaving the network Advanced issues in ad hoc networks o Wireless sensor networks (Chapter 12 of the textbook and papers) o Cooperation (Reference papers) o Simulating ad hoc networks as part of project (optional: ns3, OMNET, OPNET) o Energy Harvesting Project presentations by students	
Study Goals	By the end of this course students should be able to: - Model the ad-hoc networks using Graphs. - Describe the working principles of medium access control protocols for ad-hoc networks - Explain the working principles, advantages and disadvantages of different classes of routing protocols for ad-hoc networks - Choose various components to form a coherent ad hoc networking architecture - Develop a simulator to evaluate the MAC and routing protocols for ad hoc networks - Assess the suitability of ad-hoc networks for different communication needs and scenarios	
Education Method	The course will be taught in lecture form. The presence of students at all lectures is required for optimum result. Students are required to participate actively in various forms of activities and peer-learning. New forms of teaching aids are used.	
Literature and Study Materials	1. Textbook: Ad Hoc Wireless Networks, Architectures and Protocols by C. Siva Ram Murthy and B.S.Manoj, Prentice-Hall Pearson, 2004. 2. Lecture notes consist of slides presented at the lectures (Slides are only teaching aids and are not substitutes for textbooks,	

	research papers, etc). 3. Some recent journal papers 4. Optional Reference Books 4.1. Distributed Algorithms, Nancy A. Lynch, Morgan Kaufmann, 1996 (for networking algorithms) 4.2. Ad Hoc Mobile Wireless Networks, Principles, Protocols and Applications by Subir Kumar Sarkar, C Puttamadappa, and T. G Basavaraju, Auerbach Publications, 2008. This book is available online in the library. 4.3. Wireless Ad Hoc and Sensor Networks, A Cross-Layer Design Perspective by Jurdak, Raja, Springer, 2007. This book is available online in the library. 4.4. Ad-hoc Networks: Fundamental Properties and Network Topologies, by Ramin Hekmat, Springer. 5. OPNET/ns-2 web pages, tutorials and video lectures
Books	Ad Hoc Wireless Networks, Architectures and Protocols by C. Siva Ram Murthy and B.S.Manoj, Prentice-Hall Pearson, 2004. However, I also use other materials from Internet and other books listed above.
Assessment	1. There will be totally three written/oral tests/examinations for this course (including the final exam). 2. The students will carry out a project in a group and submit a short report. 3. Participation in off-track discussions on Facebook/Brightspace/FeedbackFruits and wikis. Final score is based on marks obtained during tests, project, assignment (in groups) and bonus marks. All the details will be given in the first class. Breakup: 2 Tests + Final Exam = 55% Project 40% Self assessment + Reflection 3% Activities on Feedback Fruits or any Online platform 2% Resit: written/oral exam [+ carry over from projects]. === Changes due to COVID19 in 2021 is below; in case there is a similar situation, we use the following methods. === There are three chapters Chapter 1: Network modelling Chapter 2: MAC protocol Chapter 3: Routing + Lab: Simple experiments using laptops (individual) + simulations (group-wise) Marks Breakup -------------- Homework/Assignments/Group works Part A1: Group-wise assignment. 3 times (1 per chapter) -- 15 marks Part A2: Group-wise Q&A. 3 times (1 per chapter) -- 30 marks Part B1: Individual experiment + report Part B2: Group-wise simulations + report + demo Part B1 + Part B2 - 50 Marks Part C: Self-assessment + peer activities -- 5 marks Resit: Part A1 & A2 will not be repeated. Only Part B1 & B2 are allowed for the Resit this year, because of COVID19; and the projects should be done individually in Resit. === (More information will be given in the first class) Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).
Permitted Materials during Tests	Different conditions for different test/exams. Conditions will be informed 1 week before the exams/test.

CESE4060	Wireless IoT and Local Area Networks	5
Responsible Instructor	Dr. P. Pawelczak	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Students are advised to follow the course Wireless Communications (ET4358) before taking this Wireless Networking course. An advantage is to have entry-level programming skills (Matlab, Python, C/C++). Nonetheless, students with little knowledge of programming will be helped.	
Course Contents	The following modules will be discussed during the lectures: Introduction (example topics): - What is wireless networking - Where to search for (academic) wireless network literature and resources Medium Access Control (example topics): - WiFi: hidden/exposed terminal problem, Carrier Sense Multiple Access - Bluetooth standard: in-depth look into the channel hopping, protocol specifications WiFi (example topics): - Review of IEEE 802.11 standards - Protocol format - ISM band regulation - Adaptive Modulation and Coding - WiFi Matlab class (assignment) IoT networking standards (example topics): - LoRa: protocol specifications, energy consumption, modulation format, network design Review of wireless tools (example topics): - Introduction to wireless packet sniffing and analysis using Wireshark (assignment) - Simple simulations of WiFi network with NS3 RFID networking (example topics): - Principles of backscatter - Protocol formats: EPC C1G2 - RFID hackathon (assignment) Cognitive radio (example topics): - Basics of spectrum management - White Space Databases - Theory of spectrum sensing	
Study Goals	At the end of the course students will be able to: (i) to understand how practical wireless systems work and get a deeper understanding of how the theoretical concepts of wireless communications apply to practice; (ii) employ their own analysis methodology to assess new wireless network systems (especially at the physical layer); (iii) understand rapid prototyping of new wireless systems (for instance, with software defined radio).	
Education Method	Lecture presentations, mini-project assignments, assigned paper reading and its critical analysis and presentation.	
Computer Use	Each student should have its own laptop (preferably with a Linux distribution, where Linux must not be installed on a virtual machine). We will be using Matlab, and/or NS3 and/or GNURadio and/or Wireshark for the assignments.	
Literature and Study Materials	WiFi Matlab WLAN toolbox: https://nl.mathworks.com/help/wlan/; Wireshark learn page: https://www.wireshark.org/#learnWS; tutorial on NS3 network simulator: https://www.nsnam.org/documentation/; specific chapters from books provided at the beginning of each lecture.	
Prerequisites	Background in programming (Matlab, Python, Bash)	
Assessment	Points from the mini-project assignments. A research paper analysis from conferences such as IEEE INFOCOM, ACM MobiCom, ACM SIGCOMM will be required to pass the course. Repair option: Individual Assignment = Re-submit deliverable Individual Report = Re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

CESE4065	Advanced Practical I.o.T. and Seminar	5

Responsible Instructor: Dr. R.R. Venkatesha Prasad

Contact Hours / Week x/x/x/x: 2/0/0/0 and 0/0/0/2

Education Period: 1
4

Start Education: 1
4

Exam Period: none

Course Language: English

Expected prior knowledge: fundamental understanding of wireless communications, familiarity with wireless communications and embedded systems and knowledge of Android programming/Python/C++/Matlab.

Parts: Each seminar: 2x 45 minutes (2 parts) + 10 minute break

Summary

his course is an involved hands-on course for self-motivated students. Students usually work in a group of two (max 3). Students are expected to have sufficient programming and hardware development skills. The course is project-oriented thus students are expected to deliver a working model with demonstrations.

IMPORTANT NOTICE REGARDING ADJUSTMENTS DURING THIS COVID19 SITUATION

The course will be offered ONLINE. Depending on the COVID19 situation there may be on-campus meetings/discussions [Safety of everyone is the highest priority].

This course is done in a group of two (max 3). The instructor/TA will help connect a potential groupmate in the first class using Brightspace/Feedback Fruits.

Activities: How to in Groups?

Part 1: Seminar
We guide the groups in selecting a paper. Students need to prepare slides and deliver a seminar using online tools; it is a group activity. For each seminar, all the students MUST be present. Absenting without permission will be taken seriously.

Part 2: Project
(1). Students can share the responsibilities if they can organize in such a way that one works on H/w and another on software so that they develop the project together. Meeting each other may be possible. However such face-to-face meetings should be done by strictly adhering to the COVID-19-related instructions from the university/government.

(2). Students may exchange the components/boards within the group but carefully adhering to the 1.5m rule. The decision to do such cooperation is left to the judgement of the students and it is their responsibility. Please note safety is first.

For Students not in NL or not able to be in Delft:
(3). Instructor/TA would be offering an opportunity to do a hardware project if students can buy simple hardware like Arduino boards/Raspberry Pi, etc. They should try to use the method as in (1) above.
(4). Instructor/TA may offer Algorithm/simulations/android programmes/Contiki-based networks or Open testbed projects. In these cases, online group activities are possible.

NOTE: In some extreme cases, the Instructor/TA may allow one person to project after assessing the situation and after discussing it with the student.

Supply of Hardware:
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1. Instructor/TA is able to provide take-home components (limited numbers) like Arduino boards and some sensors. We may provide some other instruments/sensors/boards depending on the selected project. These could be collected from TA while adhering to sanitization and social distancing rules.

2. Students may also use their own components.

Course Contents: Course will be composed of a series of seminars related to the broad topic of the Internet of Things. Students will present their results on investigations regarding the possible extension of the ideas presented in the assigned papers.

Study Goals: To be able to design components of Internet of Things and showcase an application or product through an implementation of a project. Specifically, to be able to bring entrepreneurial aspect of the project and also to be able to evaluate the project in depth. To be able to criticize and assess system-level components of the Internet of Things environment discussed in the scientific literature.

Education Method: Seminar will be composed of (i) seminar presentation on a selected research paper (from top journals/conferences) presented individually by students and, (ii) work on a research project. Students will be provided with a list of projects that will be assigned to them. Project will be summarized in the form of a written report (report must include critical analysis). Within a project any hardware/software platform can be used and demonstrated. User experience/study, where applicable, also needs to be executed.

Selected paper needs to be critically evaluated and a proposal to extend the assigned paper will need to be presented in a form of a presentation. Paper extension should focus on a system level idea.

Presentation skills, thinking and reflection abilities are looked into carefully.

The teams are composed of two (or three) students generally.

NOTE: The total amount of work would be on the higher side of 150 hours; since this is an advanced course, students are expected to already have very good knowledge of hardware platforms, coding, and design environment.
In case of lacking in some of these skills, we expect students to acquire them outside this budgeted 150Hrs.
However, the number of hours of workload mentioned here is ONLY a guide, the efforts depend on the project, goals, and the collaboration with the team member(s).

Literature and Study Materials: This is a project based course. Thus, Internet and other appropriate manuals (for chipsets, etc.) would be useful.

For papers to read and present, we expect students to look into:
Infocom, Mobicom, IPSN, Sensys, Sensapp, Mobihoc conference papers.

Journals: IEEE Trans on Networking, Trans on Mobile computing, JSAC, etc.
ACM Trans on CPS, etc.

Assessment	Part 1: Assessment based on presentation quality, slides, Q/A, constructive criticisms, ideas for improvements, etc. Part 2: Project execution in a group, demonstration, Q/A and a report describing the outcome of the assigned project. Part 1: 30% of the whole marks; Part 2: 60% of the whole marks. In the assessment, a focus on the practicality and entrepreneurial aspect of the idea will be prevailing. A working model, demonstration, and a report, are expected. Individual Q/A after demonstrations will be part of the assessment. Part 3: Up to 10% marks would be awarded if the report is detailed above the minimum expected and the work is ready for a submission to a conference. Report will not have individual component, however, Q/A may have individual component. Please NOTE: There would be no resit since this is a hands-on course being executed in a team. As this is a project-based course, there is no official resit scheduled. Instead, an opportunity will be given to improve the work. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).
Tags	Circuits Group work Programming Project Software
maximum aantal deelnemers	Max 30 students, which translates to 10-15 groups (Groups consisting of 2 or 3 students).

CESE4120	Smart Phone Sensing	5
Responsible Instructor	M.A. Zuñiga Zamalloa	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Requirement 1: Students MUST either (1.1) have passed a JAVA programming course, or (1.2) have passed a C/C++ programming course and be familiar with JAVA, or (1.3) know Objective C (programming language for MACs). This requirement is equivalent to having passed the course CSE1100 in our first year Bachelor curriculum "Object-Oriented Programming" Requirement 2: Students MUST (2.1) have passed a basic course on Probability Theory. This requirement is equivalent to having passed the course CSE1210 in our first year Bachelor curriculum "Probability and Statistics". We will be refreshing some concepts on Probability, but we will not be refreshing concepts on Object Oriented Programming.	
Course Contents	The course provides an introduction to the current research trends in the area of smartphones. The course will be based on a programming project, where students will form groups of two to develop a smartphone application. This is not a programming course; students are expected to have already programming experience. To develop a smartphone application, a user needs to be familiar with (1) the signals and data that smartphones can gather, and (2) the mathematical tools necessary to process this data. This course will provide a solid background for the above two points. During the lectures we will analyze the latest research papers on this emerging field. We will dissect these papers to understand how techniques from algorithms, signal processing and machine learning are used to develop some exciting applications. The students will then use these basic technical tools to develop their own apps.	
Study Goals	The goals of this course are twofold. First, to expose students to the increasingly important area of mobile computing. Students will learn how mobile phones can be used to solve problems in areas ranging from health care and indoor localization to song recognition and traffic management. Second, to provide students with a basic set of tools to develop their own applications. For students aiming for industry, the course should enhance their ability to use theoretical tools to solve practical problems. For students involved in research activities, the course will provide them with the necessary background to use smartphones as a distributed sensing and processing unit that could be used to solve particular problems in their areas. At the end of this course, the student will be able to: (1) Explain the current applications, methods, and research trends in the area of indoor positioning with smartphones. (2) Apply key mathematical tools in the development of smartphone applications. (3) Analyze how a sensing and computing problem can be solved via the use of smartphones, and identify the steps required to design a solution. (4) Create a non--trivial and innovative smartphone application.	
Education Method	Lectures + Lab The project work, including the written report, covers the entire duration of the course period, and will take approximately 120 hours, of which 14 hours are spent on lectures, 10 hours preparing reports, 10 hours reading research papers, and the remaining part programming the App (the time spent in the Lab belong to this latter part).	
Literature and Study Materials	Research Papers and web tutorials	
Assessment	The final grade of the course consists of the following components: - Written Report 1: two intermediate reports (10% each, 2 pages each) (weighting 20%) - Written Report 2: final report (4 pages) (weighting 10%) - Oral Exam: final project demonstration (weighting 70%) The first two reports are due on the third and sixth week; and the final report, project, and oral exam are due on the ninth week. Final grade calculation = 0.2 * Written Report 1 + 0.1 * Written Report + 0.7 * Oral Exam A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Report 1: Resit about a new related project - Written Report 2: Resit about a new related project - Oral Exam: Resit about a new related project Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Enrolment / Application	1. You need to enrol in Brightspace 2. The first lecture will be compulsory 3. This course can only accommodate 60 students, with CESE students having a preference when demand exceeds capacity. If your program marks this course as required, you are guaranteed a spot.	

CS4090	Quantum Communication and Cryptography	5
Responsible Instructor	Dr. S.D.C. Wehner	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Linear Algebra, Probability & Statistics, Q101 (AP3421 Fundamentals of quantum information) or CSE3130 (Introduction to Quantum Computer Science)	
Course Contents	This class will introduce you to the fascinating field of quantum communication! We will look at the state of the art of quantum networks, and explore techniques for building quantum repeaters that promise to deliver qubits over long distances. We also briefly look at one of the most famous application of quantum cryptography, quantum key distribution.	
Study Goals	The student will acquire: A good understanding of the fundamental concepts of quantum communication The ability to understand, design and analyze quantum communication protocols Insight into the differences between classical and quantum communication and cryptography	
Education Method	- Flipped classroom (Online videos and materials, prepared for flip classroom teaching) - Weekly Quiz and Discussion Session (In person) - Weekly Tutorials on Homework Problems (In person)	
Literature and Study Materials	Book on tools and quantum cryptography: Introduction to Quantum Cryptography, Cambridge University Press, 2023 Lecture Notes Slides Review Articles Auxilliary: Nielsen and Chuang Quantum computation and information, Cambridge University Press.	
Assessment	The final grade of the course consists of the following components: Homework (weighting 60%) Final Project (weighting 30%) Weekly Quiz (weighting 10%) Final grade calculation: 0.6 * Homework + 0.3 * Final Project + 0.1 * Quiz A passing final grade for the course can only be earned when for the homework and final project at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance to TER Implementation Regulations Art 5, sub 5., for: Homework: re-submit deliverable Final Project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Tags	Abstract Adventurous Algorithmics Challenging Group Dynamics/Project Organisation Information & Communication Integrated Intensive Involved Lineair Algebra Mathematics Physics Quantum Signals Technology Telecommunication	

CS4430	Network Security	5
Responsible Instructor	Dr.ir. H.J. Griffioen	
Responsible Instructor	G. Smaragdakis	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The course provides an overview of the most important concepts, methods, and best computer and network security practices. In this course, students will obtain the knowledge and hands-on experience to secure networking and communication systems. The course will primarily focus on technologies, protocols, attacks, and defenses. More precisely, starting from a review of common vulnerabilities and attack scenarios, the course will discuss the fundamentals of security engineering and their application in system design and finally review tools and methods to assess and test communication infrastructure from a security perspective. As a result, students will gain theoretical knowledge and hands-on experience in network attacks and defense methods. Knowledge activation and the transfer from conceptual understanding towards practical experience will be further facilitated by students implementing their own attack or defense tools on selected topics and conducting measurements on the effectiveness of attack and defense schemes. Topics covered in this course are: (1) physical layer security (2) link layer security (3) network layer security (4) transport layer security (5) application layer security (6) Internet security (7) cyber threats and countermeasures	
Study Goals	At the end of this course, the student is able to: 1. Design secure computer networks 2. Detect vulnerabilities in computer networks 3. Apply mitigation techniques for network attacks 4. Measure the effectiveness of network defence mechanisms 5. Explain and investigate security incidents 6. Design and implement network defence mechanisms	
Education Method	Lectures and lab assignments	
Literature and Study Materials	- Lecture slides - "Network Security Essentials: Applications and Standards Paperback", William Stallings, Pearson - "Computer Security: Principles and Practice", William Stallings, Pearson	
Assessment	The final grade of the course consists of the following components: Component 1: - Individual Assignment 1: lab assignment on Link Layer Security (weighting 10%) - Individual Assignment 2: lab assignment on Network Layer Security (weighting 10%) - Individual Assignment 3: lab assignment on Transport Layer Security (weighting 10%) - Individual Assignment 4: lab assignment on Application Layer Security (weighting 10%) Component 2: - Individual Assignment 5: lab assignment on Internet Security (weighting 25%) - Individual Assignment 6: lab assignment on Cyberthreat Intelligence (weighting 25%) Component 3: - Individual Exam (weighting 10%) Final grade calculation = 0.1 * Individual Assignment 1 + 0.1 * Individual Assignment 2 + 0.1 * Individual Assignment 3 + 0.1 * Individual Assignment 4 + 0.25 * Individual Assignment 5 + 0.25 * Individual Assignment 6 + 0.1 * Individual Exam A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Individual Assignment 1, 2, 3, 4: Repair opportunity - Individual Assignment 5, 6: Repair opportunity - Individual Exam: Resit opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

EE4396	Mobile Networks	5
Responsible Instructor	Dr. R. Litjens	
Contact Hours / Week x/x/x/x	0/0/0/5	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Not formally required, but recommended to have the basic knowledge taught in 'Fundamentals of Wireless Communications' (ET4358). If you didn't take this course, key aspects will be repeated during the class. In case you are lacking a comfortable background, talk to me in the first week of class and I'll help you catch up.	
Course Contents	The course addresses the technology and management of contemporary and future mobile cellular communication networks. With a focus on radio access, as it dominates both the performance and cost aspects of mobile networking, the course addresses 5G, 4G (LTE/LTE-A), 3G (UMTS/HSPA) and 2G (GSM/GPRS/EDGE) technologies, all of which are operational network technologies. An outlook to 6G will also be given. Fundamental aspects of cellular networking are explained and discussed, including e.g. the intrinsic tradeoffs between coverage, capacity and quality and key aspects that influence performance (traffic, propagation, technology, deployment, spectrum. Cellular network planning, dimensioning and performance analysis are treated, as well as radio resource management mechanisms (adaptive beamforming, channel-aware single/multi-user scheduling, handover control, link adaptation, admission, congestion control,) and the current trend towards (potentially AI/ML-based) self-management. Cutting-edge radio features including single/multi-user massive MIMO beamforming, which is at the heart of 5G, are also covered. Considering the on-going technological developments in this exciting domain, the course material is updated every year. The principal lecturer has a 25+-year track record of research and consultancy in the field. The course is enriched with notes about actual deployments, historical anecdotes, popular misconceptions, fascinating factoids, illustrative demonstrations and (potentially) guest appearances.	
Study Goals	The objective of the course is to provide the student with a thorough understanding of contemporary mobile cellular networking technologies, their fundamental properties and tradeoffs, as well as the key challenges and approaches related to network dimensioning, planning and (self-)optimization towards efficient provisioning of coverage, capacity and quality.	
Education Method	Lectures, demonstrations, in-class discussions, assignment(s).	
Literature and Study Materials	Primarily: the lecture slides! For the enthusiastic student, supplementary background materials are recommended in the form of e.g. (white/scientific) papers and book chapters. A number of books on wireless/mobile communications in general or about 2/3/4/5G technologies specifically are recommended as background material; specifics will be given in class / in the lecture slides. As said, this is merely recommended for the most eager students. In principle, the lecture slides cover all relevant content.	
Assessment	Closed-book written exam (75%) and one assignment (25%). In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Co-Instructor	Prof.dr.ir. P.F.A. Van Mieghem	

EE4630	Telecommunication Network Architectures	3
Responsible Instructor	Ir. R.A.C.J. Noldus	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course presents the fundamentals of cellular mobile networks, including network architecture and services, with a focus on end-to-end and core network aspects. The course forms a companion course for Cellular Networks Radio Access Networks (ET4396). Students will be introduced to the concepts circuit switched and packet switched, the two fundamental communication methodologies that form the backbone of the mobile network. The course aims to educate students on the evolution of the network from 2G to 5G, whereby for each network generation ('G'), the new architectural principles are studied. For example, we discuss the GSM core network (2G), GPRS (2.5G), UMTS (3G), Evolved Packet Core (EPS; 4G), the IP Multimedia Subsystem (IMS), Multimedia telephony (MMTel), Intelligent Networks (IN), Circuit switched fallback (CSFB), Voice over LTE (VoLTE), Voice call continuity (VCC), Voice over Wifi and 5G network architecture. For the 5G network architecture, the concept of 'network slicing' will be presented, which is used for flexible resource allocation, for Internet of Things (IoT) applications and for 'verticals' such as Automotive, Smart Energy, Media & Entertainment, Vehicle communication and other forms of machine-type communication. The course also deals with the concept of virtualization and cloud computing. Networks are increasingly built as a collection of virtual network functions (VNF), deployed on industry standard computing platforms, comprising standard hardware and standard OS. Keywords: - Network architecture and network protocols - Core network, Radio access network (RAN) - Circuit switched, Packet switched - Location/mobility management/roaming - Security (authentication, ciphering,) - QoS/QoE/bearer architecture, QCI definitions, service management, policy control - IMS/VoLTE/Messaging/Voice-over-Wifi/Enriched calling - Virtualization, Cloud computing, Software Defined Networks (SDN) - Network slicing - Multi access Edge Computing (MEC) - Intelligent Networks - Verticals/applications, including e.g. Automotive, Smart Energy, IoT, Machine-to-machine communication - Web Real-time Communication (WebRTC)	
Study Goals	After this course, students are able to explain the meaning, working and rationale behind key cellular network concepts (such as those given by list of keywords above). Furthermore, students will be able to understand the challenges and associated trade-offs in the deployment and management of cellular networks. Students will also be able to analyse network architecture and evaluate network architecture on its merits or on its drawbacks.	
Education Method	This course consists of 8 classes with each class consisting of 2 hours. After the 4th lecture, a mid-term rehearsal takes place; and after the 8th lecture, an end-term rehearsal take place. These rehearsals help the students in preparing for the exam. The students are expected to study the lecture material after every lecture and to ask questions the next lecture, on any topic that requires further explanation. Homework will be provided after each lecture. Students may at any time interact with the lecturer via mail, to ask clarification on any topic of the course.	
Literature and Study Materials	Slides Articles and book chapters	
Assessment	Examination, closed book. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Co-Instructor	Prof.dr.ir. P.F.A. Van Mieghem	

EE4C06	Networking	5
Responsible Instructor	Prof.dr.ir. P.F.A. Van Mieghem	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	PART 1: Basics, concepts and computations of networks 1. Basics of networking & introduction to Network Science - what is a network? Representation of a graph, basics of graph theory, overview of the relatively new theory of complex networks, called Network Science. - important characterizers of a network (network metrics) - basic network/graph models - examples of real-world networks (airline transportation, the web and Internet, social networks, brain networks, etc.) and applications of network science 2. Concepts of networking - routing - Quality of Service (QoS) - traffic management and scheduling - network robustness (failure, cascading effects,...) - overlay networking and new aspects of networking such as interdependent networks PART 2: Applications and examples of networks (as listed below) will be taught (some of those by a guest lecturer). Ranging from year to year, a selection among the following will be covered: 1. Electrical networks (smart grids) 2. Networks on Chip (NoC) 3. Optical networks 4. Computer Networks (the Internet) 5. Mobile communication networks 6. Sensor networks 7. Biological networks 8. Social networks	
Study Goals	The course on Networking aims to provide a general and basic introduction to the art of networking, that tries to unravel the operation and behavior of networks, both man-made (infrastructures such as the Internet and power grids) as well as networks appearing in nature (such as the human brain, biological networks and social human interactions). The course on Networking will introduce concepts of the new Network Science, that basically studies the interplay between, on the one hand, the processes (also called functions or services) on the network and on the other hand, the underlying topology, that is mostly changing over time as an evolving organism, rather than as given or fixed object. Network Science combines many disciplines such as graph and network theory, probability theory, physical processes, control theory and algorithms. After this course, students are expected to represent/abstract real-world infrastructural network (e.g. a communication system) as a complex network, understand the basic methods to analyze properties of networks and dynamic processes on networks. Students will also understand why processes on networks and design of networks are so complex. Finally, students may appreciate the fascinatingly rich structure and behavior of networks and may realize that much in the theory of networks still lies open to be discovered.	
Education Method	Lectures, slides & homework. On the slides, references to articles (i.e. numbered paragraphs) or to pages in the book "Graph Spectra for Complex Networks" (see Literature and Study Material) are made. The information in the book thus complements the slides.	
Literature and Study Materials	Graph Spectra For Complex Networks, second edition (2023) Cambridge University Press ISBN: 978-009-36680-9 See: https://www.nas.ewi.tudelft.nl/people/Piet/bookGraphSpectraCN.html	
Assessment	The examination consists of multiple choice questions. The examination is executed on a computer. Each student logs in with his own student ID. The kind of questions is the same for each student, but the parameters and the order of multiple choice questions is individual. All information can be used (except consulting humans), i.e. the exam is "open book". Further details are given in Brightspace. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

ET4034	Telecom Business Architectures and Models	4
Responsible Instructor	Dr. E.F.M. van Boven	
Responsible for assignments	Prof.dr.ir. P.F.A. Van Mieghem	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	none	
Summary	Edgar van Boven, (KPN, TUD EEMCS)-- Past, present & future of communications -- Assignment 1 Eric Smeitink (KPN, TUD EEMCS)--Strategy, Spectrum & 5G Ton van der Knaap, (Stedin)-- What, why and how of business architecture; experiences from a grid operator and a telecom operator compared Samuel Pronk, (Krauthammer)-- Real Options-- Assignment 2 Ramin Hekmat, (KPN)-- Security and privacy in contemporary communication networks-- Assignment 3 Wouter de Vries (KPN)-- Artificial Intelligent Processes for People Tim Daeleman, (Prodapt Consulting)and Erik Lemmens (KPN)-- Digital Telecom Platforms Jose Almodovar, (TNO)--Standardization of Mobile Communications Frank Mertz (KPN)--5G Networks Developments and Opportunities Remco Helwerda (KPN) -- Digitalization Marco van der Pal, (Cap Gemini)--The IoT landscape architecture Ruud van de Bovenkamp (KPN)-- Fiber-to-the-Home Access Networks -- Assignment 4 Mark Dirksen (KPN)-- Business Continuity Management Pieter Veenstra, (Titanium)-- Service Architecture and 5G Roaming Security Edgar van Boven (KPN, TUD EEMCS)-- Value Case pitch practicing -- Assignment 5	
Course Contents	The essence of the course The ET4034 course Telecom, Architectures and Business Models was designed in 2001 and scheduled since 2002. The aim of the course is giving students from any faculty of the Delft University of Technology the opportunity to learn about a large variety of aspects of the telecom domain, not only its technologies, but also the diverse and challenging daily practice of organisations active in the communications sector. The course is provided by a team of 15 lecturers interactively sharing their knowledge about communications infrastructure in the context of overarching societal trends, service & network architecture, long term developments and new technology evolving from standardization initiatives. As a consequence of continuous innovation and societal developments, the course content is being updated each year including a well-considered choice of central themes that enhance and connect the course lectures. In 2022, the themes Internet of Things, Artificial Intelligence and 5G were highlighted, to be updated in 2023.	
Course Contents Continuation	EE5010 Internships or ET4399 projects in telecom related industries.	
Study Goals	Overview and understanding the Communications Sector, Services, Telecom infrastructures and Value Cases serving society	
Education Method	Online lectures, one powerpoint pitch presentation given by student teams and four off-line written assignments in teams of 2 or 3 students	
Literature and Study Materials	See information on Brightspace.tudelft.nl	
Reader	Course lectures, background information and video recordings available on ET4034 Brightspace	
Prerequisites	Finalised BSc as ET4034 is meant for Master students. PhD students are also welcome and can obtain credits when successfully passing the course assignments via the Graduate School.	
Assessment	The examination of the course Together forming the basis for the examination, the ET4034 course contains five assignments of which four written documents and one opportunity to strengthen ones pitching skills presenting a self-developed Value Case to a live audience. Students are invited to work in couples or trio's on the five assignments. The four written assignments each have a weight factor 1 and the Value Case pitch presentation has a weight factor 0,25 in the end mark of the ET4034 course. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Exam Hours	Not applicable, the course assignments are four documents written offline together with a course partner. The Value Case powerpoint presentation is a pitch that takes circa 7 minutes.	
Maximum number of participants	No limitation	
Self test	Not applicable	
maximum aantal deelnemers	No limitation	

ET4358	Fundamentals of Wireless Communications	5
Responsible Instructor	Dr.ir. G.J.M. Janssen	
Instructor	Dr.ir. J.H. Weber	
Instructor	Dr. P. Pawelczak	
Instructor	Dr. R. Litjens	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basics on Telecommunication Techniques and Digital Communication Networking.	

Course Contents	The Fundamentals of Wireless Communications course provides a selection of fundamental topics of modern wireless communication systems. The course is prerequisite for the courses on Wireless IoT and Local Area Networks (ET4394) and Mobile Networks (ET4396).
	
	The following aspects will be covered in this course.
	
	1. Radio propagation:
	- review of radio propagation, link-budget model,
	- path-loss and shadow fading,
	- multipath propagation:
	   * time domain: rms-delay spread, signal dispersion, inter-symbol interference,
	   * frequency domain: frequency selectivity, coherence bandwidth,
	- stochastic propagation models.
	
	2. Modern modulation techniques:
	- review of basic modulation techniques,
	- Orthogonal Frequency Division Multiplexing (OFDM),
	- Direct Sequence Spread Spectrum (DS-SS),
	- Frequency Hopping.
	
	3. Diversity:
	- diversity domains: frequency/time/spatial/polarization diversity
	- transmit and receive diversity,
	- diversity techniques: selection diversity, maximal ratio combining, equal gain combining,
	- MIMO (multiple input multiple output), fundamentals, capacity, space-time coding.
	
	4. Channel Coding:
	- characterization of error types, and basics of error control coding,
	- code design: trade-off between efficiency, reliability, complexity/delay,
	- block & convolutional codes,
	- puncturing, interleaving,
	- Turbo coding.
	
	5. Multiple Access:
	- multiplexing: FDMA, TDMA, CDMA,(O)FDMA, ALOHA,
	- delay modeling in multiplexing: Little's theorem, review of continuous time Markov chains, M/M/1 system.
	
	6. Cellular System Design:
	- basic cellular networking concepts
	- coverage-capacity-quality trade-off
	- network planning aspects: site density, sectorisation, frequency reuse
	- link budget analysis
	- traffic engineering: Erlang-B model	

Study Goals	You have gained understanding and insight in the basic concepts and principles applied in wireless communication systems, as described under Course Contents. You are able to explain the relations between the different concepts and trade-offs that can be made and can apply this knowledge by solving related problems by means of calculations.	
Education Method	Lectures, instruction lectures and homework exercises.	
Literature and Study Materials	Slides, notes, and papers, to be provided by the instructors. The following books are recommended as background material: - "Wireless Communications", Andrea Goldsmith, Cambridge University Press, 2005, ISBN 9780521837163, - http://web.mit.edu/dimitrib/www/datanets.html The following book is recommended as refresher material: - "Digital and Analog Communication Systems", L.W. Couch II, Prentice Hall, 2013, ISBN 978-0-273-77421-1.	
Assessment	The exam Fundamentals of Wireless Communications is a closed-book written exam. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Permitted Materials during Tests	During the written exam you are allowed to use the the following materials: - the booklet Tables & Graphs for use at the exam Wireless Communications ET4358, as provided on Brightspace, - 1 page A4 of your own hand-written notes/equations, written on both sides, (not type-written, no copy of notes and no worked out exercises or examples), - a non-programmable electronic calculator.	
Remarks	This course belongs to the set of elective track core courses of the MScEE track Wireless Communication & Sensing (WiCoS) and it is an elective course for other MSc programs. Actual course information is available on Brightspace.	

IN4341	Performance Analysis	5
Responsible Instructor	Prof.dr.ir. P.F.A. Van Mieghem	
Instructor	Dr. C. De Bacco	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course applies probability theory and the theory of stochastic processes to the design and performance evaluation of complex networks such as man-made networks as telecommunication, computer and embedded networks and biological networks. The computation with random variables is reviewed. Markov processes and queuing theory will be introduced to the current important concept of "Quality of Service (QoS)" provisioning and to the computation of the blocking probabilities in telephony (both fixed as mobile). Several applications (e.g. the robustness of networks, epidemics in networks, the Internet shortest path routing) are also included. More details are found on brightspace.	
Study Goals	The course intends to provide students with mathematical techniques, in particular probabilistic methods and graph theory, to compare the performance of different network designs and protocols.	
Education Method	Lectures and homework after each class The content of the lectures follows chapters in the book "Performance Analysis of Complex Networks and Systems" (see Literature and Study Material).	
Literature and Study Materials	Performance Analysis of Complex Networks and Systems. ISBN: 978-1-107-05860-6 See http://www.nas.ewi.tudelft.nl/people/Piet/bookPA.html	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 100%). Written and closed book. A formularium is provided that can be consulted at the examination. Final grade calculation: 1 * Written exam A passing final grade for this course is obtained when the final grade for the exam is at least a 5.8. Resit/ Repair opportunities: Written exam: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

SC42101	Networked and Distributed Control Systems	4
Responsible Instructor	Prof.dr.ir. T. Keviczky	
Instructor	Dr. M. Mazo Espinosa	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course starts with the modelling of so-called networked control systems (NCS). Networked control systems are systems in which the communication between plant and controller takes place via a (e.g. wireless) network. Such network-based communication leads to imperfections in sensor and control signals, such as time-varying and uncertain sampling intervals and delays, packet dropouts, scheduling constraints, etc. The modelling framework introduced in the course is subsequently used to support stability analysis, using characterizations based on linear matrix inequalities (LMI). Such stability analysis allows to study trade-offs between requirements on the controller, the network and plant properties. The second part of the course deals with the aspect of the distributed control of networked systems. In particular, distributed optimization methods and various decomposition techniques (primal, dual, augmented Lagrangian / proximal point method, ADMM), links to consensus algorithms, and their application in networked multi-vehicle distributed robotics problems. Online optimization-based control approaches such as distributed model predictive control for multivehicle cooperation, distributed LQR and decomposition based methods that are applicable to collections of mobile agents. The methods will be illustrated on application examples including cooperative rendezvous, distributed formation control, spacecraft formation flight, and robotic networks.	
Study Goals	The student must be able to: 1. create models for networked control systems with network-induced uncertainties / effects / imperfections, including time-varying and uncertain sampling intervals, delays, packet dropouts, and scheduling constraints 2. analyse the stability of NCS (involving the above effects), e.g. by applying LMI-based stability characterizations 3. describe and apply decomposition techniques for distributed optimization problems 4. describe and apply consensus algorithms to multi-agent coordination problems 5. solve cooperative control problems by implementing a distributed model predictive control approach 6. analyse and evaluate the stability and convergence of distributed optimization and control methods 7. develop evaluative judgments on the quality of your own work and of the performance of others (including peers) by implementing and providing feedback	
Education Method	Lectures.	
Computer Use	Matlab/Simulink is used to carry out the exercises of this course. Python is also allowed for some exercises/assignments.	
Literature and Study Materials	Course material: Lecture slides including additional reading material in the form of papers and textbooks are made available online.	
Assessment	1. This course is graded by four written assignment reports, each representing 25% of the final grade. 2. In order to complete the course, each assignment must be completed with at least a mark of 5,0 and the corresponding assignment report submitted before the deadline. In the event of a late submission, or a lower than 5,0 mark for any of the reports, no final mark will be given.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Delft Center for Systems and Control	

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer and Embedded-Systems Engineering**

Specialisation courses Control 2024	
Introduction 1	Choose at least 15 EC on specialisation courses, from any of the 4 specialisation lists.

RO47019	Intelligent Control Systems	4
Responsible Instructor	Dr. C. Della Santina	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Fundamentals of Control theory Fundamentals of Coding in Python and MatLab	
Course Contents	Theory and application of intelligent control system i.e., the combination of machine learning and control theory. Topics: * Introduction to nonlinear control and ICS * Learning nonlinear dynamical systems * Sum of Basis functions * Artificial neural networks, deep learning, and other advanced neural architecture in the context of ICS * Symbolic regression * Gaussian processes * Reinforcement learning * Imitation learning * Adaptive control * Iterative learning control We will aim for a balance of practical intuition and theoretical understanding: we will discuss examples of applications and go deeper into the theory when feasible.	
Study Goals	 At the end of this course, you will be able to: LO1 Explain the fundamentals of 'intelligent control' techniques - namely iterative control, Reinforcement learning, Model learning for control, etc. LO2 Implement 'intelligent control' techniques in Python. LO3 Reason on the expected performance of a given intelligent controller when applied to a given control problem. LO4 Propose an original, intelligent control scheme that can provide motor intelligence to a nonlinear and uncertain system (e.g., robot within a partially known environment).	
Education Method	Lectures (recorded and in class) and one assignment.	
Course Relations	Course title until 20/21 "Knowledge Based Control Systems" Course code until 21/22 "SC42050"	
Literature and Study Materials	 Lecture notes: R. Babuska. Knowledge-Based Control Systems. Slides and other course material (video lectures, software, demos, notes) can be downloaded from BrightSpace.	
Assessment	The examination will be as follows: (i) practical assignment and (ii) written exam, open book. You must score at least 5.5 on both assignments to get a pass. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be a remote open book with several fraud prevention measures. In accordance with the master program regulations: * The students will have the opportunity to resit if they fail the written examination. * If your practical assignment is graded between 5 and 5.75, you will have two weeks to integrate the assessors feedback. If this is done satisfactorily, you can be rewarded with a mark that is not higher than 6/10. The exact instructions for the addition will be announced after the grading of the PA is concluded.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding course registration, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) exam registration is obligatory. More information regarding exam registration, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *In some cases, there are entry requirements or additional criteria for admission to a course. These can be found in the OER/TER of the program, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Elective	Yes	
Tags	Artificial intelligence Complex Mathematics Intensive Lineair Algebra Mathematics Modelling Programming Project Proofs	
Department	ME Department Cognitive Robotics	
Contact	Cosimo Della Santina, C.DellaSantina@tudelft.nl	

SC42001	Control System Design	5
Responsible Instructor	Dr.ir. A.J.J. van den Boom	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	State-space description of single-input, single-output linear dynamic systems, interconnections, block diagrams. Linearization, equilibria, stability, Lyapunov functions and the Lyapunov equation Dynamic response, relation to modes, the matrix exponential. Realization of transfer function models by state space descriptions, coordinate changes, canonical forms. Controllability, stabilizability, uncontrollable modes and pole-placement by state-feedback. Application of LQ regulator. Observability, detectability, unobservable modes, state-estimation observer design Output feedback synthesis and separation principle. Reference signal modeling, integral action for zero steady-state error; Analysis in robust stability and robust performance. Basics of model predictive control (MPC). Different model-structures. Prediction models in state-space setting. Standard predictive control scheme. Relation standard form with GPC, LQPC and other predictive control schemes. Solution of the standard predictive control problem. Stability of MPC.	
Study Goals	By taking this course, the student - will be able to master the introduced theoretical concepts in systems theory and feedback control design - will be able to practically apply these concepts to design projects and tasks - and will be able to relate the learned concepts and techniques to other more specialized ones, to potentially integrate them by taking adjacent courses. - will be able to translate a predictive control problem into a standards setting and solve the predictive control problem. More specifically, the student will be able to: - Translate differential equation models into state-space and transfer function descriptions - Rationalize differences between state-space and transfer function approaches - Linearize a system, determine its equilibrium points, analyze directly its local stability, leverage Lyapunov theory to study general stability properties - Describe the effect of eigenvalue/pole locations to the dynamic system response in time/frequency domain. Contrast step and impulse responses. Analyze transients and steady-state - Investigate model controllability. Formulate and apply the procedure of pole-placement by state-feedback, as well as LQ optimal state-feedback control - Derive observability properties. Formulate and apply the procedure of state estimation and build converging observers - Formulate the separation principle and employ it for the design of output feedback - Build reference models and achieve zero steady-state error using integral control. - set up a predictive control problem. - Solve the standard predictive control problem; - Master the main analytical details in stability proofs of predictive control schemes;	
Education Method	Lectures 4/0/0/0	
Literature and Study Materials	Textbook (its use is strongly recommended): K.J. Astrom, R.M. Murray, Feedback Systems: An Introduction for Scientists and Engineers; second edition; Princeton University Press, Princeton and Oxford, 2009 Available online for download: http://www.cds.caltech.edu/~murray/books/AM08/pdf/fbs-public_24Jul2020.pdf The link for downloading the lecture notes for model predictive control will be available in the first quarter.	
Assessment	Written examination.	
Department	ME Department Delft Center for Systems and Control	

SC42015	Control Theory	6
Responsible Instructor	Prof.dr.ir. T. Keviczky	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Students following this course are expected to have a strong background in the following topics: - Linear algebra (basic concepts up to generalized eigenvectors and Jordan canonical form) - Calculus (smoothness, Jacobian, gradient) - Differential equations (existence, uniqueness, Laplace transform) - Classical control (transfer functions, block diagram algebra, Bode/Nyquist, etc.)	
Course Contents	- State-space description of multivariable linear dynamic systems, interconnections, block diagrams - Linearization, equilibria, stability, Lyapunov functions and the Lyapunov equation - Dynamic response, relation to modes, the matrix exponential and the variation-of-constants formula - Realization of transfer matrix models by state space descriptions, coordinate changes, normal forms - Controllability, stabilizability, uncontrollable modes and pole-placement by state-feedback - LQ regulator, robustness properties, algebraic Riccati equations - Observability, detectability, unobservable modes, state-estimation observer design - Output feedback synthesis (one- and two-degrees of freedom) and separation principle - Disturbance and reference signal modeling, the internal model principle	
Study Goals	The student is able to apply the developed tools both to theoretical questions and to simulation-based controller design projects. More specifically, the student must be able to: - Translate differential equation models into state-space and transfer matrix descriptions - Linearize a system, determine equilibrium points and analyze local stability - Describe the effect of pole locations to the dynamic system response in time- and frequency-domain - Verify controllability, stabilizability, observability, detectability, minimality of realizations - Sketch the relevance of normal forms and their role for controller design and model reduction - Describe the procedure and purpose of pole-placement by state-feedback and apply it - Apply LQ optimal state-feedback control and analyze the controlled system - Reproduce how to solve Riccati equations and describe the solution properties - Explain the relevance of state estimation and build converging observers - Apply the separation principle for systematic 1dof and 2dof output-feedback controller design - Build disturbance and reference models and apply the internal model principle	
Education Method	Lectures and Exercise Sessions	
Computer Use	The exercises will be partially based on Matlab in order to train the use of modern computational tools for model-based control system design.	
Course Relations	This course is intended primarily for MSc students that are enrolled in the Systems & Control MSc program! For students who are interested in Systems & Control within their own MSc program, the SC42001 Control System Design course is recommended instead. There is a large overlap between the two courses in terms of topics, but SC42001 emphasizes broader concepts of control, it is more application-oriented, and does not discuss the technical proofs and mathematical background so extensively.	
Literature and Study Materials	B. Friedland, Control System Design: An Introduction to State-space Methods. Dover Publications, 2005 K.J. Astrom, R.M. Murray, Feedback Systems: An Introduction for Scientists and Engineers, Princeton University Press, Princeton and Oxford, 2009 http://www.cds.caltech.edu/~murray/amwiki/index.php?title=Main_Page	
Assessment	Written mid-term examination (15% grade) and written final examination (85% grade) will determine the total final grade. The mid-term result is only valid for the academic year when it was obtained. For the resit examination there will be a written examination (100% grade) for which the mid-term result will not count. If on-campus exams will not be possible, then all exams will be organized online (via Ans).	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Delft Center for Systems and Control	

SC42025	Filtering & Identification	6
Responsible Instructor	Dr. M. Kok	
Instructor	Dr.ir. K. Batselier	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The objective of this course is to apply filtering and identification methods. The focus is both on deriving these methods and on using these methods on real-life data sequences. The course focuses on stochastic and recursive least squares, on (extended) Kalman filtering, on prediction error methods and on subspace identification. The course also includes a review of specific topics from linear algebra, dynamical system theory and statistics, that are relevant for filtering and system identification.	
Study Goals	At the end of the course you should be able to: 1) Apply filtering and identification methods to solve previously unseen estimation problems. As sub-learning goals you should be able to: a. Relate basic linear algebra, signal processing techniques and probability theory to physical properties of a (dynamical) system. b. Derive the solution of the stochastic and recursive least squares problems from both a Bayesian perspective and in terms of unbiased / minimum variance properties. c. Derive the Kalman filter from both a Bayesian perspective and in terms of unbiased / minimum variance properties. d. Derive subspace and prediction error identification methods. e. Motivate the steps that should be taken in the derivations in b-d. 2) Apply filtering and identification methods to estimate the state and identify the model using real-life data sequences. As sub-learning goals you should be able to: a. Implement a (stochastic / recursive) least squares problem, an (extended) Kalman filter and identification methods in Matlab. b. Apply and motivate practical aspects of filtering and identification.	
Education Method	Lectures 0/6/0/0	
Literature and Study Materials	Book Filtering and System Identification: A Least Squares Approach by Michel Verhaegen and Vincent Verdult. ISBN: 13-9780521875127	
Assessment	Written exam (open book) and practical exercises. If needed due to covid-19 restrictions, the exam will be online.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Remarks	The software package Matlab is needed to solve the practical exercise.	
Department	ME Department Delft Center for Systems and Control	

SC42056	Optimization for Systems and Control	3
Responsible Instructor	Prof.dr.ir. B.H.K. De Schutter	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	basic experience with Matlab	
Course Contents	Essentially, almost all engineering problems are optimization problems. If a civil engineer designs a bridge, then one of the main objectives is to obtain the cheapest design or the design that can be implemented most rapidly, where of course several specifications and constraints such as size, strength, safety, etc. have to be taken into account. When developing a new type of engine, we look for the most economical design, the cheapest design, or the design with the highest performance. A process engineer wants a production unit to deliver a final product of maximal quality, with minimal expenditure of energy or with maximal output flow. When composing a portfolio, a financial engineer tries to maximize the expected profits, subject to the given risk constraints. So we encounter optimization problems in almost every engineering field. How can we solve such an optimization problem? That is the topic that will be addressed in this course. We will in particular discuss several classes of optimization problems and next focus on how to select most efficient numerical algorithms to solve a given optimization problem. The examples and case studies of this course are primarily oriented towards systems and control. More specifically, the following topics will be treated: * Linear Programming * Quadratic Programming * Nonlinear Optimization * Constraints in Nonlinear Optimization * Convex Optimization * Global Optimization * Matlab Optimization Toolbox * Multi-Objective Optimization * Integer Optimization	
Study Goals	After successfully completing this course the students should be able to select the most efficient and best suited optimization algorithm for a given optimization problem. They should have insight into the basic operation of some of the most important and relevant optimization algorithms. They should be able to reduce the complexity of the problem using simplifications and/or reformulations so as to augment the efficiency of the solution approach. Finally, they should be able to solve the resulting optimization problem in practice.	
Education Method	lectures + 2 mandatory assignments Attending the lectures is not mandatory, provided the recorded lectures and lecture materials are processed via self-study.	
Assessment	Written exam (individual, fully closed-book, no calculators nor formulas sheets allowed, counts for 70% of the final marks) + 2 assignments (group assignments, assessed through written reports, count for 30% of the final marks). To successfully pass the course the final weighted average score for the exam and the assignments together should be at least 5.75. Important: Partial marks for the exam or the assignments do not carry over from one academic year to the next.	
Permitted Materials during Tests	none	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Delft Center for Systems and Control	
Contact	Questions are preferably asked during, before, or after the lectures, or via the Discussion forum on Brightspace.	

SC42061	Nonlinear Control Systems	3
Responsible Instructor	Dr.ir. M. Jafarian	
Contact Hours / Week x/x/x/x	0.6.0.0	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	- Linear state space models (e.g. SC42015 Control Theory) - Basic knowledge of (multivariable) calculus, differential equations and linear algebra	
Course Contents	- Fundamentals of nonlinear systems (basic properties, solvability, types of equilibrium points) - Analysis of planar systems (graphical representations, periodic orbits, elementary bifurcations) - Stability theory for autonomous nonlinear systems (Lyapunov's methods, invariance principle) - Advanced stability theory (time-varying systems, input-to-state stability, passivity) - Observers (local, global and high-gain, extended Kalman filter)	
Study Goals	Students can: - determine if a system is nonlinear, time-invariant, autonomous, etc. - determine time intervals over which a system is solvable - sketch and interpret graphical representations of 2d systems - check for the existence or absence of periodic orbits in 2d systems, compute bifurcation points, and classify elementary bifurcations - determine if equilibrium points of autonomous nonlinear systems are (asymptotically, exponentially, ...) stable - determine if equilibrium points of certain classes of non-autonomous systems are (asymptotically, exponentially, ...) stable - construct observers for certain classes of nonlinear systems	
Education Method	Lectures, problem sets and instruction sessions.	
Literature and Study Materials	- H. K. Khalil: "Nonlinear Control", Pearson, 1st Global Edition, 2015. ISBN 9781292060507. - Additional handouts	
Assessment	Written exam, depending on the COVID-19 situation either on campus or online.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Delft Center for Systems and Control	

SC42075	Modeling and Control of Hybrid Systems	3
Responsible Instructor	Prof.dr.ir. B.H.K. De Schutter	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	core Systems & Control topics (including state-space models, discrete-time and continuous-time models, stability, linearization), optimization basics (linear programming), Matlab	
Course Contents	Recent technological innovations have caused a considerable interest in the study of dynamical processes of a mixed continuous and discrete nature. Such processes are called hybrid systems and are characterized by the interaction of time-continuous models (governed by differential or difference equations) on the one hand, and logic rules and discrete-event systems (described by, e.g., automata, finite state machines, etc.) on the other. A hybrid system also arises in practice when continuous physical processes are controlled via embedded software that intrinsically has a finite number of states only (e.g., on/off control). This course provides an introduction to modeling, analysis, and control of hybrid systems. The main topics considered are: * General introduction, examples of hybrid systems and motivation * Modeling frameworks (automata, hybrid automata, piecewise-affine systems, complementarity systems, mixed logic dynamical systems, Petri nets) * Properties and analysis of hybrid systems (well-posedness, Zeno behavior, stability, liveness, safety, ...) * Control of hybrid systems (switching controllers, model predictive control) * Verification and tools	
Study Goals	After successfully completing the course, you will be familiar with the main aspects and characteristics of hybrid systems, you have insight in trade-off modeling power versus decision power in the context of analysis and control of hybrid systems, and you will have obtained an overview of modeling, analysis, and control methods of tractable classes of hybrid systems. In addition, you will be able to apply hybrid systems modeling and control to simple case studies.	
Education Method	lectures + mandatory assignment Attending the lectures is not mandatory, provided the recorded lectures and lecture materials are processed via self-study.	
Assessment	Written exam (individual, closed-book, no calculators, counts for 60% of the final marks) + assignment (group assignment, assessed through written report, counts for 40% of the final marks) To successfully pass the course the final weighted average score for the exam and the assignments together should be at least 5.75. Important: partial marks for exam or assignment do not carry over from one academic year to the next.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Delft Center for Systems and Control	
Contact	Questions are preferably asked during, before, or after the lectures, or via the Discussion forum on Brightspace.	

SC42095	Digital Control	3
Responsible Instructor	A. Dabiri	
Instructor	M.A. Sharifi Kolarijani	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	SC42015, SC42000 or similar. Knowledge of classical control techniques (systematic and realistic PID design, frequency domain approaches) as well as state space theory is required.	
Course Contents	Computer control. Sampling of continuous-time signals. The sampling theorem. Aliasing. Discrete-time systems. State-space systems in discrete-time. The z-transform. Selection of sampling-rate. Analysis of discrete-time systems. Stability. Controllability, reachability and observability. Disturbance models. Reduction of effects of disturbances. Design methods. Approximations of continuous design. Digital PID-controller. State-space design methods. Observers. Pole-placement. Optimal design methods. Linear Quadratic control. Prediction. LQG-control. Implementation aspects of digital controllers.	
Study Goals	The student must be able to: 1. describe the essential differences between continuous time and discrete-time control 2. transform a continuous time description of a system into a discrete-time description 3. calculate input-output responses for discrete-time systems 4. analyse the system characteristics of discrete-time systems 5. employ a pole-placement method on a discrete-time system 6. implement an observer to calculate the states of a discrete time system 7. apply optimal control on discrete-time systems 8. describe the functioning of the Kalman-filter as a dynamic observer	
Education Method	Lectures and computer exercises	
Computer Use	Matlab/Simulink is used to carry out the exercises of this course.	
Literature and Study Materials	Course material: Lecture notes are made available on Brightspace Textbooks: K.J. Astrom, B. Wittenmark 'Computer-controlled Systems', Prentice Hall ,1997, 3rd edition. L. Keviczky et. al 'Control Engineering', Springer, 2019. L. Keviczky et. al 'Control Engineering: MATLAB Exercises', Springer, 2019. References from literature: B.C. Kuo 'Digital Control Systems', Tokyo, Holt-Saunders, 1980 G.F. Franklin, J.D. Powell 'Digital Control of Dynamic Systems', 1989, 2nd edition, Addison-Wesley	
Assessment	Project assignment	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Remarks	Old course code: WB2305	
Department	ME Department Delft Center for Systems and Control	

SC42101	Networked and Distributed Control Systems	4
Responsible Instructor	Prof.dr.ir. T. Keviczky	
Instructor	Dr. M. Mazo Espinosa	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course starts with the modelling of so-called networked control systems (NCS). Networked control systems are systems in which the communication between plant and controller takes place via a (e.g. wireless) network. Such network-based communication leads to imperfections in sensor and control signals, such as time-varying and uncertain sampling intervals and delays, packet dropouts, scheduling constraints, etc. The modelling framework introduced in the course is subsequently used to support stability analysis, using characterizations based on linear matrix inequalities (LMI). Such stability analysis allows to study trade-offs between requirements on the controller, the network and plant properties. The second part of the course deals with the aspect of the distributed control of networked systems. In particular, distributed optimization methods and various decomposition techniques (primal, dual, augmented Lagrangian / proximal point method, ADMM), links to consensus algorithms, and their application in networked multi-vehicle distributed robotics problems. Online optimization-based control approaches such as distributed model predictive control for multivehicle cooperation, distributed LQR and decomposition based methods that are applicable to collections of mobile agents. The methods will be illustrated on application examples including cooperative rendezvous, distributed formation control, spacecraft formation flight, and robotic networks.	
Study Goals	The student must be able to: 1. create models for networked control systems with network-induced uncertainties / effects / imperfections, including time-varying and uncertain sampling intervals, delays, packet dropouts, and scheduling constraints 2. analyse the stability of NCS (involving the above effects), e.g. by applying LMI-based stability characterizations 3. describe and apply decomposition techniques for distributed optimization problems 4. describe and apply consensus algorithms to multi-agent coordination problems 5. solve cooperative control problems by implementing a distributed model predictive control approach 6. analyse and evaluate the stability and convergence of distributed optimization and control methods 7. develop evaluative judgments on the quality of your own work and of the performance of others (including peers) by implementing and providing feedback	
Education Method	Lectures.	
Computer Use	Matlab/Simulink is used to carry out the exercises of this course. Python is also allowed for some exercises/assignments.	
Literature and Study Materials	Course material: Lecture slides including additional reading material in the form of papers and textbooks are made available online.	
Assessment	1. This course is graded by four written assignment reports, each representing 25% of the final grade. 2. In order to complete the course, each assignment must be completed with at least a mark of 5,0 and the corresponding assignment report submitted before the deadline. In the event of a late submission, or a lower than 5,0 mark for any of the reports, no final mark will be given.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Delft Center for Systems and Control	

SC42110	Dynamic Programming and Stochastic Control	5
Responsible Instructor	M.A. Sharifi Kolarijani	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Solid knowledge of undergraduate probability (e.g., conditional distribution, expectation, variance), linear algebra (e.g., eigendecomposition and Jordan forms), and calculus (e.g., gradient and extrema of a function). Mathematical maturity and the ability to provide precise and rigorous arguments are also important. A class in analysis will be helpful, although this prerequisite will not be strictly enforced.	
Course Contents	The course covers the basic models and solution techniques for stochastic optimal control problems and sequential decision-making under uncertainty. The course starts with a class of stochastic processes known as Markov chains. We then extend the model to include control variables (Markov Decision Processes, or in short MDPs) and study optimal control of a dynamical system over a finite number of stages. We also discuss an important class of problems known as optimal stopping problems. Different applications from engineering and finance to operations research and management science will be covered.	
Study Goals	After successfully completing this course, you will be able to - model real-world control and decision-making problems under uncertainty and in dynamic environments using the framework of discrete-time Markov decision process; - compute relevant technical concepts including hitting probabilities, expected hitting times, invariant distributions, and the long-run proportion of time spent in a given state; - formalize the optimal decision-making problem by deriving the respective dynamic programming principle; - use the dynamic programming algorithm to find useful properties of the desired solution and solve the optimal decision-making problem.	
Education Method	Lectures and tutorials	
Literature and Study Materials	- The lecture notes of the course (available on Brightspace) - Dynamic Programming and Optimal Control, 3rd edition, D. Bertsekas, Athena Scientific, 2005 - Introduction to Probability, D. Bertsekas and J. Tsitsiklis, Athena Scientific, 2002	
Assessment	Written exam (If on-campus exams will not be possible, then all exams will be organized online via Ans)	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *In some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Delft Center for Systems and Control	
Contact	The students can reach the course staff via email regarding administrative issues.	

SC42125	Model Predictive Control	4
Responsible Instructor	Dr.ing. S. Grammatico	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Model Predictive Control (MPC) is perhaps the most effective optimal control strategy for constrained dynamical systems. The basic concept of MPC is to exploit a dynamic model to forecast the system behavior, and optimize the forecast to determine the control input as the best decision at the current time. With emphasis on constrained linear systems, the course will present the theoretical fundamentals of MPC, such as Lyapunov stability, optimality and robustness, and its computational methods, such as quadratic programming.	
Study Goals	- Derive state prediction matrices from discrete-time linear models - Design the cost function, state and input constraints - Design MPC controllers with guaranteed recursive feasibility and asymptotic stability via appropriate terminal cost and constraint set - Formulate and solve constrained-linear-quadratic MPC problems via quadratic programming - Implement and simulate closed-loop systems controlled by MPC on Matlab - Design MPC controllers with integral action for reference tracking	
Education Method	Lectures, instructions.	
Assessment	Final project and oral exam.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Elective	Yes	
Tags	Mathematics Matlab Optimalisation	
Department	ME Department Delft Center for Systems and Control	

SC42130	Fault Diagnosis and Fault Tolerant Control	4
Responsible Instructor	Dr. R. Ferrari	
Contact Hours / Week x/x/x/x	0.0.0.4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Control Systems Design, System Identification, Signal Processing, Statistics and Probability	
Summary	In critical systems such as transportation systems, power plants, distribution networks or life-support systems it is not enough to design a control system that meets the required performance levels. It is also of paramount importance to guarantee that such requirements, or a subset of them, are met also during non ideal conditions such as in the presence of physical failures or cyber attacks. Most of all, safety must be guaranteed at all times. This can be obtained by designing into the control system additional functionalities that will allow to detect the occurrence of faults or cyber attacks, and deploy adequate countermeasures, which may include the reconfiguration of the control system.	
Course Contents	Faults and failures in dynamical systems; Cybersecurity and cyber attacks to Industrial Control Systems; Components and Services model; Availability and Reliability; FTA (Fault tree Analysis); FMEA (Failure Modes and Effects Analysis); Signal based methods for Diagnosis; Model based methods for Diagnosis; Residuals computation and evaluation; Learning of fault model; Fault Tolerant Control (Model Matching, Virtual Sensor and Actuator, Adaptive Control)	
Study Goals	after the course the student should be able to: 1) analyse the structure of a system and identify main components, the services provided by each and carry out a Fault Modes and Effects Analysis; 2) analyse and model faults or cyber attacks that can occur in a given dynamical system; 3) design an algorithm for detecting, isolate and identify them; 4) design a policy for reconfiguring the control system in order to accommodate them. 5) analyze the vulnerability of an industrial control system to cyber attacks and design prevention and detection algorithms	
Education Method	in person lectures, in-class exercises, homework	
Literature and Study Materials	Recorded lectures, Slides, Example exercises with partial solutions, Suggested textbook: Blanke, M., Kinnaert, M., Lunze, J., Staroswiecki, M., & Schröder, J. (2006). Diagnosis and fault-tolerant control (Vol. 691). Berlin: Springer.	
Assessment	Assignments 30%, final exam (written) 70%. The grade for the assignments will be the average grade of all the assignments. To pass the course, the final grade must be a 5.75 or higher. The grade for the exam is not allowed to be lower than a 5.0.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Elective	Yes	
Tags	Design Group work Mathematics Matlab Mechatronics Modelling Numeric Methods Process Signals and Systems	
Department	ME Department Delft Center for Systems and Control	

SC42145	Robust Control	3
Responsible Instructor	Prof.dr.ir. J.W. van Wingerden	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	· Recap on background in linear systems theory and classical feedback control · Multivariable system control: Nyquist, interaction, decoupling · Directionality in multiloop control, gain and interaction measure · Stabilizing controllers and the concept of the generalized plant · Parametric uncertainty descriptions, approximations · The general framework of robust control · Robust stability analysis · Nominal and robust performance analysis · The H-infinity control problem · The structured singular value: Definition of mu · Mu synthesis, DK-iteration, role of uncertainty structure. · Design of robust controllers, choice of performance criterion and weights	
Study Goals	The student is able to reproduce theory and apply computational tools for robust controller analysis and synthesis. More specifically, the student must be able to: 1. substantiate relation between frequency-domain and state-space description of dynamical systems 2. define stability and performance for multivariable linear time-invariant systems 3. construct generalized plant for complex system interconnections 4. describe parametric and dynamic uncertainties 5. translate concrete controller synthesis problem into abstract framework of robust control 6. reproduce definition of the structured singular value 7. master application of structure singular value for robust stability and performance analysis 8. design robust controllers on the basis of the H-infinity control algorithm 9. apply controller-scalings iteration for robust controller synthesis	
Education Method	Lectures + Assisted lectures+Computer sessions	
Literature and Study Materials	S. Skogestad, I. Postlethwaite, Multivariable Feedback Control, 2nd Edition. John Wiley & Sons, 2005. References from literature: K. Zhou, J.C. Doyle, K. Glover, Robust and optimal control, Prentice Hall, 1996 D.-W. Gu, P.Hr. Petkov, M.M. Konstantinov, Robust Control Design with Matlab. Springer Verlag, London, UK, 2005	
Assessment	Controller design exercise (report and code)	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Remarks	Old course code: SC42010	
Department	ME Department Delft Center for Systems and Control	

SC42150	Statistical Signal Processing	3
Responsible Instructor	M. Khosravi	
Instructor	G. de Albuquerque Gleizer	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	The course unit assumes that you are familiar with complex numbers, Z-transform, Fourier transform, basics of linear time-invariant systems, probability and statistics at the undergraduate level, and fundamentals of linear algebra. Also, programming skills in MATLAB/Python are necessary for the computer assignments.	
Course Contents	1. Theory of probability and point estimation 2. Stochastic signals 3. Linear filtering	
Study Goals	After the course, students should be able to: - Design and analyze parameter estimators - Analyze stochastic signals in both time and frequency domains. - Apply filtering techniques in the time and frequency domains. - Design linear time-invariant filters based on given frequency and time domain specs. - Estimate the information-carrying components from noisy measurement signals.	
Education Method	Lecture with tutorial consisting of theoretical and MATLAB/python exercises	
Assessment	Written exam(s) and homework. The final grade is a weighted average of the mark obtained on the examination(s) (80%) and that on the compulsory homework (MATLAB/python) assignments (20%).	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Delft Center for Systems and Control	

SC42155	Modelling of Dynamical Systems	3
Responsible Instructor	Dr. R. Ferrari	
Instructor	Dr. M. Mazo Espinosa	
Contact Hours / Week x/x/x/x	4.0.0.0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	The course will cover the basics of modeling of both physical systems (in the form of ordinary differential equations), computing and scheduling systems (as finite state machines), and combinations thereof (as hybrid models). Physical systems modeling. The course will cover the basics of Lagrangian modelling, I.e. employing energy balances and the Euler-Lagrange equations to arrive to closed form state-space dynamical equations for system analysis and control design. Bond-graph modelling will be covered as a powerful tool for modelling engineering systems, especially when different physical domains are involved. The basic ideas of model reduction, a technique to reduce the complexity of dynamical models, may also be covered in the course. The fundamental differences and connections between continuous vs discrete time modelling will be discussed. Similarly, frequency domain modelling is also discussed with practical motivations for frequency vs time domain modelling. Finite State Models. A brief introduction to models traditionally employed for computation, e.g. finite state machines, automata, and petri-nets, will be provided. Particular attention will be given to the applicability of these models to provide high-level dynamic descriptions of discrete-event systems and for behaviour specifications of designs. Hybrid systems. The course will finalize with a short introduction to hybrid systems through the formalisms of: Hybrid automata and jump-flow systems, and presenting a brief number of interesting subclasses of hybrid systems, e.g. PWA systems, and timed -automata.	
Study Goals	The purpose of the course is to introduce the students to basic concepts and results in physical modeling and the theory of nonlinear control systems. After successful completion of the course, the student is able to construct models of physical systems from the knowledge of the basic laws of physics (first principles) construct models of systems using the theory of bond graphs. construct models of systems using the Euler-Lagrange method (perform a model reduction for linear systems using a balanced realization) construct finite state models (automata, FSMs, Petri-Nets) of simple computing and scheduling systems analyze basic properties of finite state systems: blocking, determinism, and language expressed by a model. describe cyber-physical systems as hybrid dynamical models. select the appropriate modelling approach for a given problem.	
Education Method	Lectures 0/4/0/0	
Literature and Study Materials	Lecture notes and additional reader	
Assessment	Assignments (35%) and a mini-project (65%). The grade for the assignments will be the average grade of all the assignments. To pass the course, the final grade must be a 5.75 or higher. The grade for the mini-project is not allowed to be lower than a 5.0.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Tags	Embedded systems Group work Lineair Algebra Mathematics Matlab Mechatronics Modelling Numeric Methods Signals and Systems Stochastics	
Department	ME Department Delft Center for Systems and Control	

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer and Embedded-Systems Engineering**

Thesis Project 2024

CESE5000	Thesis Project	45

Responsible Instructor	Dr.ir. A.J. van Genderen
Responsible Instructor	Prof.dr. K.G. Langendoen
Contact Hours / Week x/x/x/x	x/x/x/x
Education Period	1 2 3 4 Summer Holidays
Start Education	2
Exam Period	none
Course Language	English
Course Contents	CESE5000 Final Project is the final part of the Master's degree programme. During this project, you will be required to demonstrate your ability to solve a research or engineering problem. The project must be carried out using the techniques of project management. You will begin by making a project plan in cooperation with your Masters thesis advisor. Several aspects of the project are defined within the plan, including the assignment, the frequency of interaction with the advisors, the milestones of the project and the resources and facilities offered by the faculty. You will be required to adhere to your plan throughout the project. It is obviously possible to adjust your plan under certain circumstances and after discussion with your daily supervisor. It is possible to do the project in industry. In that case it is important to note that you should not make arrangements with any company without prior consultation with your master coordinator and without having a TU Delft supervisor. At the end of the project, you will submit your Masters thesis, which must be written in English, and make an oral presentation of your work to the Thesis Committee. The Thesis Committee will announce the final mark, which is based on the quality of work, project performance, the thesis, the presentation and the subsequent discussion. More information about the graduation process: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/graduation-msc/. Or go to Brightspace: https://brightspace.tudelft.nl/d2l/home/492869 Important: you should register your thesis project in MaRe: https://mare.ewi.tudelft.nl and use this, together with your supervisor, to further administer your progress during the thesis project.
Study Goals	1. The student is able to design a research or engineering project: - The student is able to orientate on an interesting scientific or engineering question by means of a scientific literature study; - The student is able to use, explain and justify adequate research and design methodologies; - The student is able to apply theory to the performed project; - The student is able to use techniques for interpretation and verification and bases his/her conclusions on results; - The student is able to do reliable work with scientific significance. 2. The student is able to execute a research or engineering project: - The student has a critical attitude towards his/her own results, literature and specialists; - The student makes an original contribution to the project; - The student takes initiative (together with the supervisor) to give his/her own input within the research project; - The student interacts sufficiently with peers and superiors; - The student is able to make and execute a project plan. 3. The student is able to write a research report about the project: - The student is able to write a research report that shows sufficient coherence of content; - The students is able to structure the research report and sufficiently present the content (text and figures); - The student expresses argumentation using correct spelling and grammar; 4. The student is able to present and defend the research or engineering project: - The student is able to present the content using sufficient detail to support conclusions; - The student is able to logical structure the presentation and use visual aids; - The student is able to adequately formulate and express himself/herself as well as sufficiently address the audience; - The student is able to argument and answer the questions asked by the committee.
Education Method	Supervised research, individual project
Prerequisites	The student must have completed at least 60 EC of the Master's degree programme and be in possession of a Declaration of Credits Obtained (can be requested from EEMCS service desk) that shows that this is true. To be able to get a Declaration of Credits Obtained, the individual exam program (IEP) should have been approved by the Board of Examiners. After you have registered your thesis project in MaRe (https://mare.ewi.tudelft.nl) you should upload your Declaration of Credits Obtained there. Note: In some cases, the thesis supervisor may impose additional conditions for starting this project.
Assessment	The thesis committee assesses the thesis and the defense on the following criteria: - quality of work: novelty, volume, grasp, methodology, publishable; - personal performance: autonomy, planning, creativity, attitude; - quality of thesis report: clarity, organisation, argumentation; - oral presentation and defense: clarity, focus, relevance, discussion. More information on thesis grading: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/graduation-msc/assessment/ Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer and Embedded-Systems Engineering**

Suggestion free elective space: Internship & Projects 2024	
Introduction 1	Free elective courses worth at least 15 credits. The free elective space may be used for additional specialisation courses, a project or internship, language courses or courses from other Masters programmes. Please note: Students may choose only one project or internship (10-15 EC) from the list above. Students with a Bachelor degree from a Dutch HBO institution who have had 30 credits or more worth of work experience in their prior education, may not include the Industry Internship in their IEP. Students who wish to carry out their Thesis project outside TU Delft (i.e. in a company or other organisation) in any case may not include the Industry Internship in their IEP. The Research Internship and the Thesis project must be in different research groups. We want to ensure that the student has a broad experience in topics, supervisor, and culture. RULES: 1) an intership is an elective (15 EC) part, a thesis project is a compulsory (45 EC) part of the CESE program; both can be done either at a research group or with industry. 2) if a student decides to do both, one of them must be done at a research group (i.e. both with industry is forbidden). 3) if a student does both at a research group, the topic and supervisor must be different. 4) for research labs in industry (like TNO) the nature of the topic decides whether the internship/thesis has the industry or research label.

CESE5010	Industry Internship	15
Responsible Instructor	Dr.ir. A.J. van Genderen	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	Master students of Computer and Embedded Systems Engineering can follow CESE5010 as part of their free electives. The industry internship period should be from 10 weeks up to three months, for which you will receive 15 credits. Industry internships longer than 3 months will NOT be approved! The industry internship should be of sufficient level to be part of your MSc curriculum performed at a company or institute, either in the Netherlands or abroad. An industry internship in a university outside the Netherlands can be accepted. An industry internship at TU Delft will not be accepted. Instead, you can do a research internship CESE5020 at TU Delft. When carrying out Industry Internship CESE5010 at a company or institute, the following conditions apply: The student participates in the company or institute culture in general, including sharing the company-established balance between on-site and online work. The student takes part in the day-to-day running of the company or institute similar to regular employees. This includes having regular consultation and interaction with colleagues and manager(s), and use of and access to the company-specific resources and tools applicable to the internship. Such resources and tools can be physical/on-site (e.g. measurement equipment, company machines and infrastructure) as well as virtual (online, software tools, databases, ...). For more info, enrol on "Internship CESE" on Brightspace.	
Study Goals	There are three types of educational goals for the Internship: I. Profession related aspects Insight into the profession of the computer and embedded systems engineer Insight into the duties and the responsibility of the engineer in a company II. Social-psychological aspects Learning to operate in another culture (both within and outside the company) Gaining insight into the position of the company with respect to comparable companies III. Professional knowledge and skills related aspects Learning to apply the gained knowledge and skills in a situation different from the (own) university situation Learning to acquire new knowledge and skills that are necessary to meet the requirements for the Internship	
Education Method	Detailed information on projects, procedures and application is available on the Brightspace organization "Internship CESE". The internship application form should be submitted to the Internship Office prior (!) to starting the internship. This application form contains a project description that the student writes with input from the company supervisor and a TU Delft supervisor. An approved internship application form indicates that you can do the internship for study credits.	
Literature and Study Materials	Enroll in Brightspace organizaton "Internship CESE" for more information about the Internship. You can find the step-by-step guide explaining all procedures, forms and internship offers. For additional information e.g. about grants and funding for internships abroad enroll in Brightspace course Study Abroad EEMCS.	
Prerequisites	In order to ensure that students start the projects with a solid skill set, students can apply for the Industry Internship after the third quarter of the first year of the MSc CESE programme. In order to start the Internship the following requirements must be fulfilled: 1. The Individual Exam Programme (IEP) must have been submitted to and approved by the Board of Examiners. 2. You have obtained 45EC before the start of your Internship. Detailed information on projects, procedures and application is available on the Brightspace organizaton "Internship CESE". The internship application form should be submitted to the Internship Office prior (!) to starting Internship. Note that is NOT allowed to do both an Industry Internship and a Thesis Project at a company.	
Assessment	Pass/Fail (no grade) For the assessment you need to submit: The company assessment form The TU Delft assessment form An internship and evaluation report. You can find guidelines for the report on Brightspace. In order to give the company supervisor sufficient time to read and check the report, yet to ensure that students receive timely feedback and final grade, your company supervisor is asked to assess your internship at the end of your internship period, within two weeks after submission of the reports, and send his assessment form to the TU Delft supervisor. Your TU Delft supervisor needs to assess your internship within two weeks after receiving the assessment of the company supervisor. The TU Delft supervisor will then send the TU Delft assessment form and the company supervisor assessment form to the EEMCS Internship Office; internship-eemcs@tudelft.nl. The assessment forms can be found on Brightspace CESE5010. Step by step: - During your internship period you write the scientific report. - At the end of your internship, you have up to one week to finalize your evaluation report. - At the end of your internship, you send the scientific report to your company supervisor. - The company supervisor sends his assessment within 2 weeks after receiving the scientific report to the TU Delft supervisor. - You send both reports to the internship office. - The TU Delft supervisor sends his assessment within 2 weeks after receiving the report and the company assessment and send both assessments to the internship office. The final assessment is based on the Company supervisor Assessment Form, the TU Delft supervisor Assessment Form and the Internship Report and will be graded with a pass or fail. Your mark will be processed as soon as the Internship Office EEMCS has received: Two digital assessment forms signed by the supervisors and send by the TU Delft supervisor to the internship office (you need	

	to provide your supervisors with the assessment forms yourself. You can find them on Brightspace CESE5010). A digital copy of both reports sent by you to the internship office repair option: Improve and resubmit report Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).

CESE5020	Research Internship (10-15 EC)	10
Responsible Instructor	Dr.ir. A.J. van Genderen	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	None (Self Study)	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	This Research Internship offers students the opportunity to enhance lab skills (such as a specific training), academic skills (such as writing a paper for publishing or participating in a conference with a posterpresentation or paper) or for studying special topics (such as capita selecta with (guest) lecturers). The supervisor needs to be part of the scientific staff within the Faculty of Electrical Engineering, Mathematics and Computer Science, this means they need to be a Full Professor, Assistant Professor or Associate Professor.	
Study Goals	To be formulated by the student in the Research Internship proposal (see education method) and to be approved beforehand by the supervisor and the MSc coordinator.	
Education Method	Before the start of the Research Internship, a proposal of the project needs to be provided according to a specified format; the format of this Research Internship proposal can be downloaded from the Brightspace page of the Research Internship. The following information is required: name of the TUD supervisor, proposed number of credits, a motivation for the extra project, learning goals and a description of the planned activities and output. This project proposal needs to be approved by the TUD supervisor, and the CESE MSc coordinator. When submitting the Individual Exam Programme to the Board of Examiners, the Research Internship proposal needs to be attached. After the assignment, a final report needs to be produced. The Board of Examiners may ask to see the report to check the study load and quality at the time of the final MSc exam. At most one Research Internship may be performed.	
Prerequisites	Note the following rule from the TER since Sep. 2024: The research Internship and the thesis project must have a different Supervisor.	
Assessment	The Research Internship and the resulting report is evaluated and graded by the supervisor. The grade is Pass/Fail. repair option: Improve and resubmit report Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

IFEEMCS520201	Advanced Interdisciplinary Artificial Intelligence Project	15
Responsible Instructor	J. Zatarain Salazar	
Responsible Instructor	Dr. P.K. Murukannaiah	
Contact Hours / Week x/x/x/x	10/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	FROM SEPTEMBER 2025, THIS COURSE WILL BE PART OF IFM4040 JOINT INTERDISCIPLINARY PROJECT (JIP) AND WILL NO LONGER BE OFFERED UNDER THIS COURSE CODE. ENROLLMENT FOR THE COURSE STARTS MARCH 17TH 2025 VIA BRIGHTSPACE (IFM4040) This course is aimed at MSc students from all TU Delft faculties who have already obtained AI-related knowledge and skills in previous courses, and who want to apply their skills to a real-world AI challenge. Prospective students, for example, will have completed an AI minor or have a good amount of AI knowledge from their bachelor (e.g. Mechanical Engineering), on topics such as data handling and machine learning OR possess demonstrated expertise in the domain relevant to a specific AI project on offer. This is an indicative guideline and we ask students that want to enroll for this course, to send a motivation letter and give an indication of their experience and knowledge on AI.	
Course Contents	FROM SEPTEMBER 2025, THIS COURSE WILL BE PART OF IFM4040 JOINT INTERDISCIPLINARY PROJECT (JIP) AND WILL NO LONGER BE OFFERED UNDER THIS COURSE CODE. ENROLLMENT FOR THE COURSE STARTS MARCH 17TH 2025 VIA BRIGHTSPACE (IFM4040) Artificial Intelligence (AI) is an interdisciplinary challenge. This Advanced Interdisciplinary Artificial Intelligence Project (AI2P) (IFEEMCS520201) is a 15EC MSc elective course aims to provide students with a unique experience to learn, develop and apply AI to a real-world challenge, working in an interdisciplinary project team. In this course you work with a group (4-6 students) on a large interdisciplinary project that is run by TU Delft AI Labs. [link: https://www.tudelft.nl/ai/tu-delft-ai-labs] Further, some of the projects may offer business cases from the industry. You and your fellow students will be offered to join workshops, tutorials and seminars organized by cutting edge researchers from TU Delft. You will also share kick-offs, reviews, and events with students from JIP (Joint Interdisciplinary Project). You find the projects for next academic here on Project Forum [link: https://projectforum.tudelft.nl/course_editions/109]	
Study Goals	Upon finishing this course, the student should be able to: 1. position an AI challenge in the broader literature, e.g., by performing a literature review on the AI challenge in a domain; 2. set-up a research strategy for an interdisciplinary AI project; 3. develop the AI solution according to the research strategy; 4. effectively collaborate with the team and supervisor; 5. evaluate and reflect on the research / developed AI solution; 6. assess the impact of deploying AI-based solutions and interventions on individuals, organizations, and society; and 7. report and present their research on a scientific level.	
Education Method	The students may work and discuss in groups. Each student is responsible for gathering, sharing and implementing information within the overall project. This allows for collaboration and communication as well as self-directed learning. At least once a week, the students meet with their supervisor. Peer feedback as well as feedback from the supervisor is organized on essential steps in the research project, which include: 1. Project proposal 2. Mid-term review, and 3. A scientific report, describing the problem, methods, results, and conclusion 4. A scientific presentation	
Assessment	The scientific report and the project presentation are assessed by the project supervisor and an external examiner. The assessment based on the clarity of presentation, the soundness of the methods used, and the conclusions drawn. Resit next quarter: A repair option is allowed only for students that received a fail mark (<5.8). Overall, the resit mark will be capped to 5.8.	

IFM4040	Joint Interdisciplinary Project	15
Responsible Instructor	Prof.dr.ir. J. Hellendoorn	
Instructor	Ir. B.J.E. de Bruin	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	English	
Expected prior knowledge	To enter JIP, a student should be a 2nd-year master student; have completed the first year of the masters degree programme and will enrol in the second year of the masters degree programme at TU Delft. be full time available (40 hours per week) in Q5; meet the application criteria: JIP application form, JIP motivation letter + top 3 company cases, CV Students will be selected by the JIP core staff based on motivation, study progress and recommendation.	
Course Contents	The Joint Interdisciplinary Project aims to prepare students to contribute to impactful technological solutions for industrial/governmental R&D and support one or more of the Global Sustainable development Goals or ToPsectoren in the Netherlands. Interdisciplinary and Intercultural teams of students work a dedicated ten weeks on a complex problem to realise an innovative and viable (primarily)technical solution. Team deliverables include a report, prototype, concept model and a business case to instigate continued developments. A company coach and JiP staff provide team guidance. Along the way, the companies and TU Delft offer students academic and industry expertise. The projects demand sound engineering working knowledge and experience with interdisciplinary and systems theory, knowledge and mindsets of innovation and entrepreneurial behaviour. The project brief is provided by renowned companies like Airbus, Embraer, Huisman,VanOord, Royal Haskoning DHV, Tahmo, Nationale Politie, Erasmus MC, and KLM.	
Study Goals	Meta-cognitive abilities attributable to interdisciplinary learning LO 1: Demonstrate disciplinary grounding, showing synthesis and integration of different disciplinary scientific and professional (technological) knowledge systems and critical evaluation skills of the approaches used to solve complex problems (Ilo;K) Scientific and intellectual development LO 2: Demonstrate the ability to carry out the analysis and evaluation of ethical, scientific and societal consequences of the proposed innovation (ILO: I) Research and design capabilities LO3: Demonstrate the ability to create reasonable and relevant output, in accordance with the academic standards of well conducted research, design, modelling and technical approaches for the problem addressed within JIP. (ILO: I,J) Collaboration and communication in an interdisciplinary team LO 4: Demonstrate behavioural competences and skills relevant for interdisciplinary teamwork and effective communication with different stakeholders (ILO: K) Self-adjustment and reflection capabilities LO 5: Demonstrate to carry out regular reflections on professional and personal development and being able to improve upon those reflections (ILO: K)	
Education Method	Full-time project work in an interdisciplinary team of about five students. The project work is interspersed with some just in time seminars and workshops about specialized subjects, methods or practical situations that play an essential role in interdisciplinary project work.	
Assessment	Students will meet (online) during the module and discuss their project with professionals, the company, and academia. The JIP core team will also guide the team process. The module includes three review sessions: Problem statement review the teams are required to present the scope of their problem in the form of a two-pager report. The main activity during this review is the team presentation of the academic staff, company coaches, JIP core staff and teams within the same cluster. The formative and 360 grades feedback means giving a head start in the execution of the project. The input is based on a rubric addressing interdisciplinary teamwork, scientific and technical rigour, responsible /ethical engineering, innovation and communication skills. Midterm review The teams present multiple solution concepts in a four-page report and a presentation to the academic staff, company coaches, JIP staff and students within the same cluster. Teams should demonstrate that the feedback based on the rubric has been considered and incorporate into the work leading up to the mid-term review. However, the Midterm review uses the same criteria as the problem definition, re-evaluated at a deeper level. In addition to the review sessions, the JiP team monitors individual growth. The monitoring is done by reviewing a team blog, which shows the team's progress to peers in JIP and beyond. The team blog is meant to get feedback and keep track of the decisions students have made while working together concerning the project and team process. The individual process is realised by a 360-degree team feedback tool and team sessions to jointly evaluate the team process and improve the individual professional and collaborative attitude. (4 Team blogs/4 Personal Reflection blogs/ 4 time 360-degree team feedback) Summative assessment on module level Final review the presentation is behind closed doors and is open for academic staff, JIP staff and the company coach(es). Note that; the grade for the project report/presentation/prototype/model/design (80% of the final grade) will only be given if the two earlier reports (the two-pager on the problem statement review and the four-pager of the midterm review) are handed in as well. The items are assessed according to a rubric addressing interdisciplinary teamwork, scientific and technical rigour, responsible /ethical engineering, innovation and communication skills. The assessors (company, academic and Jip staff) will score the rubric independently, after which a final grade is calibrated and discussed with the entire committee. The Problem def/mid-term reviews and the growth that has occurred during the process are incorporated in the final grade. The committee consists of minimally, the company coach, one academic staff member with expertise on the topic, 1 (academic chair), 1 Jip staff. Often an additional expert from the field and academic staff member is added.	

	The minimum grade for the summative assessment is =/> 5.8. Process grade is 20% The team blog/personal blog is evaluated according to a rubric on interdisciplinary teamwork, professional development, academic attitude, personal leadership. The deliverables are subject to a knockout system. Meaning the grade is insufficient if not delivered. Students cannot finalise the project without handing in the obligatory work.

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer and Embedded-Systems Engineering**

Language Courses 2024	
Introduction 1	Up to 6 credits may be spent on language courses. Placement tests showing the necessity to take one or more of these courses must be taken and submitted to the master coordinator.

TPM018A	English Grammar for the University	2
Module Manager	Drs. D.W. Laponder	
Instructor	S.F. Johnson	
Instructor	Drs. M.A. Swennen	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. K.I. van der Linden	
Responsible for assignments	Drs. D.W. Laponder	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	0/2/0/2	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	The course is designed for students who want to improve their English grammar with the aim of enhancing their writing skills. We use a university grammar that addresses the limitations many students face when interacting in English. The focus of this course is on improving your grammar range and accuracy to help you express complex thoughts and ideas more clearly.	
Study Goals	During this course, students will: 1. expand their basic understanding of sentence structure in academic discourse 2. learn how to use punctuation in academic writing 2. demonstrate an understanding of more complex grammatical structures 3. write a variety of sentence types needed to write an academic text 4. write paragraphs to practise targeted grammar structures 5. demonstrate an understanding of grammar through text analysis of academic articles	
Education Method	This is a 7-week course with a weekly 90-minute class. Homework typically involves grammar theory as well as text and written assignments. In-class activities tend to focus on text analysis and writing assignments.	
Literature and Study Materials	Grammar for Academic Purposes 2 - Steve Marshall.	
Practical Guide		
Prerequisites		
Assessment	Assignments and a Final Test in week 7. Students have to complete weekly assignments with set deadlines. Students who submit their weekly assignments late or not at all may be asked to leave the course. An 85% attendance record is required. This means students may miss no more than one session (preferably none). In the event of a missed class, the assignments for that particular week still need to be completed. Students receive their course credits when they complete four out of five weekly assignments at pass level and pass the Final Test in week 7.	
Enrolment / Application	Enrolment is through Brightspace.	
Remarks	The course will be taught if there is a minimum enrolment of 12 students.	
Elective	Yes	
Tags	English proficiency	
Targetgroup	Bachelor's and Master's students. No Placement Test is required to take the course.	
Category	BSc and MSc level	

TPM302B	Spoken English for Academic Purposes Advanced	2
Module Manager	Drs. M.A. Swennen	
Module Manager	Drs. K.I. van der Linden	
Instructor	S.F. Johnson	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. K.I. van der Linden	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	3/0/3/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Course Contents	This speaking course focuses on expanding and improving the language students need to participate at the university. A particular focus point of the course is the vocabulary needed at a higher level of English, both in an academic and a more general context. A range of sources for language input is offered, such as podcasts, videos, websites, or (academic) articles. The sessions revolve around student interaction.	
Study Goals	The first aim of the course is to improve and expand the language skills needed at a higher level of English. These include paraphrasing and sharing information with others, using polite language, giving feedback, and presenting information. Another study course goal is to expand and improve the students' general vocabulary.	
Education Method	Group and pair work, combined with (individual) assignments to do at home. Feedback on course components is given by the course lecturer and other students.	
Literature and Study Materials	To be announced in Brightspace.	
Practical Guide	This is a course for both BSc and MSc students who would like to expand and improve their English language skills.	
Assessment	Class participation and weekly assignments. An 80% attendance rate is obligatory, which means you may only miss one session. To keep your place in the group you must attend the first session. The assignments must be submitted each week before the deadline set by the instructor. You can drop the course with no penalty before the second session. Be aware of your "commitment" from the start, as you will still receive a grade [NV: Niet Verschenen--Not Attending], in Osiris even if you stop coming to class after session 2.	
Enrolment / Application	Before enrolling in a group you must first take our Placement Test. This is not an actual test but a tool to ensure that you are taking the course that is the most suitable for your level of English. When you have taken our Placement Test you will receive a course recommendation. With this recommendation you can sign up for this course. You can only register for courses that are recommended to you, or do not require a Placement Test. For more information on this 'test' please visit our website: www.itav.tudelft.nl Enrolment for the course (and the Placement Test) is via Brightspace. You do not need to register for this course in Osiris. Please note that you must also choose a group to reserve a place in the class. Type 'TPM302B' in Course search (Catalog). Please make sure you select the correct academic year and quarter. Register in the course AND in a group that suits your schedule.	
Remarks	The course will only be offered if there is a sufficient number of participants. Please note that it is not a presentation course.	
Elective	Yes	
Tags	English proficiency Group work Small groups	
Category	BSc and MSc level	

TPM305A	Writing a Masters Thesis in English	2
Module Manager	Drs. D.W. Laponder	
Instructor	S.F. Johnson	
Instructor	Drs. M.A. Swennen	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. K.I. van der Linden	
Responsible for assignments	Drs. D.W. Laponder	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This course is designed for students who have started their MSc thesis or are about to start on their graduation project. The main focus is on the writing process, which means that matters such as grammar, academic style and precision of formulation are considered at length. Particular attention is also paid to the importance of structure and organising ideas. Aspects of the writing process will be backed up by weekly reading and writing assignments.	
Study Goals	The course is primarily designed for students working on their Masters thesis. As the name of the course indicates, the focus throughout is on writing. Samples of your academic writing will be corrected in considerable detail and you will be taught how to structure your thesis. Plenty of tips will be given on ways of perfecting your English and invigorating your writing. Your systematic mistakes will be pointed out to you so that you can become critical about your own writing and go on to do your own editing.	
Education Method	Learning by doing and learning from lectures, corrections, peer review and tutor feedback.	
Books	Academic Writing for Graduate Students, Essential Tasks and Skills ,3rd edition by John M. Swales & Christine B. Feak. ISBN: 978-0-472-03475-8	
Assessment	Continuous assessment. Participants are to submit a number of text analysis and writing assignments in which they demonstrate their writing skills and their understanding of text composition and academic English. You may not submit texts that were previously submitted for any other course. Students who submit their weekly assignments late, repeatedly miss deadlines or lack commitment may be asked to leave the course. An 85% attendance record is required. This means students may miss no more than one session (preferably none). Students will receive their course credits when their work shows sufficient progress and weekly deadlines and all other conditions are met.	
Special Information	Enrolment is through Brightspace.	
Remarks	The course will only be offered when there is sufficient demand. If there are not enough enrolments, the course may be cancelled.	
Elective	Yes	
Tags	English proficiency	
Category	MSc level	

TPM306A	Writing Academic English using AI	2
Module Manager	Drs. K.I. van der Linden	
Instructor	Drs. M.A. Swennen	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	0/x/0/x	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	The Writing Academic English using AI course is designed to elevate students' writing proficiency through the strategic use of AI tools. Students learn the conventions of academic writing, such as text structure, paragraph structure, sentence structure, grammatical variation, and formal-vs-informal words. They will practise editing and proofreading with AI, focusing on improving grammar, style, and coherence. They also learn to critically evaluate the AI-generated responses. Additionally, the course will cover ethical considerations in the use of AI in writing, ensuring students understand issues related to plagiarism and data privacy.	
Study Goals	This course aims to familiarise students with academic writing conventions and show them how to use large language models to improve their writing.	
Education Method	This is a 7-week course with one lesson a week.	
Literature and Study Materials	Course material will be announced in Brightspace.	
Books	Course material will be announced in Brightspace.	
Prerequisites	Before enrolling in this course you must first take our Placement Test. When you have taken our Placement Test you will receive a course recommendation. With this recommendation you can sign up for this course. You can only register for courses that are recommended to you, or do not require a Placement Test. Enrolment for the course (and the Placement Test) is via Brightspace. You do not need to register for this course in Osiris. Please note that you must also choose a group to reserve a place in the class. Type 'TPM306A' in Course search (Catalog). Please make sure you select the correct academic year and quarter. Register in the course AND in a group that suits your schedule.	
Assessment	Continuous assessment. Participants submit various writing assignments in which they demonstrate their writing skills and their ability to critically evaluate the output given by AI tools.	
Remarks	The course will only be offered if there is a sufficient number of participants.	
Elective	Yes	
Tags	English proficiency	
Category	MSc level	

WM1115TU	Dutch Elementary 1	3
Module Manager	Drs. G. Hoezen	
Responsible for assignments	Drs. G. Hoezen	
Co-responsible for assignments	Dr. S.J. van Boxtel	
Contact Hours / Week x/x/x/x	3/3/3/3	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	Dutch English	
Required for	After this course, participants can continue learning in Dutch Elementary 2 (course code WM1116TU)	
Expected prior knowledge	No prior knowledge on the Dutch language needed. Basic knowledge of English is required. If you already know about 1000 words in Dutch or more, we advise you to sit a level test. Contact delftsemethode@tudelft.nl.	
Parts	Preparing lessons at home by reading, listening and repeating Dutch texts, speaking during conversation class. Continual testing through listening tests and gap filing tests.	
Course Contents	Dutch Elementary 1 is a basic Dutch language course aimed at every day communication. You will learn the language through reading, analysing and listening to texts on various topics as preparation for the lessons. The lessons themselves will be mainly devoted to speaking. Regular testing is also part of this course.	
Study Goals	The goal of this course is to acquire basic Dutch language skills on CEF level A1.	
Education Method	Interactive group sessions, individual work through online course materials.	
Literature and Study Materials	Bondi Sciarone e.a., "Nederlands voor buitenlanders. De Delftse Methode". Text book. 5th revised edition.	
Assessment	The final grade is compiled of grades for speaking skills during class and several tests and assignments during the course period: listening tests and written gap filling tests. Attendance also contributes to the final grade. If this final grade is not sufficient, students sit a written exam, a speaking test or a combination of both. These tests take place after the regular course period.	
Permitted Materials during Tests	No materials admitted during tests.	
Enrolment / Application	--- for MSc/BSc students --- There is a two-step-enrolment procedure. 1: Enrolment in Preliminary program for Dutch Elementary 1 course This programme takes place one week before the actual course. It is meant for receiving information about the learning method. On top of that, you will self-study the first two texts and sit the introductory test. If you pass, you can enrol for the actual course Dutch Elementary 1. 2: Enrolment in the actual course Dutch Elementary 1 (WM1115TU) You can enrol for one of the groups. --- for others (PhDs, employees) --- Go to the website www.dm.tudelft.nl for registration.	
Special Information	Brightspace, www.dm.tudelft.nl or delftsemethode@tudelft.nl	
Remarks	Regular courses take place at the campus in groups of max. 15 participants. There is an option to take this course as a self-study programme with a final exam + a speaking test. Contact the secretarys office for registration. delftsemethode@tudelft.nl More information: www.dm.tudelft.nl	
Elective	Yes	
Tags	Broad Diverse Small groups	
Targetgroup	Students and others who want to learn basic Dutch and who have little or no prior knowledge of Dutch. People who already know some Dutch (approx. 1000 words or more) can contact delftsemethode@tudelft.nl for a level test.	
Self Test	If you want to do self-study, you can register for a written test + speaking test by sending an email to delftsemethode@tudelft.nl	
Category	MSc level	

WM1116TU	Dutch Elementary 2	3
Module Manager	Drs. G. Hoezen	
Responsible for assignments	Drs. G. Hoezen	
Co-responsible for assignments	Dr. S.J. van Boxtel	
Contact Hours / Week x/x/x/x	3/3/3/3	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	Dutch	
Required for	After this course, participants can continue learning in Dutch Intermediate 1 (course code WM1117TU)	
Expected prior knowledge	A completed course Elementary 1 (WM1115TU). It is also possible to enrol if you have a similar level: CEF A2. In that case you have to sit a level test. Contact delftsemethode@tudelft.nl.	
Parts	Preparing lessons at home by reading, listening and repeating Dutch texts, speaking during conversation class. Continual testing through listening tests and gap filing tests.	
Course Contents	Dutch Elementary 2 is the second part of a basic Dutch language course aimed at every day communication. You will learn the language through reading, analysing and listening to texts on various topics as preparation for the lessons. The lessons themselves will be mainly devoted to speaking. Regular testing is also part of this course.	
Study Goals	The goal of this course is to acquire basic Dutch language skills on CEF level A2.	
Education Method	Interactive group sessions, individual work through online course materials.	
Literature and Study Materials	Bondi Sciarone e.a., "Nederlands voor buitenlanders. De Delftse Methode". Text book. 5th revised edition.	
Assessment	The final grade is compiled of grades for speaking skills during class and several tests and assignments during the course period: listening tests and written gap filling tests. Attendance also contributes to the final grade. If this final grade is not sufficient, students sit a written exam, a speaking test or a combination of both. These tests take place after the regular course period.	
Permitted Materials during Tests	No materials admitted during tests.	
Enrolment / Application	--- for MSc/BSc students --- Enrolment through Brightspace in the course Dutch Elementary 2 (WM1116TU) --- for others (PhDs, employees) --- Go to the website www.dm.tudelft.nl for registration.	
Special Information	Brightspace, www.dm.tudelft.nl or delftsemethode@tudelft.nl	
Remarks	Regular courses take place at the campus in groups of max. 15 participants. There is an option to take this course as a self-study programme with a final test + a speaking test.. Contact the secretarys office for registration. delftsemethode@tudelft.nl More information: www.dm.tudelft.nl	
Elective	Yes	
Tags	Broad Diverse	
Targetgroup	Students and others who want to improve their Dutch knowledge. If you are not sure whether is course is on the right level for you, please contact delftsemethode@tudelft.nl for a level test.	
Category	MSc level	

WM1117TU	Dutch Intermediate 1	3
Module Manager	Drs. G. Hoezen	
Responsible for assignments	Drs. G. Hoezen	
Co-responsible for assignments	Dr. S.J. van Boxtel	
Contact Hours / Week x/x/x/x	3/3/3/3	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	Dutch	
Required for	After this course, participants can continue learning in Dutch Intermediate 2. This is a paid course, also for students and will not give you ECTS credits.	
Expected prior knowledge	A completed course Elementary 2 (WM1116TU). It is also possible to enrol if you have a similar level: CEF A2+. In that case you have to sit a level test. Contact delftsemethode@tudelft.nl.	
Parts	Preparing lessons at home by reading, listening and repeating Dutch texts, speaking during conversation class. Continual testing through listening tests and gap filing tests.	
Course Contents	Dutch Intermediate 1 is the second part of a basic Dutch language course aimed at every day communication. You will learn the language through reading, analysing and listening to texts on various topics as preparation for the lessons. The lessons themselves will be mainly devoted to speaking. Regular testing is also part of this course.	
Study Goals	The goal of this course is to acquire basic Dutch language skills on CEF level A2.	
Education Method	Interactive group sessions, individual work through online course materials.	
Literature and Study Materials	Wesdijk en Blom Sciarone e.a., "Tweede Ronde, herziene editie. Nederlands voor buitenlanders. De Delftse Methode". Text book.	
Assessment	The final grade is compiled of grades for speaking skills during class and several tests and assignments during the course period: listening tests and written gap filling tests. Attendance also contributes to the final grade. If this final grade is not sufficient, students sit a written exam, a speaking test or a combination of both. These tests take place after the regular course period.	
Permitted Materials during Tests	No materials admitted during tests.	
Enrolment / Application	--- for MSc/BSc students --- Enrolment through Brightspace in the course Dutch Intermediate 1 (WM1117TU) --- for others (PhDs, employees) --- Go to the website www.dm.tudelft.nl for registration.	
Special Information	Brightspace, www.dm.tudelft.nl or delftsemethode@tudelft.nl	
Remarks	Regular courses take place at the campus in groups of max. 15 participants. We will also offer some groups in an online format. These groups have max. 9 participants. Check the website or Brightspace to see the current situation. There is an option to take this course as a self-study programme with a final test + speaking test. Contact the secretarys office for registration. delftsemethode@tudelft.nl More information: www.dm.tudelft.nl	
Elective	Yes	
Tags	Broad Diverse	
Targetgroup	Students and others who want to improve their Dutch knowledge. If you are not sure whether is course is on the right level for you, please contact delftsemethode@tudelft.nl for a level test.	
Category	MSc level	

Dr.ir. Z. Al-Ars

Unit Elektrotechn., Wisk. & Inform.
Department Computer Engineering

Telephone +31 15 27 89097
Room 36.HB 09.080

Dr. S.M. Alavi

Unit Elektrotechn., Wisk. & Inform.
Department Electronics

Telephone +31 15 27 83826
Room 36.HB 18.270

G. de Albuquerque Gleizer

Department Team Sander Wahls

Telephone +31 15 27 83371
Room 34.C-2-300

Unit Mech, Maritime & Materials Eng
Department Team Manuel Mazo Jr

Telephone +31 15 27 83371
Room 34.C-2-300

M. Babaie

Department QCD/Babaie Lab

Telephone +31 15 27 86162

Unit Elektrotechn., Wisk. & Inform.
Department Electronics

Telephone +31 15 27 86162

Dr.ir. K. Batselier

Unit Mech, Maritime & Materials Eng
Department Team Kim Batselier

Telephone +31 15 27 82725

Dr.ing. R.K. Bishnoi

Department Computer Engineering

Telephone +31 15 27 86101
Room 36.HB 09.140

Dr.ir. A.J.J. van den Boom

Unit Mech, Maritime & Materials Eng
Department Team Ton van den Boom

Telephone +31 15 27 84052
Room 34.C-3-180

Prof.dr. P.A.N. Bosman

Unit Elektrotechn., Wisk. & Inform.
Department Algorithmics

Dr.ir. P.A. Bouter

Department Algorithmics

Unit Elektrotechn., Wisk. & Inform.
Department Algorithmics

Dr. E.F.M. van Boven

Unit Elektrotechn., Wisk. & Inform.
Department Network Architectures and Services

Telephone +31 15 27 87853
Room 36.HB 09.240

Dr. S.J. van Boxtel

Unit	Techniek, Bestuur & Management
Department	Instituut voor Talen en Academische Vaardigheden
Telephone	+31 15 27 81893
Room	31.c0.090

Ir. S.C. Bregman

Department	Team Jan-Willem van Wingerden
Telephone	+31 15 27 84993
Room	34.C-1-300

Ir. B.J.E. de Bruin

Department	Cognitive Robotics

Department	Cognitive Robotics

Dr. S.S. Chakraborty

Department	Programming Languages
Telephone	+31 15 27 85867
Room	28.3.E260

R. Corvino

Department	Embedded Systems

Dr. M.A. Costea

Department	Programming Languages

Dr. S.D. Cotofana

Unit	Elektrotechn., Wisk. & Inform.
Department	Computer Engineering
Telephone	+31 15 27 86267
Room	36.HB 10.040

A. Dabiri

Department	Team Azita Dabiri
Telephone	+31 15 27 82259

Dr. C. De Bacco

Department	Network Architectures and Services

Prof.dr.ir. B.H.K. De Schutter

Unit	Mech, Maritime & Materials Eng
Department	Delft Center for Systems and Control
Telephone	+31 15 27 85113

Dr. J.E.A.P. Decouchant

Department	Data-Intensive Systems
Telephone	+31 15 27 83804

Dr. C. Della Santina

Department	Learning & Autonomous Control
Telephone	+31 15 27 81630
Room	34.F-2-320

Dr. E. Demirovi

Department	Algorithmics
Telephone	+31 15 27 87177
Room	28.4.E400

Prof.dr. A. van Deursen

Unit	Elektrotechn., Wisk. & Inform.
Department	Software Engineering
Telephone	+31 15 27 82486
Room	28.1.W600

Dr. S. Du

Department	Electronic Instrumentation
Telephone	+31 15 27 83203
Room	36.HB 15.310

S. Dumani

Department	Algorithmics
Telephone	+31 15 27 88941
Room	28.2.W600

Ir. J.B. Dönszelmann

Department	Computer Science & Engineering-Teaching Team

Dr. Z. Erkin

Unit	Elektrotechn., Wisk. & Inform.
Department	Cyber Security
Telephone	+31 15 27 86283
Room	29.02.250

C. Errando Herranz

Department	QID/Herranz Lab

Department	Quantum Circuit Architectures and Technology

Dr. S. Feld

Department	QCD/Feld Group
Telephone	+31 15 27 82462

Department	Quantum Circuit Architectures and Technology
Telephone	+31 15 27 82462
Room	36.HB 11.060

Dr. R. Ferrari

Unit	Mech, Maritime & Materials Eng
Department	Team Riccardo Ferrari
Telephone	+31 15 27 83519
Room	34.C-2-260

Dr.ir. M.C.R. Fieback

Department	Computer Engineering
Telephone	+31 15 27 86191
Room	36.HB 10.250

Unit	Elektrotechn., Wisk. & Inform.
Department	Computer Engineering
Telephone	+31 15 27 86191
Room	36.HB 10.250

Prof.dr. P.J. French

Unit	Elektrotechn., Wisk. & Inform.
Department	Bio-Electronics
Telephone	+31 15 27 84729
Room	36.HB 16.310

Dr.ing. C.P. Frenkel

Department	Electronic Instrumentation
Telephone	+31 15 27 83447

C. Galuzzi

Department	Computer Engineering

C. Gao

Department	Electronics

Prof.dr.ir. G. Gaydadjiev

Department	Computer Engineering
Telephone	+31 15 27 86168
Room	36.HB 10.030

Department	Computer Engineering
Telephone	+31 15 27 86168
Room	36.HB 10.030

Dr.ing. A.B. Gebregiorgis

Department	Computer Engineering
Telephone	+31 15 27 86553
Room	36.HB 11.280

Dr. J.C. van Gemert

Unit	Elektrotechn., Wisk. & Inform.
Department	Pattern Recognition and Bioinformatics
Telephone	+31 15 27 88434
Room	28.6.E340

Dr.ir. A.J. van Genderen

Unit	Elektrotechn., Wisk. & Inform.
Department	Computer Engineering
Telephone	+31 15 27 86217
Room	36.HB 10.030

M.R.P. van Gog

Department	Onderwijs en Studentenzaken
Telephone	+31 15 27 86300

Dr.ing. S. Grammatico

Unit	Mech, Maritime & Materials Eng
Department	Team Sergio Grammatico
Telephone	+31 15 27 83593

Dr.ir. H.J. Griffioen

Department	Cyber Security
Telephone	+31 15 27 87662

Department	Cyber Security
Telephone	+31 15 27 87662

Dr. N.M. Gürel

Department	Pattern Recognition and Bioinformatics

Prof.dr.ir. S. Hamdioui

Unit	Elektrotechn., Wisk. & Inform.
Department	Computer Engineering
Telephone	+31 15 27 83643
Room	36.HB 10.160

Prof.dr.ir. J. Hellendoorn

Department Cognitive Robotics

Telephone +31 15 27 89007
Room 34.F-2-440

Unit Mech, Maritime & Materials Eng
Department College van Bestuur

Telephone +31 15 27 89007
Room 34.F-2-440

Drs. G. Hoezen

Unit Techniek, Bestuur & Management
Department Instituut voor Talen en Academische Vaardigheden

Telephone +31 15 27 82915
Room 31.C0.110

Dr. G. Iosifidis

Department Networked Systems

Dr. R. Ishihara

Unit QuTech
Department QID/Ishihara Lab

Telephone +31 15 27 88498
Room 36.HB 11.070

Unit Elektrotechn., Wisk. & Inform.
Department Quantum Circuit Architectures and Technology

Telephone +31 15 27 88498
Room 36.HB 11.070

S. Izadkhast

Unit Elektrotechn., Wisk. & Inform.
Department Electrical Engineering Education

Telephone +31 15 27 85788
Room 36.LB 01.250

Dr.ir. M. Jafarian

Department Team Matin Jafarian

Telephone +31 15 27 82422

Dr.ir. G.J.M. Janssen

Unit Elektrotechn., Wisk. & Inform.
Department Signal Processing Systems

Telephone +31 15 27 86736
Room 36.HB 17.060

S.F. Johnson

Unit Techniek, Bestuur & Management
Department Instituut voor Talen en Academische Vaardigheden

Telephone +31 15 27 82793

Prof.dr.ir. T. Keviczky

Unit Mech, Maritime & Materials Eng
Department Team Tamas Keviczky

Telephone +31 15 27 82928
Room 34.C-3-310

M. Khosravi

Department Team Khosravi

Telephone +31 15 27 87563

M.A. Kitsak

Department	Network Architectures and Services
Telephone	+31 15 27 85386

Dr. M. Kok

Unit	Mech, Maritime & Materials Eng
Department	Team Manon Kok
Telephone	+31 15 27 81529

Dr.ir. J.H. Krijthe

Department	Pattern Recognition and Bioinformatics
Telephone	+31 15 27 88433
Room	28.5.E820

Prof.dr.ir. F.A. Kuipers

Unit	Elektrotechn., Wisk. & Inform.
Department	Networked Systems
Telephone	+31 15 27 81347
Room	28.2.E160

G.K. Kuzmanov

Department	Computer Engineering

Dr. G. Lan

Department	Embedded Systems
Room	28.2.W720

Prof.dr. K.G. Langendoen

Unit	Elektrotechn., Wisk. & Inform.
Department	Embedded Systems
Telephone	+31 15 27 87666

Drs. D.W. Laponder

Department	Instituut voor Talen en Academische Vaardigheden
Telephone	+31 15 27 89799

Dr. K. Liang

Department	Cyber Security
Telephone	+31 15 27 87656

Prof.dr.ir. H.X. Lin

Unit	Elektrotechn., Wisk. & Inform.
Department	Mathematical Physics
Telephone	+31 15 27 87229
Room	36.HB 05.060

Drs. K.I. van der Linden

Unit	Techniek, Bestuur & Management
Department	Instituut voor Talen en Academische Vaardigheden
Telephone	+31 15 27 86384
Room	31.c0.140

Dr. R. Litjens

Unit	Elektrotechn., Wisk. & Inform.
Department	Network Architectures and Services
Telephone	+31 15 27 86622

Dr. M. Mazo Espinosa

Unit	Mech, Maritime & Materials Eng
Department	Team Manuel Mazo Jr
Telephone	+31 15 27 88131
Room	34.C-2-250

Dr.ir. N.P. van der Meijs

Unit	Elektrotechn., Wisk. & Inform.
Department	Signal Processing Systems
Telephone	+31 15 27 86258
Room	36.HB 17.310

Dr. N. Mohan

Department	Networked Systems

Y. Murakami

Department	Discrete Wiskunde en Optimalisering
Telephone	+31 15 27 88617

Dr. P.K. Murukannaiah

Department	Interactive Intelligence
Telephone	+31 15 27 82291
Room	28.4.W900

F.A. Muñoz Muñoz

Department	High Voltage Technology Group
Telephone	+31 15 27 81994

Ir. R.A.C.J. Noldus

Unit	Elektrotechn., Wisk. & Inform.
Department	Network Architectures and Services
Telephone	+31 15 27 86654
Room	36.HB 09.280

Dr. F.A. Oliehoek

Unit	Elektrotechn., Wisk. & Inform.
Department	Sequential Decision Making
Telephone	+31 15 27 87494
Room	28.4.W600

Dr.ir. T.H. Oosterkamp

Dr. D. Palagin

Department	Numerical Analysis
Room	36.HB 03.270

Dr. P. Pawelczak

Unit	Elektrotechn., Wisk. & Inform.
Department	Embedded Systems
Telephone	+31 15 27 87491
Room	28.2.W680

O.E. Popa

Department	Ethiek & Filosofie van Techniek
Telephone	+31 15 27 88609

Drs. P.C. Post

Unit	Techniek, Bestuur & Management
Department	Instituut voor Talen en Academische Vaardigheden
Telephone	+31 15 27 83175
Room	31.c0.040

Dr.ir. J.A. Pouwelse

Unit Elektrotechn., Wisk. & Inform.
Department Data-Intensive Systems

Telephone +31 15 27 82539
Room 36.HB 07.290

Dr. F. Sebastiano

Department QCD/Sebastiano Lab

Telephone +31 15 27 83342

Unit Elektrotechn., Wisk. & Inform.
Department Quantum Circuit Architectures and Technology

Telephone +31 15 27 83342
Room 36.HB 11.080

M.A. Sharifi Kolarijani

Department Team Amin Sharifi Kolarijani

Telephone +31 15 27 85331
Room 34.C-3-200

Unit Mech, Maritime & Materials Eng
Department Team Peyman Mohajerin Esfahani

Telephone +31 15 27 85331
Room 34.C-3-200

Ing. A.M.J. Slats

Unit Elektrotechn., Wisk. & Inform.
Department Electrical Engineering Education

Telephone +31 15 27 88787
Room 36.LB 01.260

G. Smaragdakis

Department Cyber Security

Telephone +31 15 27 82555

Prof.dr.ir. D. Spinellis

Department Software Engineering

Room 28.1.W900

Dr. J. Sun

Department Pattern Recognition and Bioinformatics

P.P. Sundaramoorthy

Department Group De Breuker

Telephone +31 15 27 83817

Department Signal Processing Systems

Telephone +31 15 27 83817

Department Astrodynamics & Space Missions

Telephone +31 15 27 83817

Unit Elektrotechn., Wisk. & Inform.
Department Support Microelectronics

Telephone +31 15 27 83817

Drs. M.A. Swennen

Unit Techniek, Bestuur & Management
Department Instituut voor Talen en Academische Vaardigheden

Telephone +31 15 27 82912
Room 31.C0.160

Dr.ir. M. Taouil

Unit	Elektrotechn., Wisk. & Inform.
Department	Computer Engineering
Telephone	+31 15 27 83591
Room	36.HB 09.070

Dr. D.M.J. Tax

Unit	Elektrotechn., Wisk. & Inform.
Department	Pattern Recognition and Bioinformatics
Telephone	+31 15 27 84232
Room	28.5.W860

Dr. S.H. Tindemans

Unit	Elektrotechn., Wisk. & Inform.
Department	Intelligent Electrical Power Grids
Telephone	+31 15 27 84487
Room	36.LB 03.190

N. Tömen

Department	Pattern Recognition and Bioinformatics
Telephone	+31 15 27 88247
Room	28.6.E360

Dr.ir. H. Vahedi

Department	DC systems, Energy conversion & Storage
Telephone	+31 15 27 89086

Prof.dr.ir. P.F.A. Van Mieghem

Unit	Elektrotechn., Wisk. & Inform.
Department	Network Architectures and Services
Telephone	+31 15 27 82397
Room	36.HB 09.270

Dr. R.R. Venkatesha Prasad

Unit	Elektrotechn., Wisk. & Inform.
Department	Networked Systems
Telephone	+31 15 27 86272

Ir. S.E. Verwer

Unit	Elektrotechn., Wisk. & Inform.
Department	Algorithmics
Telephone	+31 15 27 88435
Room	28.6.E100

T.J. Viering

Department	Pattern Recognition and Bioinformatics
Telephone	+31 15 27 88212
Room	28.6.E160

Dr.ir. R. Vismara

Unit	Elektrotechn., Wisk. & Inform.
Department	Photovoltaic Materials and Devices
Telephone	+31 15 27 88902
Room	36.LB 03.460

Dr.ir. J.H. Weber

Unit	Elektrotechn., Wisk. & Inform.
Department	Discrete Wiskunde en Optimalisering
Telephone	+31 15 27 81698
Room	36.HB 04.040

Prof.dr. M.M. de Weerdt

Unit	Elektrotechn., Wisk. & Inform.
Department	Algorithmics
Telephone	+31 15 27 84516
Room	28.4.E100

Dr. S.D.C. Wehner

Unit	QuTech
Department	QID/Wehner Group
Telephone	+31 15 27 87746

Unit	Elektrotechn., Wisk. & Inform.
Department	Quantum Computer Science
Telephone	+31 15 27 87746

Drs. J.A. Weitenberg

Department	Instituut voor Talen en Academische Vaardigheden
Telephone	+31 15 27 87914

Prof.dr.ir. J.W. van Wingerden

Unit	Mech, Maritime & Materials Eng
Department	Team Jan-Willem van Wingerden
Telephone	+31 15 27 81720
Room	34.C-2-340

Dr.ir. J.S.S.M. Wong

Unit	Elektrotechn., Wisk. & Inform.
Department	Computer Engineering
Telephone	+31 15 27 81099
Room	36.HB 10.050

Dr. N. Yorke-Smith

Unit	Elektrotechn., Wisk. & Inform.
Department	Algorithmics
Telephone	+31 15 27 83895
Room	28.4.E220

J. Zatarain Salazar

Department	Beleidsanalyse
Telephone	+31 15 27 81670

M.A. Zuñiga Zamalloa

Unit	Elektrotechn., Wisk. & Inform.
Department	Networked Systems
Telephone	+31 15 27 82538
Room	28.2.E080